

UNIVERSITY OF GRONINGEN
FACULTY OF SCIENCE AND ENGINEERING

MATHEMATICS: PROBABILITY & STATISTICS

Bayesian Quadrature: Theory and Comparative Simulation Study



Bachelor's Project Mathematics

Date: 12th of December 2025

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Abstract

Bayesian quadrature (BQ) treats numerical integration as a statistical inference problem by modelling the integrand as a Gaussian process. Applying the integration operator to the Gaussian process and conditioning on observed function values yields a posterior distribution for the value of the integral, rather than merely a point estimate. This thesis presents a formulation of BQ based on the closure of Gaussian processes under bounded linear operators.

A systematic comparison is conducted between Monte Carlo (MC) integration, fixed-grid BQ, and uncertainty-sampling BQ under three scenarios: estimating the kernel and hyperparameters once, re-estimating hyperparameters only, and re-estimating both at each iteration. The methods are evaluated on a one-dimensional integral, the marginal likelihood of a Gaussian Bayesian linear regression model, and a synthetic trimodal marginal likelihood with a Gaussian mixture model as likelihood.

The BQ methods outperform MC integration only in the well-behaved one-dimensional setting. In more complex applications, kernel misspecification, integrand structure, and instability in hyperparameter estimation reduce the effectiveness of BQ, whereas MC integration remains robust. These results highlight that BQ is a powerful but assumption-sensitive method, whose performance critically depends on the suitability of the chosen prior structure encoded through the kernel.

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1 Introduction

Numerical integration, i.e., quadrature, plays a central role in statistics and related fields. For example, computing marginal likelihoods or marginalising over model parameters often requires evaluating integrals of the form

$$I = \int_{\mathbb{X}} f(\mathbf{x}) \, d\nu(\mathbf{x}), \quad (1)$$

which typically cannot be expressed in closed analytical form. Classical quadrature techniques perform well for smooth integrands but suffer from the curse of dimensionality. In contrast, Monte Carlo (MC) integration provides a dimension-free convergence rate of order $\mathcal{O}(n^{-1/2})$, regardless of the smoothness of f . However, this is the slowest rate attainable under any reasonable model of the integrand (Hennig et al., 2022). MC integration belongs to the class of stochastic numerical integration methods. Both classical and stochastic quadratures use only limited information about the integrand f and the integral I .

Probabilistic integration offers an alternative perspective, treating numerical integration as a statistical inference problem. Larkin (1972) and O’Hagan (1991) were the first to formalise the modern framework. In this approach, the integrand f is modelled as a random function, typically a Gaussian process (GP), and is then passed through the integration operator. This method is known as Bayesian quadrature (BQ) or Bayesian Monte Carlo (Ghahramani and Rasmussen, 2002). In contrast to classical and stochastic quadratures, BQ produces a posterior distribution over the value of the integral I , rather than merely a point estimate.

BQ relies on the closure property of Gaussian processes under bounded linear operators. However, this property is often omitted, stated informally, or proven only for simplified cases in the BQ literature; other papers present highly technical treatments with limited empirical evaluation (Matsumoto and Sullivan, 2024). When an integration operator is applied to a GP, closed-form expressions for the posterior mean and variance of the integral can be obtained under suitable conditions.

This thesis compares MC integration with several BQ methods that differ in their adaptive sampling strategies and kernel hyperparameter estimation schemes. Kernel hyperparameters are typically not chosen manually but estimated via maximum likelihood (Rasmussen and Williams, 2008). While BQ papers often employ the squared-exponential, Matérn, or rational quadratic kernels, studies that consider broader kernel families generally rely on more advanced BQ techniques. Moreover, hyperparameters are frequently estimated only once, or the estimation protocol is not explicitly stated. To address this, we compare MC integration with BQ under both fixed-grid and adaptive sampling schemes, in three scenarios: estimating the kernel and hyperparameters once, re-estimating hyperparameters only, and re-estimating both kernel and hyperparameters.

First, we present the mathematical background necessary to understand the closure properties of Gaussian processes, together with an introduction to GPs and BQ. Second, kernel and hyperparameter estimation via maximum likelihood is discussed, accompanied by simulation results and adaptive sampling strategies. Finally, MC integration and the different BQ methods are compared across three settings: a one-dimensional integral example, the marginal likelihood of a Gaussian Bayesian linear regression model with Gaussian priors, and a synthetic marginal likelihood example where the likelihood is given by a Gaussian mixture model.

2 Mathematical Background

In this section, we present the mathematical background necessary for understanding Gaussian processes and Bayesian quadrature, as well as their theoretical foundations.

2.1 Expectation and Covariance as Linear Functionals

In this subsection, we introduce expectation and covariance operators. An overview of probability theory is provided in Appendix 6.3.

2.1.1 Expectation Operator

The expectation of a random variable $g(X) : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow (\bar{\mathbb{R}}, \bar{\mathcal{B}})$ defined in equation (92) can be viewed as a linear functional $\mathbb{E} : L^1(\mathbb{R}^d, \mathbb{P}_X) \rightarrow \mathbb{R}$. See sections 6.4 and 6.5 in the appendix for a short overview about linear operators and L^p spaces, respectively. Since $L^1(\mathbb{R}^d, \mathbb{P}_X)$ is a Banach space with $\|g\|_1 = \int_{\mathbb{R}} |g| d\mathbb{P}_X$ as norm, the linear functional is bounded because:

$$|\mathbb{E}[g]| = \left| \int_{\mathbb{R}} g(x) d\mathbb{P}_X(x) \right| \leq \int_{\mathbb{R}} |g(x)| d\mathbb{P}_X(x) = \|g\|_1 < \infty. \quad (2)$$

Therefore, $\mathbb{E}$ belongs to the (topological) dual space of $L^1(\mathbb{R}^d, \mathbb{P}_X)$, i.e., $\mathbb{E} \in (L^1(\mathbb{R}^d, \mathbb{P}_X))^*$.

However, we can generalize the expectation operator $\mathbb{E}$ on $\mathbb{R}$ to general separable Banach spaces $\mathbb{B}$ (Steinwart, 2025). We require separability because it implies that the Borel σ -algebra $\mathcal{B}(\mathbb{B})$ generated by the norm topology on $\mathbb{B}$ is equal to the smallest σ -algebra on $\mathbb{B}$ for which all bounded linear functionals on $\mathbb{B}$ are measurable. Note that $\mathbb{R}^d$, $d \in \mathbb{N}$, is a separable Banach space and a Hilbert space is a special case of a Banach space. Let $\mathbf{X} : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow \mathbb{B}$ be a random element taking values in a separable Banach space $\mathbb{B}$. A random element is the generalisation of random vectors in $\mathbb{R}^d$ to general Banach spaces, and it is Bochner-measurable. The random element $\mathbf{X}$ is Bochner-integrable if and only if the following is satisfied:

$$\int_{\Omega} \|\mathbf{X}\|_{\mathbb{B}} d\mathbb{P} < \infty. \quad (3)$$

This is a generalisation of integrability for real-valued functions. Hence, the space $\mathcal{L}^p(\Omega, \nu)$ for real-valued measurable functions $f : \Omega \rightarrow \mathbb{R}$ such that $\|f\|_p < \infty$ for $1 \leq p < \infty$ defined in equation 148 is generalised to random elements on Banach spaces by:

$$\mathcal{L}^p(\Omega, \mathbb{B}, \mathbb{P}) := \left\{ \text{measurable } \mathbf{X} : \Omega \rightarrow \mathbb{B} \mid \|\mathbf{X}\|_{\mathbb{B}, p} := \left(\int_{\Omega} \|\mathbf{X}(\omega)\|_{\mathbb{B}}^p d\mathbb{P}(\omega) \right)^{\frac{1}{p}} < \infty \right\} \quad (4)$$

Therefore, $\mathcal{L}^1(\Omega, \mathbb{B}, \mathbb{P})$ is a semi-normed linear space containing all Bochner-integrability random elements $\mathbf{X} : \Omega \rightarrow \mathbb{B}$. The corresponding L^p space is constructed analogously as in section 6.5. Other results such as Hölder's inequality can also be generalized (Potter, 2016). Now, we can define the expectation operator of a random variable on a separable Banach space (van Neerven, 2008).

Definition 2.1. (Expectation Operator on Banach Space – Bochner Integral)

Let $\mathbf{X} : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow \mathbb{B}$ be a random element. If $\mathbf{X}$ is Bochner-integrable, then the expectation operator $\mathbb{E} : L^1(\Omega, \mathbb{B}, \mathbb{P}) \rightarrow \mathbb{B}$ exists and is defined by:

$$\mathbb{E}[\mathbf{X}] = \int_{\Omega} \mathbf{X} d\mathbb{P}. \quad (5)$$

This integral is called the ($\mathbb{B}$ -valued) Bochner integral.

Observe that the expectation operator in the definition above is bounded and hence continuous because:

$$\|\mathbb{E}[\mathbf{X}]\|_{\mathbb{B}} = \left\| \int_{\Omega} \mathbf{X} d\mathbb{P} \right\|_{\mathbb{B}} \leq \int_{\Omega} \|\mathbf{X}\|_{\mathbb{B}} d\mathbb{P} = \mathbb{E}[\|\mathbf{X}\|_{\mathbb{B}}], \quad \forall \mathbf{X} \in \mathcal{L}^1(\mathbb{B}, \mathbb{P}). \quad (6)$$

Therefore, $\mathbb{E} \in (L^1(\Omega, \mathbb{B}, \mathbb{P}))^* = \mathfrak{B}(L^1(\Omega, \mathbb{B}, \mathbb{P}), \mathbb{B})$.

An important result concerning the interchange of the integration operator and a bounded linear operator is presented in the following theorem (Potter, 2016).

Theorem 2.1. (Interchange Integration and Bounded Linear Operator)

Let $\mathbb{B}_1$ and $\mathbb{B}_2$ be separable Banach spaces.

If $T : \mathbb{B}_1 \rightarrow \mathbb{B}_2$ is a bounded linear operator and $\mathbf{X} \in L^1(\Omega, \mathbb{B}_1, \mathbb{P})$, then the map $T \circ \mathbf{X} = T(\mathbf{X}) : \Omega \rightarrow \mathbb{B}_2$ satisfies the following:

1. $T(\mathbf{X}) \in L^1(\mathbb{B}_1, \mathbb{P})$;
2. $\int_{\Omega} T[\mathbf{X}] \, d\mathbb{P} = T \left[\int_{\Omega} \mathbf{X} \, d\mathbb{P} \right] \iff \mathbb{E}[T[\mathbf{X}]] = T[\mathbb{E}[\mathbf{X}]]$.

Notice that this implies that we can interchange an integral and a bounded linear operator of for example random vectors. The following proposition presents an important bounded linear operators: the integration operator with integration kernel κ (Levermore, 2009).

Proposition 2.1.

Let $(X, \mathcal{A}_X, \nu_X)$ and $(Y, \mathcal{A}_Y, \nu_Y)$ be σ -finite measure spaces, let $1 \leq q \leq \infty$, and let $1 \leq p, \tilde{p} \leq \infty$ be such that $\frac{1}{p} + \frac{1}{\tilde{p}} = 1$. Assume $\kappa : X \times Y \rightarrow \mathbb{R}$ is $(\mathcal{A}_X \otimes \mathcal{A}_Y)$ -measurable satisfying $\kappa(x, \cdot) \not\equiv 0, \forall x \in X$. If $x \mapsto \|\kappa(x, \cdot)\|_{\tilde{p}} \in L^q(X, \nu_X)$, then $T : L^p(Y, \nu_Y) \rightarrow L^q(X, \nu_X)$ defined by:

$$(T[f])(x) := \int_Y \kappa(x, y) \cdot f(y) \, d\nu_Y(y), \quad \forall x \in X, f \in L^p(Y, \nu_Y), \quad (7)$$

is a well-defined bounded linear operator.

A proof is provided in Appendix 6.8.1. Furthermore, κ in equation (7) is called the integration kernel and $k(x, \cdot) \in L^{\tilde{p}}(Y, \nu_Y), \forall x \in X$.

If the integration kernel in equation (7) is set to $\kappa(x, \cdot) \equiv 1, \forall x \in X$, then we obtain the following (simple) integration operator $T : L^p(Y, \nu_Y) \rightarrow \mathbb{R}$ given by:

$$(T[f])(x) = T[f] = \int_Y f(y) \, d\nu_Y(y) \in \mathbb{R}. \quad (8)$$

This integral is well-defined since:

$$\int_Y |f(y)| \, d\nu_Y(y) = \|f\|_1 \leq \|f\|_p < \infty, \quad \forall 1 \leq p \leq \infty, \quad (9)$$

when $(Y, \mathcal{A}_Y, \nu_Y)$ is finite, i.e., $\nu_Y(Y) < \infty$, and $\nu_Y(Y) \geq 1$. A special case of this result for probability spaces is presented in the following proposition.

Proposition 2.2.

Let $(\Omega, \mathcal{A}, \nu)$ be a measure space with $\nu(\Omega) = 1$. If $f \in L^p(\Omega, \nu)$ for some $1 \leq p < \infty$, then $f \in L^1(\Omega, \nu)$ and the following inequality holds:

$$\|f\|_1 = \int_{\Omega} |f| \, d\nu \leq \left(\int_{\Omega} |f|^p \, d\nu \right)^{1/p} = \|f\|_p. \quad (10)$$

A proof is provided in Appendix 6.8.1. Observe that since $\nu(\Omega) = 1$, $(\Omega, \mathcal{A}, \nu)$ is a probability space and ν is usually denoted by $\mathbb{P}$ in this case. Proposition 2.2 shows that any function on $L^1(\Omega, \nu)$, where $(\Omega, \mathcal{A}, \nu)$ is probability space, is integrable. However, this result also holds when $(\Omega, \mathcal{A}, \nu)$ is a finite measure space, since equation (180) becomes $\|f\|_1 = \|f \cdot g\|_1 \leq \|f\|_p < \infty$ and equation (181) becomes $\|f\|_1 \leq \|f\|_{\infty} \cdot \nu(\Omega) < \infty$.

Theorem 2.2. (Fubini's Theorem)

Let $(\Omega_1, \mathcal{A}_1, \nu_1)$ and $(\Omega_2, \mathcal{A}_2, \nu_2)$ be σ -finite measure spaces. Assume that $f : (\Omega_1 \times \Omega_2, \mathcal{A}_1 \otimes \mathcal{A}_2) \rightarrow (\bar{\mathbb{R}}, \bar{\mathcal{B}})$ is an integrable function, i.e., f is in $L^1(\Omega_1 \times \Omega_2, \nu_1 \otimes \nu_2)$.

Then, the order of integration can be interchanged:

$$\int_{\Omega_1 \times \Omega_2} f \, d(\nu_1 \otimes \nu_2) = \int_{\Omega_1} \left(\int_{\Omega_2} f \, d\nu_2 \right) d\nu_1 = \int_{\Omega_2} \left(\int_{\Omega_1} f \, d\nu_1 \right) d\nu_2. \quad (11)$$

See [De Snoo and Winkler \(2018\)](#) for the proof. Note that $\int_{\Omega_2} f \, d\nu_2 \in L^1(\Omega_1, \nu_1)$ and $\int_{\Omega_1} f \, d\nu_1 \in L^1(\Omega_2, \nu_2)$ ([Potter, 2016](#)). When f is non-negative and measurable but not necessarily integrable, then equation (11) still holds. In this case, the result is known as Tonelli's Theorem.

In chapter 2.3, Fubini's theorem will be applied to functions $f \in L^1(\Omega \times \mathbb{X}, \mathbb{P} \otimes \nu)$, where $(\Omega, \mathcal{E}, \mathbb{P})$ is a probability space and $(\mathbb{X} \subseteq \mathbb{R}^d, \mathcal{B}^d, \nu)$. Hence, the integrability assumption in Theorem 2.2 is always satisfied, and the order of integration in equation (11) can be interchanged.

Corollary 2.1.

Let $(\Omega, \mathcal{E}, \mathbb{P})$ be a probability space and $(S, \mathcal{S}, \nu)$ a σ -finite measure space. If $X : \Omega \times S \rightarrow \mathbb{R}$ is integrable with respect to $\mathbb{P} \otimes \nu$, then:

$$\mathbb{E} \left[\int_S X(\cdot, s) \, d\nu(s) \right] = \int_S \mathbb{E}[X(\cdot, s)] \, d\nu(s). \quad (12)$$

Proof. $X(\omega, s)$ is a random variable, i.e., a measurable function. Hence, by Fubini's Theorem with $f(\omega, s) := \mathbf{X}(\omega, s)$ and using the fact that the expectation is defined as an integral, the order of integration can be interchanged. $\square$

2.1.2 Covariance Operator

Definition 2.1 generalizes the notion of an expectation operator to general Banach spaces. In this subsection, we generalize the notion of a covariance operator to a special type of Banach spaces, namely Hilbert spaces. First, we are going to define covariance on $\mathbb{R}^d$. Note that $\mathbb{H} = \mathbb{R}^d$ is a Hilbert space with the standard Euclidean inner product $\langle \cdot, \cdot \rangle$. For example, for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$, $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^d u_i \cdot v_i = \mathbf{u}^T \mathbf{v}$. Let $(\Omega, \mathcal{E}, \mathbb{P})$ be a probability space, and let $\mathbf{X} : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow \mathbb{R}^d$ be a random vector. Suppose that the Bochner integral of $\mathbf{X}$:

$$\mathbf{m} := \mathbb{E}[\mathbf{X}] = \int_{\mathbb{H}=\mathbb{R}^d} \mathbf{x} \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}), \quad (13)$$

exists. That is, the first moment of $\mathbf{X}$ is finite. If, additionally, the following integral:

$$\boldsymbol{\Sigma} := \text{Cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \mathbf{m})(\mathbf{X} - \mathbf{m})^T] = \int_{\mathbb{H}=\mathbb{R}^d} (\mathbf{x} - \mathbf{m})(\mathbf{x} - \mathbf{m})^T \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \in \mathbb{R}^{d \times d}, \quad (14)$$

exists, then $\boldsymbol{\Sigma} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the covariance operator. So, in finite-dimensional Euclidean space, the covariance operator is the covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$. Hence, the second moment of $\mathbf{X}$ is finite, i.e., $\mathbf{X} \in L^2(\Omega, \mathbb{P})$. Next, post-multiply $\boldsymbol{\Sigma}$ by a vector $\mathbf{v} \in \mathbb{R}^d$:

$$\begin{aligned} \boldsymbol{\Sigma} \mathbf{v} &= \int_{\mathbb{R}^d} (\mathbf{x} - \mathbf{m})(\mathbf{x} - \mathbf{m})^T \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \, \mathbf{v} = \int_{\mathbb{R}^d} (\mathbf{x} - \mathbf{m})(\mathbf{x} - \mathbf{m})^T \mathbf{v} \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \\ &= \int_{\mathbb{R}^d} (\mathbf{x} - \mathbf{m}) \langle (\mathbf{x} - \mathbf{m}), \mathbf{v} \rangle \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \\ &= \mathbb{E}[(\mathbf{X} - \mathbf{m}) \langle (\mathbf{X} - \mathbf{m}), \mathbf{v} \rangle]. \end{aligned} \quad (15)$$

Consequently, for $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^d$:

$$\begin{aligned} \text{Cov}(\mathbf{v}_1^T \mathbf{X}, \mathbf{v}_2^T \mathbf{X}) &= \text{Cov}(\langle \mathbf{v}_1, \mathbf{X} \rangle, \langle \mathbf{v}_2, \mathbf{X} \rangle) = \text{Cov}(\langle \mathbf{X}, \mathbf{v}_1 \rangle, \langle \mathbf{X}, \mathbf{v}_2 \rangle) \\ &= \mathbb{E}[(\mathbf{v}_1^T \mathbf{X} - \mathbb{E}[\mathbf{v}_1^T \mathbf{X}]) (\mathbf{v}_2^T \mathbf{X} - \mathbb{E}[\mathbf{v}_2^T \mathbf{X}])] \\ &= \mathbb{E}[(\mathbf{v}_1^T (\mathbf{X} - \mathbf{m})) (\mathbf{v}_2^T (\mathbf{X} - \mathbf{m}))] = \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{v}_1 \rangle \langle \mathbf{X} - \mathbf{m}, \mathbf{v}_2 \rangle] \\ &= \int_{\mathbb{R}^d} \langle \mathbf{x} - \mathbf{m}, \mathbf{v}_1 \rangle \langle \mathbf{x} - \mathbf{m}, \mathbf{v}_2 \rangle \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \\ &= \int_{\mathbb{R}^d} \langle \mathbf{v}_1, \mathbf{x} - \mathbf{m} \rangle \langle \mathbf{x} - \mathbf{m}, \mathbf{v}_2 \rangle \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \\ &= \int_{\mathbb{R}^d} \mathbf{v}_1^T (\mathbf{x} - \mathbf{m})(\mathbf{x} - \mathbf{m})^T \mathbf{v}_2 \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \\ &= \mathbf{v}_1^T \boldsymbol{\Sigma} \mathbf{v}_2 \in \mathbb{R}_{\geq 0}. \end{aligned} \quad (16)$$

Note that the symmetry property of the real-valued inner product is used multiple times. Furthermore, this implies that the covariance operator Σ is positive semi-definite. It is important to distinguish between the covariance function $\text{Cov}(\cdot)$ and the covariance operator Σ . The covariance function $\text{Cov} : L^2(\omega, \mathbb{P}) \rightarrow L^2(\omega, \mathbb{P})$ is a symmetric bilinear form, while $\Sigma : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a linear operator. In addition, the map $B : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ defined by $B(\mathbf{u}, \mathbf{v}) := \mathbf{u}^T \Sigma \mathbf{v}$ is a symmetric bilinear form associated with Σ , as discussed in Appendix Section 6.4.

Next, we generalize the covariance to an arbitrary separable Hilbert space $(\mathbb{H}, \langle \cdot, \cdot \rangle_{\mathbb{H}})$. Let $\mathbf{X} : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow \mathbb{H}$ be a $\mathbb{H}$ -valued random variable with finite second moment $\mathbb{E}[\|\mathbf{X}\|_{\mathbb{H}}^2] < \infty$, i.e. $\mathbf{X} \in L^2(\Omega, \mathbb{H}, \mathbb{P})$. Its mean is a special case of the Bochner integral in equation (5):

$$\mathbf{m} := \mathbb{E}[\mathbf{X}] = \int_{\Omega} \mathbf{X} \, d\mathbb{P} \in \mathbb{H}. \quad (17)$$

The covariance of $\mathbf{X}$ is defined as the symmetric bilinear form $\text{Cov} : \mathbb{H} \times \mathbb{H} \rightarrow \mathbb{R}$ given by:

$$\begin{aligned} \text{Cov}(\mathbf{u}, \mathbf{v}) &:= \text{Cov}(\langle \mathbf{X}, \mathbf{u} \rangle, \langle \mathbf{X}, \mathbf{v} \rangle) \\ &= \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{u} \rangle \langle \mathbf{X} - \mathbf{m}, \mathbf{v} \rangle] = \int_{\mathbb{H}} \langle \mathbf{x} - \mathbf{m}, \mathbf{u} \rangle \langle \mathbf{x} - \mathbf{m}, \mathbf{v} \rangle \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}), \end{aligned} \quad (18)$$

where $\mathbb{P}_{\mathbf{X}}$ is the distribution of $\mathbf{X}$ on $\mathbb{H}$. $\text{Cov}(\cdot, \cdot)$ is not a linear functional, but $\mathbf{u} \mapsto \text{Cov}(\mathbf{u}, \mathbf{v})$ is a bounded linear functional.

Lemma 2.1.

The map $F_{\mathbf{u}} : \mathbb{H} \rightarrow \mathbb{R}$ defined by $F_{\mathbf{u}}(\mathbf{v}) := \text{Cov}(\mathbf{u}, \mathbf{v})$, for arbitrary $\mathbf{u} \in \mathbb{H}$, is a bounded linear functional on $\mathbb{H}$.

A proof is provided in Appendix 6.8.1. Hence, by Riesz-Fréchet Representation Theorem, i.e., Theorem 6.4, there exists a unique element $\mathbf{y} := \Sigma \mathbf{u} \in \mathbb{H}$ such that for arbitrary $\mathbf{u} \in \mathbb{H}$:

$$F_{\mathbf{u}}(\mathbf{v}) = \text{Cov}(\mathbf{u}, \mathbf{v}) = \langle \Sigma \mathbf{u}, \mathbf{v} \rangle, \quad \forall \mathbf{v} \in \mathbb{H}. \quad (19)$$

This defines a covariance operator Σ on $\mathbb{H}$.

Proposition 2.3.

The linear operator $\Sigma : \mathbb{H} \rightarrow \mathbb{H}$ defined by:

$$\Sigma[\mathbf{u}] = \Sigma \mathbf{u} := \mathbb{E}[(\mathbf{X} - \mathbf{m})(\mathbf{X} - \mathbf{m}, \mathbf{u})] = \mathbb{E}[(\mathbf{X} - \mathbf{m})(\mathbf{X} - \mathbf{m}, \mathbf{u})]. \quad (20)$$

In particular, Σ is a linear, bounded, self-adjoint and positive semi-definite operator.

If $\mathbb{H} = \mathbb{R}^d$, then Σ coincides with the usual covariance matrix. Further, note that similar to equation (16):

$$\langle \Sigma \mathbf{u}, \mathbf{v} \rangle = \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{u} \rangle \langle \mathbf{X} - \mathbf{m}, \mathbf{v} \rangle] = \text{Cov}(\langle \mathbf{X}, \mathbf{u} \rangle, \langle \mathbf{X}, \mathbf{v} \rangle). \quad (21)$$

[Bogachev \(1998\)](#) generalizes the notion of covariance operators to Banach spaces and even locally convex spaces.

Moreover, for any $\mathbf{u} \in \mathbb{H}$, the real-valued random variable $\langle \mathbf{X}, \mathbf{u} \rangle$ has variance:

$$\text{Var}(\langle \mathbf{X}, \mathbf{u} \rangle) = \text{Cov}(\langle \mathbf{X}, \mathbf{u} \rangle, \langle \mathbf{X}, \mathbf{u} \rangle) = \langle \Sigma \mathbf{u}, \mathbf{u} \rangle. \quad (22)$$

In particular, if $\mathbb{H} = \mathbb{R}^d$, then $\text{Var}(\mathbf{u}^T \mathbf{X}) = \mathbf{u}^T \Sigma \mathbf{u}$. Additionally, the equation shows that the covariance operator Σ is positive-semidefinite, because:

$$\langle \Sigma \mathbf{u}, \mathbf{u} \rangle \geq 0, \quad \forall \mathbf{u} \in \mathbb{H}. \quad (23)$$

2.2 Gaussian Measure

In this subsection, we define the Gaussian measure, i.e., Gaussian distribution, in both the univariate and multivariate setting. In addition, the most important properties of the Gaussian distribution for Gaussian processes and Bayesian quadrature are summarized.

2.2.1 Univariate Gaussian Distribution

An absolutely continuous random variable $X : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B})$ is Gaussian if $\mathbb{P}_X = G := \mathcal{N}_1(m, \sigma^2)$, where G is a Gaussian measure with mean $m \in \mathbb{R}$ and variance $\sigma^2 \in \mathbb{R}_{>0}$. Therefore, a Gaussian random variable is a random variable with a Gaussian distribution (Bogachev, 1998). Frequently, the fact that X follows a Gaussian distribution is denoted by $X \sim \mathcal{N}_1(m, \sigma^2)$.

The following definition of a Gaussian distribution measure G follows Rosenthal (2009) and Tao (2011).

Definition 2.2. (Gaussian Measure)

A measure $G : (\mathbb{R}, \mathcal{B}) \rightarrow [0, \infty]$ on the space $(\mathbb{R}, \mathcal{B})$ is called a Gaussian measure with mean $m \in \mathbb{R}$ and variance $\sigma^2 \in \mathbb{R}_{>0}$, if:

$$\begin{aligned} G(B) &= \int_B \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left\{-\frac{(x-m)^2}{2\sigma^2}\right\} dx \\ &= \int_{\mathbb{R}} p_X(x) \cdot \mathbb{I}_B(x) dx, \quad B \in \mathcal{B}, \end{aligned} \tag{24}$$

where $p_X : (\mathbb{R}, \mathcal{B}) \rightarrow [0, \infty)$ is the Gaussian PDF of X :

$$p_X(x) := \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left\{-\frac{(x-m)^2}{2\sigma^2}\right\}. \tag{25}$$

A brief discussion of the relationship between G , the Radon–Nikodym derivative, and the expectation of a Gaussian random variable X is provided in Appendix 6.3.5.

2.2.2 Multivariate Gaussian Distribution

Our goal is to extend the univariate Gaussian distribution defined for a single random variable to the multivariate case, where we consider a random vector $\mathbf{X}$.

Let $\boldsymbol{\mu} \in \mathbb{R}^d$ be an arbitrary vector, and let $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ be a symmetric, positive semidefinite matrix. Every such matrix $\boldsymbol{\Sigma}$ can be written as:

$$\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T, \quad \mathbf{L} \in \mathbb{R}^{d \times d}. \tag{26}$$

There are several possible choices for $\mathbf{L}$, for example the Cholesky factor. The Cholesky decomposition of a symmetric, positive semidefinite matrix $\boldsymbol{\Sigma}$ is $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T$, where $\mathbf{L}$ is a lower triangular matrix called the Cholesky factor. An eigendecomposition of $\boldsymbol{\Sigma}$ given by $\boldsymbol{\Sigma} = \tilde{\mathbf{L}}\tilde{\mathbf{L}}^T = \mathbf{O}\mathbf{D}^{1/2}\mathbf{O}^T$, where $\mathbf{O}$ is an orthogonal matrix and $\mathbf{D}$ a diagonal matrix containing the eigenvalues of $\boldsymbol{\Sigma}$, with $\tilde{\mathbf{L}} := \mathbf{O}\mathbf{D}^{1/2}\mathbf{O}^T$ is another possible choice. However, the computation of the Cholesky factor $\mathbf{L}$ is considered numerically stable. In comparison, not using fast numerical approximations, computing the Cholesky factor $\mathbf{L}$ takes time $\mathcal{O}(d^3/6)$ while computing an eigendecomposition takes time $\mathcal{O}(d^3)$. Therefore, we use the Cholesky decomposition as the method of choice (Rasmussen and Williams, 2008).

The following definition of a multivariate Gaussian distribution random vector follows van der Vaart (1997).

Definition 2.3.

A random vector $\mathbf{X}$ is said to be multivariate Gaussian distributed with mean vector $\boldsymbol{\mu} \in \mathbb{R}^d$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$, i.e., $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}) =: G_d$, if it has the same distribution as the following vector:

$$\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}, \tag{27}$$

where $\mathbf{L}$ is a matrix such that $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T$, and the elements of the vector $\mathbf{Z} \in \mathbb{R}^d$ are independent standard Gaussian random variables, i.e., $Z_i \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1)$, $\forall i \in \{1, \dots, d\}$.

Hence, Gaussian measures on $\mathbb{R}^d$ can be described as the images of the standard Gaussian measure on $\mathbb{R}^d$ under the affine mapping $\mathbf{Z} \mapsto \boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$. An alternative but equivalent definition is given by Bogachev (1998).

Definition 2.4.

A probability measure $\mathbb{P}_{\mathbf{X}}$ on $(\mathbb{R}^d, \mathcal{B}^d)$ is Gaussian, if for every linear functional $f : \mathbb{R}^d \rightarrow \mathbb{R}$, the push-forward measure $\mathbb{P}_{\mathbf{X}} \circ f^{-1}$ is a Gaussian measure.

Equivalently, a random vector $\mathbf{X} \in \mathbb{R}^d$ follows a Gaussian distribution, if for every linear functional $f : \mathbb{R}^d \rightarrow \mathbb{R}$ the random vector $f(\mathbf{X})$ follows a Gaussian distribution. Furthermore, notice that $\mathbb{P}_{\mathbf{X}} = G_d$.

The Gaussian measure G_d has a PDF $f_{\mathbf{X}}$ if and only if the covariance matrix Σ is non-singular (van der Vaart, 1997; Bogachev, 1998).

Proposition 2.4. *A vector $\mathbf{X} \in \mathbb{R}^d$ is multivariate-Gaussian distributed for a non-singular Σ if and only if it has the PDF $f_{\mathbf{X}} : \mathbb{R}^d \rightarrow \mathbb{R}$ given by:*

$$p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} \cdot \sqrt{\det(\Sigma)}} \cdot \exp \left\{ -\frac{1}{2} \cdot (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}. \quad (28)$$

Note that the notation $\mathbf{X} \sim p_{\mathbf{X}}(\mathbf{x}) = \mathcal{N}_d(\mathbf{X}; \boldsymbol{\mu}, \Sigma)$ is sometimes used to as PDF notation.

2.2.3 Properties of Gaussian Distribution

An important property of the Gaussian measure is that it is closed under finite-dimensional transformations, in particular under matrix pre-multiplications and affine transformations.

Corollary 2.2.

Let $\mathbf{A} : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be an arbitrary matrix, i.e. $\mathbf{A} \in \mathbb{R}^{m \times d}$, and let $\mathbf{b} \in \mathbb{R}^d$ be an arbitrary vector. If $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$, then $\mathbf{A}\mathbf{X} + \mathbf{b} \sim \mathcal{N}_m(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\Sigma\mathbf{A}^T)$

A proof can be found in Appendix 6.8.2. Furthermore, Gaussian measures are closed under marginalisation.

Theorem 2.3. (Gaussian Distribution is Closed under Marginalisation)

Let $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)^T \in \mathbb{R}^d$ be a random vector with $\mathbf{X}_1 \in \mathbb{R}^m$ and $\mathbf{X}_2 \in \mathbb{R}^{d-m}$.

If $\mathbf{X} \sim \mathcal{N}_d(\mathbf{X}; \boldsymbol{\mu}, \Sigma)$ with:

$$\boldsymbol{\mu} := \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix} := \begin{bmatrix} \mathbb{E}[\mathbf{X}_1] \\ \mathbb{E}[\mathbf{X}_2] \end{bmatrix} \quad \& \quad \Sigma := \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} := \begin{bmatrix} \text{Cov}(\mathbf{X}_1) & \text{Cov}(\mathbf{X}_1, \mathbf{X}_2) \\ \text{Cov}(\mathbf{X}_2, \mathbf{X}_1) & \text{Cov}(\mathbf{X}_2) \end{bmatrix}, \quad (29)$$

then the marginal distribution $\mathbb{P}_{\mathbf{X}_1} := \mathbb{P} \circ \mathbf{X}_1^{-1}$ of $\mathbf{X}$ is Gaussian:

$$\mathbb{P}_{\mathbf{X}_1} = \mathcal{N}_m(\mathbf{X}_1; \boldsymbol{\mu}_1, \Sigma_{11}). \quad (30)$$

Or equivalently, $\mathbf{X}_1 \sim \mathcal{N}_m(\mathbf{X}_1; \boldsymbol{\mu}_1, \Sigma_{11})$.

Proof. Define $\mathbf{A} := \begin{bmatrix} \mathbf{I}_m & \mathbf{O} \end{bmatrix} \in \mathbb{R}^{m \times d}$, where $\mathbf{I}_m$ is the $m \times m$ identity matrix and $\mathbf{O} \in \mathbb{R}^{m \times (d-m)}$ a matrix of zeros, and define $\mathbf{b} := \mathbf{0}$. Then, by Corollary 2.2:

$$\mathbf{A}\mathbf{X} = \mathbf{X}_1 \sim \mathcal{N}_m(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\Sigma\mathbf{A}^T) \stackrel{d}{=} \mathcal{N}_m(\boldsymbol{\mu}_1, \Sigma_{11}). \quad (31)$$

□

Hence, when we are interested in only a finite subset of m variables denoted by $\mathbf{X}_1$ from a larger collection of d jointly Gaussian variables $\mathbf{X}$, the corresponding marginal distribution $\mathbb{P}_{\mathbf{X}_1}$ can be obtained directly by selecting the appropriate elements of the mean vector $\boldsymbol{\mu}$ and covariance matrix Σ . On the one hand, this illustrates that the covariance structure encodes only limited dependence information. This enables us to model arbitrarily large, potentially infinite-dimensional, collections of variables, since any finite subset remains tractable (Hennig et al., 2022).

We are not only interested in the marginal behaviour of a subset of Gaussian random variables, but also in the distribution of one subset conditioned on another.

Theorem 2.4. (Conditional Distribution of Gaussian Distribution)

Let $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)^T \in \mathbb{R}^d$ be a random vector with $\mathbf{X}_1 \in \mathbb{R}^m$ and $\mathbf{X}_2 \in \mathbb{R}^{d-m}$.

If $\mathbf{X} \sim \mathcal{N}_d(\mathbf{X}; \boldsymbol{\mu}, \Sigma)$ with $\boldsymbol{\mu}$ and non-singular Σ as in equation (29), then the conditional distribution of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$ is also Gaussian distribution:

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \sim \mathcal{N}_m(\mathbf{X}_1; \boldsymbol{\mu}_{1|2}, \Sigma_{1|2}), \quad (32)$$

i.e., $\mathbb{P}_{\mathbf{x}_1|\mathbf{x}_2=\mathbf{x}_2} = \mathcal{N}_m(\boldsymbol{\mu}_{1|2}, \boldsymbol{\Sigma}_{1|2})$, with:

$$\begin{aligned}\boldsymbol{\mu}_{1|2} &= \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2); \\ \boldsymbol{\Sigma}_{1|2} &= \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}.\end{aligned}\tag{33}$$

Note that the non-singularity assumption of the covariance matrix $\boldsymbol{\Sigma}$ is not always satisfied. However, we assume that $\boldsymbol{\Sigma}$ is non-singular.

2.3 Gaussian Process

In this subsection, Gaussian processes are introduced and discussed. In addition, their most important property for Bayesian quadrature is explained.

2.3.1 Motivation & Definition

The Gaussian process (GP) framework provides a probabilistic approach to function approximation. Suppose that our goal is to approximate an unknown function $f : \mathbb{X} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$, and that we can collect $n \in \mathbb{N}$ possibly noisy observations $y_i = f(\mathbf{x}_i) + \epsilon_i$, $i \in \{1, \dots, n\}$, where ϵ_i denotes the noise term. The collection of our observed datapoints is denoted by:

$$\mathcal{D} := \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\} =: \{\mathbf{X}, \mathbf{y}\}.\tag{34}$$

This problem of approximating an unknown function is relevant in both statistics and supervised machine learning. The objective of the latter is to learn about an input-output mapping $f : \mathbf{x}_i \mapsto y_i$ from observational data, which is typically partitioned into a training and a test dataset. When the output values are continuous or discrete, the supervised learning problem is referred to as a regression or classification problem, respectively. Roughly speaking, the aim in statistics is to infer about relationships between variables, while the aim of machine learning is to predict new values (Rasmussen and Williams, 2008). In a classical regression problem, we assume that the n observed datapoints have the following form:

$$y_i = f(\mathbf{x}_i) + \epsilon_i, \quad i \in \{1, \dots, n\},\tag{35}$$

where $\mathbf{x}_i \in \mathbb{X} \subseteq \mathbb{R}^d$ are explanatory variables, $y_i \in \mathbb{R}$ are response variables, and ϵ_i are i.i.d. noise terms with $\mathbb{E}[\epsilon_i] = 0$ and homoskedastic variance $\text{Var}(\epsilon_i) = \sigma^2 \in \mathbb{R}_{\geq 0}$. The ordinary least squares (OLS) technique assumes that f is linear, i.e., $f(\mathbf{x}_i) = \mathbf{x}_i^T \cdot \boldsymbol{\beta}$ for some $\boldsymbol{\beta} \in \mathbb{X} \subseteq \mathbb{R}^d$. The parameter $\boldsymbol{\beta}$ is estimated by minimizing the sum of squared residuals

$$\hat{\boldsymbol{\beta}}_{OLS} = \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \cdot \boldsymbol{\beta}).\tag{36}$$

Therefore, OLS yields the following estimator of the unknown function f :

$$\hat{f}(\mathbf{x}_i) = \mathbf{x}_i^T \cdot \hat{\boldsymbol{\beta}}_{OLS}.\tag{37}$$

OLS is a parametric model that assumes the unknown function f lies in a finite-dimensional linear space. However, this is a relatively restrictive assumption, because such a finite-dimensional parametric form cannot capture all non-linear, smooth, periodic, or otherwise complicated relationships. Consequently, there remains uncertainty about the correctness and usefulness of the model for describing the underlying relationship and for predicting new function values $f(\mathbf{x}_*)$, for new input values $\mathbf{x}_* \in \mathbb{R}^d \setminus \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. The GP model provides a probabilistic approach to address this limitation. Rasmussen and Williams (2008) present two common equivalent formulations of the GP model: the weight-space view and the function-space view.

The weight-space view projects the inputs $\mathbf{x}_i$ into a (possibly high-dimensional) feature space through a set of basis functions $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^k$. In addition, it puts a Gaussian prior distribution on the weight vector $\boldsymbol{\beta}$. Hence, the setup is as follows:

$$f(\mathbf{x}_i) = \phi(\mathbf{x}_i)^T \cdot \boldsymbol{\beta}, \quad \boldsymbol{\beta} \sim \mathcal{N}_d(\mathbf{0}, \boldsymbol{\Sigma}),\tag{38}$$

where Σ is a prior covariance matrix. Observe that in the weight-space view, randomness of f is induced via the randomness in weights β . Using Bayes's theorem, this setup yields Gaussian posterior distributions for both the weights and any predicted function value:

$$\beta \mid \mathbf{X}, \mathbf{y} \text{ and } f(\mathbf{x}_*) \mid \mathbf{x}_*, \mathbf{X}, \mathbf{y} \text{ are Gaussian distributed.} \quad (39)$$

This is the Bayesian linear regression formulation of the GP model.

On the other hand, the function-space view places a prior directly over functions rather than over parameters. That is, it places a prior probability measure on the space of possible functions that f could be, i.e., on $\mathcal{F} := \{f : \mathbb{X} \rightarrow \mathbb{R}\}$, where higher probabilities are given to functions that are considered more likely a priori. For example, if we believe that f is a periodic function, then we choose a prior on f that assigns relatively high probabilities to periodic functions. However, this prior imposes a preference bias about structural properties of f such as smoothness and periodicity. Furthermore, this approach introduces the problem that there exists an uncountable set of possible functions for f . To computationally deal with uncountably many possible functions, we require that the joint behaviour of f is well-defined only at a finite number of function evaluations. In particular, we require that any finite collection of function evaluations follows a joint Gaussian distribution. Hence, a GP can be formally defined as a stochastic process whose finite-dimensional marginal distributions are multivariate Gaussians.

In Appendix 6.6, stochastic processes including Kolmogorov's Extension Theorem are briefly explained. Let $(\Omega, \mathcal{E}, \mathbb{P})$ be a probability space. The following function:

$$\begin{aligned} f : \mathbb{X} \times \Omega &\rightarrow \mathbb{R} \\ (\mathbf{x}, \omega) &\mapsto f(\mathbf{x}, \omega) \end{aligned} \quad (40)$$

defined as a collection of random variables:

$$f := \{f(\mathbf{x}) : \mathbf{x} \in \mathbb{X}\} \quad (41)$$

is a stochastic process. Observe that for a fixed $\omega \in \Omega$, $\mathbf{x} \mapsto f(\mathbf{x}, \omega)$ is a sample functions of the stochastic process. In addition, for a fixed $\mathbf{x} \in \mathbb{X}$, $\omega \mapsto f(\mathbf{x}, \omega)$ is a random variable. Hence, a random variable $f(\mathbf{x})$ is equal to $f(\mathbf{x}, \omega)$ for a fixed $\omega \in \Omega$. Using the function space $\mathbb{R}^{\mathbb{X}} = \{\text{all (possible) functions } \phi : \mathbb{X} \rightarrow \mathbb{R}\}$ and the σ -algebra $\mathcal{B}^{\mathbb{X}}$ defined in equation (164), a stochastic process $f = \{f(\mathbf{x}) : \mathbf{x} \in \mathbb{X}\}$ can be equivalently represented as $f : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow (\mathbb{R}^{\mathbb{X}}, \mathcal{B}^{\mathbb{X}})$. Observe that $\mathbb{R}^{\mathbb{X}}$ is equal to the previously defined $\mathcal{F} = \{f : \mathbb{X} \rightarrow \mathbb{R}\}$.

Analogously to equation (165), the distribution of the stochastic process f is the push-forward measure on the function space $\mathbb{R}^{\mathbb{X}}$:

$$\mathbb{P}_f := \mathbb{P} \circ f^{-1}. \quad (42)$$

$\mathbb{P}_f$ measures how likely every possible function $\phi \in \mathbb{R}^{\mathbb{X}}$ is to be realized as a sample function of the stochastic process f . More specifically, for any measurable set of functions B , $\mathbb{P}_f(B)$ is the probability that the sample function $f(\cdot, \omega) \in \mathbb{R}^{\mathbb{X}}$ belongs to B :

$$\mathbb{P}_f(B) = \mathbb{P}(\{\omega \in \Omega \mid f(\cdot, \omega) \in B\}), \quad B \in \mathcal{B}^{\mathbb{X}}. \quad (43)$$

Note that $\mathcal{B}^{\mathbb{X}}$ is the smallest σ -algebra that makes all evaluation $\Psi_{\mathbf{x}} : \mathbb{R}^{\mathbb{X}} \rightarrow \mathbb{R}$ defined by $\Psi_{\mathbf{x}}(\phi) := \phi(\mathbf{x})$ measurable. So, $\mathbb{P}_f$ is a probability measure that gives the distribution over the functions in the function space $(\mathbb{R}^{\mathbb{X}}, \mathcal{B}^{\mathbb{X}})$.

Now, we can formally define a Gaussian process.

Definition 2.5. (Gaussian Process ($\mathcal{GP}$))

Let $(\Omega, \mathcal{E}, \mathbb{P})$ be a probability space, and let $f = \{f(\mathbf{x}) : \mathbf{x} \in \mathbb{X}\}$ be a stochastic process.

The stochastic process f is a Gaussian process with mean function $m : \mathbb{X} \rightarrow \mathbb{R}$ and a positive definite covariance kernel $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$, denoted by:

$$f \sim \mathcal{GP}(m, k), \quad (44)$$

if for any finite subset $\mathbf{X} := \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \mathbb{X}$, the finite-dimensional random vector $(f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))^T \in \mathbb{R}^n$ has the following multivariate distribution:

$$\mathbf{f}_{\mathbf{X}} := (f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))^T \sim \mathcal{N}_n(\mathbf{m}_{\mathbf{X}}, \mathbf{K}_{\mathbf{XX}}), \quad (45)$$

with mean vector $\mathbf{m}_{\mathbf{X}} = (m(\mathbf{x}_1), \dots, m(\mathbf{x}_n))^T \in \mathbb{R}^n$ and covariance matrix $\mathbf{K}_{\mathbf{XX}} = [k(\mathbf{x}_i, \mathbf{x}_j)]_{i,j=1}^n \in \mathbb{R}^{n \times n}$.

The mean function $m : \mathbb{X} \rightarrow \mathbb{R}$ and the covariance kernel $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ in definition 2.5 are defined as:

$$\begin{aligned} m(\mathbf{x}) &:= \mathbb{E}[f(\mathbf{x})] \\ &= \int_{\Omega} f(\mathbf{x}, \omega) \, d\mathbb{P}(\omega) \in \mathbb{R}, \end{aligned} \quad (46)$$

and

$$\begin{aligned} k(\mathbf{x}_1, \mathbf{x}_2) &:= \text{Cov}(f(\mathbf{x}_1), f(\mathbf{x}_2)) \\ &= \mathbb{E}[(f(\mathbf{x}_1) - m(\mathbf{x}_1)) \cdot (f(\mathbf{x}_2) - m(\mathbf{x}_2))] \\ &= \int_{\Omega} (f(\mathbf{x}_1, \omega) - m(\mathbf{x}_1)) \cdot (f(\mathbf{x}_2, \omega) - m(\mathbf{x}_2)) \, d\mathbb{P}(\omega) \in \mathbb{R}, \end{aligned} \quad (47)$$

for $\mathbf{x}, \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{X} \subseteq \mathbb{R}^d$.

Definition 2.5 implies that if f is a Gaussian process, then there exists a mean function m and a covariance kernel k . [Kanagawa et al. \(2018\)](#) state that there is a one-to-one correspondence between $f \in \mathcal{GP}(m, k)$ and pairs (m, k) of mean functions m and covariance kernels k . Hence, for any mean function m and positive definite covariance kernel k , there exists a corresponding Gaussian process $f \sim \mathcal{GP}(m, k)$. So, the specification of a particular m and k implicitly leads to the specification of the corresponding $\mathcal{GP}(m, k)$. In practice, it is important to understand the properties of the $\mathcal{GP}(m, k)$, such as smoothness and periodicity, which are induced by the specification of m and k . Often, the mean function is set to zero: $m \equiv 0$. However, care must be taken when specifying k , because the covariance matrix $\mathbf{K}_{\mathbf{X}\mathbf{X}}$ may be singular which implies non-existence of the Gaussian PDF.

The well-known stochastic process Brownian motion, i.e., Wiener process, is a Gaussian process with mean function $m \equiv 0$ and the following covariance kernel:

$$k(\mathbf{x}_1, \mathbf{x}_2) := \theta^2 \cdot (\min\{\mathbf{x}_1, \mathbf{x}_2\} - \tau), \quad (48)$$

with constants $\theta \in \mathbb{R}_{\geq 0}$ and $\tau \in \mathbb{R}$ ([Hennig et al., 2022](#)). This illustrates the power of Gaussian processes, since Brownian motion is such a fundamental and widely-applied stochastic process ([Stroock, 2023](#)).

Furthermore, the requirement of a Gaussian distribution and not any other distribution in equation (45) is because the Gaussian distribution has convenient algebraic properties such as closedness under linear transformations and conditioning, see Corollary 2.2 and Theorem 2.4. Moreover, the Gaussian distribution arises as the unique limit distribution in the central limit theorem, and it is the unique distribution that maximises the differential entropy for the limit information constraint of fixed mean and variance ([Hennig et al., 2022](#)). These properties of the Gaussian distribution make the Gaussian process analytically tractable and well-suited as a prior distribution of some function f of interest.

A GP is an infinite-dimensional Gaussian measure whose finite-dimensional marginals are multivariate normal. The existence of GPs is proved in Appendix 6.8.3 using Kolmogorov's Existence Theorem 6.5.

2.3.2 Kernels

Definition 2.5 introduces the concept of a positive definite covariance kernel $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$. A kernel is a general function $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$. We assume that $\mathbb{X} \subseteq \mathbb{R}^d$, but $\mathbb{X}$ can also be a collection of non-vectorial objects such as strings, tree and graphs ([Rasmussen and Williams, 2008](#)). The term kernel plays a central role in the theory of integral operators:

$$(T_k[f])(\mathbf{x}) = \int_{\mathbb{X}} k(\mathbf{x}, \tilde{\mathbf{x}}) \cdot f(\tilde{\mathbf{x}}) \, d\nu(\tilde{\mathbf{x}}), \quad \mathbf{x} \in \mathbb{X}. \quad (49)$$

In proposition 2.1, we showed that $T_k[f] : L^p(\mathbb{X}, \nu) \rightarrow L^q(\mathbb{X}, \nu)$, for $1 \leq p, q \leq \infty$, is a well-defined bounded linear operator when $(\mathbb{X}, \mathcal{A}, \nu)$ is a σ -finite measure space and $\mathbf{x} \mapsto \|k(\mathbf{x}, \cdot)\|_{\tilde{p}} \in L^q(\mathbb{X}, \nu)$ for a $1 \leq \tilde{p} \leq \infty$ such that $\frac{1}{p} + \frac{1}{\tilde{p}} = 1$. If ν is the measure on for example $(\mathbb{X}, \lambda_d)$ or $(\mathbb{X}, \mathbb{P})$, then it is a σ -finite measure on that space. λ_d denotes the Lebesgue measure on $\mathbb{R}^d$. If $\mathbb{X} := [a_1, b_1] \times \dots \times [a_d, b_d] \subsetneq \mathbb{X}$ with $f \in \mathcal{C}(\mathbb{X}) := \{f : \mathbb{X} \rightarrow \mathbb{R} : f \text{ is continuous}\}$, then T is compact and called the Fredholm operator. In addition, T has a unique extension $\dot{T} : L^2(\mathbb{X}, \nu) \rightarrow L^2(\mathbb{X}, \nu)$ ([Sterk and de Snoo, 2025](#)):

$$(\dot{T}_k[f])(\mathbf{x}) = \int_{\mathbb{X}} k(\mathbf{x}, \tilde{\mathbf{x}}) \cdot f(\tilde{\mathbf{x}}) \, d\nu(\tilde{\mathbf{x}}), \quad \mathbf{x} \in \mathbb{X}, \quad f \in L^2(\mathbb{X}, \nu). \quad (50)$$

For an arbitrary $\mathbb{X} \in \mathbb{R}^d$, k satisfying $\int_{\mathbb{X}} \int_{\mathbb{X}} |k(\mathbf{x}, \tilde{\mathbf{x}})|^2 d\nu(\mathbf{x}) d\nu(\tilde{\mathbf{x}}) < \infty$ is called a Hilbert-Schmidt kernel. The associated integral operator $T_k : L^2(\mathbb{X}, \nu) \rightarrow L^2(\mathbb{X}, \nu)$, defined in equation (49), is called a Hilbert-Schmidt integral operator. It is bounded and satisfies $\|T\| = \|k\|_2 = \|k\|_{L^2(\mathbb{X} \times \mathbb{X}, \nu \otimes \nu)}$ (Bredikhin, 2011). Hence, $\mathbf{x} \mapsto \|k(\mathbf{x}, \cdot)\|_2 \in L^2(\mathbb{X}, \nu)$ is satisfied with $p = \tilde{p} = q = 2$.

A kernel k is symmetric if $k(\mathbf{x}_1, \mathbf{x}_2) = k(\mathbf{x}_2, \mathbf{x}_1)$, $\forall \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{X}$. Further, for a finite subset $\mathbf{X} := \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \mathbb{X}$, we define the Gram matrix as $\mathbf{K}_{\mathbf{X}\mathbf{X}} = [k(\mathbf{x}_i, \mathbf{x}_j)]_{i,j=1}^n \in \mathbb{R}^{n \times n}$. Next, recall that a matrix $\mathbf{M} \in \mathbb{R}^{d \times d}$ is positive semi-definite, if $\mathbf{v}^T \mathbf{M} \mathbf{v} \geq 0$, $\forall \mathbf{v} \in \mathbb{R}^d$. And $\mathbf{M}$ is positive definite, if $\mathbf{v}^T \mathbf{M} \mathbf{v} > 0$, $\forall \mathbf{v} \in \mathbb{R}^d \setminus \{\mathbf{0}\}$. Rasmussen and Williams (2008) and Kanagawa et al. (2018) present three equivalent definitions of a positive definite kernel.

Definition 2.6.

A symmetric kernel $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ is called a positive (semi-)definite kernel, if one of the following equivalent definitions holds:

$$\begin{aligned} 1). \quad & \langle T_k f, f \rangle = \int_{\mathbb{X}} \int_{\mathbb{X}} k(\mathbf{x}, \tilde{\mathbf{x}}) \cdot f(\mathbf{x}) \cdot f(\tilde{\mathbf{x}}) d\nu(\mathbf{x}) d\nu(\tilde{\mathbf{x}}) \geq 0, \quad \forall f \in L^2(\mathbb{X}, \nu); \\ 2). \quad & \sum_{i=1}^n \sum_{j=1}^n c_i \cdot c_j \cdot k(\mathbf{x}_i, \mathbf{x}_j) \geq 0, \quad \forall \mathbf{x}_i, \mathbf{x}_j \in \mathbb{X}, \quad \forall c_i, c_j \in \mathbb{R}; \\ 3). \quad & \mathbf{K}_{\mathbf{X}\mathbf{X}} \text{ is positive semi-definite, } \forall \mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \in \mathbb{X}^n := \underbrace{\mathbb{X} \times \dots \times \mathbb{X}}_{n \text{ times}}. \end{aligned} \tag{51}$$

To demonstrate the equivalence between 1) and 2), we can treat f as a simple function of the form $f(\mathbf{x}) = \sum_{i=1}^n c_i \cdot \delta_{\mathbf{x}}(C_i)$, where $C_i := f^{-1}(c_i)$, $\delta_{\mathbf{x}}$ is the Dirac measure as defined in equation (97), and $c_1, \dots, c_n \in \mathbb{R}$. Next, 2) is the polynomial equivalent expression of the quadratic form used in 3), with coefficients $\alpha_{ij} := c_i \cdot c_j \in \mathbb{R}$.

2.3.3 Covariance Function

Let $f \sim \mathcal{GP}(m, k)$. Hence, $f : \mathbb{X} \rightarrow \mathbb{R}^{\mathbb{X}} = \{\text{all possible function } \phi : \mathbb{X} \rightarrow \mathbb{R}\}$ is a stochastic process. However, $\mathbb{R}^{\mathbb{X}}$ is a large function space that, for example, does not require existence of $\int_{\mathbb{X}} f(\mathbf{x}) d\nu(\mathbf{x})$. We require f to be an $\mathbb{H} := L^2(\mathbb{X}, \nu)$ -valued random variable, i.e., a Hilbert space valued random variable. This ensures existence of $T_1[f] = \int_{\mathbb{X}} f(\mathbf{x}) d\nu(\mathbf{x}) = \langle f, 1 \rangle_{L^2(\mathbb{X}, \nu)}$ and $\mathbb{E}[f^2] = \int_{\mathbb{X}} f(\mathbf{x})^2 d\nu(\mathbf{x}) = \langle f, f \rangle_{L^2(\mathbb{X}, \nu)}$. In fact, by Riesz-Fréchet Representation Theorem, i.e., Theorem 6.4, for every bounded linear functional $L[f] \in \mathbb{H} = L^2(\mathbb{X}, \nu)$, there exists some $l \in \mathbb{H}$ such that:

$$L[f] = \langle f, l \rangle_{\mathbb{H}}. \tag{52}$$

Moreover, since $f \in \mathbb{H} = L^2(\mathbb{X}, \nu)$, the inner product on this function space is defined as:

$$\langle f, h \rangle_{\mathbb{H} = L^2(\mathbb{X}, \nu)} = \int_{\mathbb{X}} f(\mathbf{x}) h(\mathbf{x}) d\nu(\mathbf{x}), \quad \forall h \in L^2(\mathbb{X}, \nu). \tag{53}$$

Often, the $\mathbb{H} = L^2(\mathbb{X}, \nu)$ subscript in the inner product notation is omitted. Hence, by equations (18) and (21), and for arbitrary $h_1, h_2 \in L^2(\mathbb{X}, \nu)$, we have:

$$\text{Cov}(h_1, h_2) = \text{Cov}(\langle f, h_1 \rangle, \langle f, h_2 \rangle) = \int_{\mathbb{X}} (T_k[h_1])(\tilde{\mathbf{x}}) \cdot h_2(\tilde{\mathbf{x}}) d\nu(\tilde{\mathbf{x}}) = \langle T_k h_1, h_2 \rangle_{L^2(\mathbb{X}, \nu)}. \tag{54}$$

Equation (21) states that $\text{Cov}(h_1, h_2) = \langle \Sigma h_1, h_2 \rangle$. Hence, since $\langle \Sigma h_1, h_2 \rangle = \langle T_k h_1, h_2 \rangle$, the covariance operator of f on $L^2(\mathbb{X}, \nu)$ is $\Sigma := T_k$. Additionally, since $\text{Cov}(h_1, h_2) \geq 0$, the covariance operator T_k is positive definite, i.e., $\langle T_k h, h \rangle \geq 0$ for all $h \in L^2(\mathbb{X}, \nu)$. Observe that this is the first definition of a positive definite kernel in Definition 2.6. So, a kernel $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ is the covariance function of a Gaussian process $f \sim \mathcal{GP}(m, k)$ if and only if k is positive definite. Furthermore, note that in equation (54), the assumption $m \equiv 0$ is made to simplify notation. However, when $m \neq 0$, f can simply be replaced by $f - m$. A proof of equation (54) is provided in Appendix 6.8.3.

If a kernel is the covariance function of a Gaussian process, then we call it a covariance kernel. The covariance kernel determines the joint covariance of any finite collection $\mathbf{f}_{\mathbf{X}} := (f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))^T$, and thus

completely specifies the second-order structure of the Gaussian process. In fact, the covariance kernel characterizes the entire process, as it encodes prior assumptions about smoothness, periodicity, and other structural properties of the functions modeled by the Gaussian process.

2.3.4 Properties of Kernels

Proposition 2.5.

Let $k_i(\mathbf{x}, \tilde{\mathbf{x}}) = \text{Cov}(f_i(\mathbf{x}), f_i(\tilde{\mathbf{x}}))$ be the covariance kernel of the Gaussian process $f_i \sim \mathcal{GP}(m_i, k_i)$. Let $\mathbf{x}, \tilde{\mathbf{x}} \in \mathbb{X}$. Then, the following functions are also covariance kernels:

- i). For any $a_i \in \mathbb{R}_{\geq 0}$, the non-negative linear combination $\sum_{i=1}^m a_i \cdot k_i(\mathbf{x}, \tilde{\mathbf{x}})$ is a covariance kernel.
- ii). If f_1 and f_2 are independent Gaussian processes, then the pointwise/Hadamard product $k(\mathbf{x}, \tilde{\mathbf{x}}) := (k_1 \odot k_2)(\mathbf{x}, \tilde{\mathbf{x}}) = k_1(\mathbf{x}, \tilde{\mathbf{x}}) \cdot k_2(\mathbf{x}, \tilde{\mathbf{x}})$ is a covariance kernel.

A proof is provided in Appendix 6.8.3. Covariance kernels satisfy many other (closure) properties. See, for example, [Rasmussen and Williams \(2008\)](#) and [Shawe-Taylor and Cristianini \(2004\)](#).

Note that if f_1 and f_2 are (independent) Gaussian processes, then their product $f := f_1 \odot f_2$ is, in general, not a Gaussian process. However, there exists a Gaussian process with covariance function equal to the product $k := k_1 \odot k_2$; see [Rasmussen and Williams \(2008\)](#). This is because there is a one-to-one correspondence between $f \in \mathcal{GP}(m, k)$ and pairs (m, k) .

Moreover, observe that a covariance kernel $k(\mathbf{x}, \tilde{\mathbf{x}}) := \text{Cov}(f(\mathbf{x}), f(\tilde{\mathbf{x}}))$, for $\mathbf{x}, \tilde{\mathbf{x}} \in \mathbb{X}$ and $f \sim \mathcal{GP}(m, k)$, measures the similarity between the random function values $f(\mathbf{x})$ and $f(\tilde{\mathbf{x}})$, while depending only on the corresponding input points $\mathbf{x}, \tilde{\mathbf{x}} \in \mathbb{X}$. For instance, when $\tilde{\mathbf{x}}$ lies close to $\mathbf{x}$ according to the Euclidean metric, we expect $f(\tilde{\mathbf{x}})$ to be similar to $f(\mathbf{x})$ and the GP prior enforces that $f(\mathbf{x})$ and $f(\tilde{\mathbf{x}})$ should be highly correlated. Therefore, observations collected near $\mathbf{x}$ provide substantial information about the value $f(\mathbf{x})$. So, under the GP prior, the covariance kernel encodes the notion of similarity or nearness in input space $\mathbb{X}$, and it determines how information is shared across different locations in $\mathbb{X}$.

When a covariance kernel is invariant under translations in the input space, i.e., when it is a function of only $\mathbf{x} - \tilde{\mathbf{x}}$, it is called stationary. Additionally, if the covariance kernel is only a function of $\|\mathbf{x} - \tilde{\mathbf{x}}\|$, then it is called isotropic. Throughout, we assume the Euclidean norm $\|\cdot\|_2$. These isotropic kernels are also known as radial basis functions.

2.3.5 Examples of Kernels

There exist many covariance kernels, since any kernel function satisfying the positive semi-definiteness condition in Definition 2.6 qualifies as one. In this subsection, we present several of the most well-known covariance kernels. Specifically, we consider the squared-exponential, Matérn, periodic kernels, and pairwise sums of two kernels. In addition, the rational quadratic kernel, the polynomial kernel, and the product of two kernels are discussed in Appendix 6.7.

We consider a scaled covariance kernel family of the form:

$$k(\mathbf{x}, \tilde{\mathbf{x}}) := \sigma_f^2 \cdot \tilde{k}(\mathbf{x}, \tilde{\mathbf{x}}), \quad \mathbf{x}, \tilde{\mathbf{x}} \in \mathbb{X}, \quad (55)$$

where $\sigma_f^2 \in \mathbb{R}$ is a scale (hyper-)parameter called the output variance, which determines the average deviation of a sample path $f \sim \mathcal{GP}(m, k)$ and from the mean function m . In this subsection, σ_f^2 is set to one. The function $\tilde{k}$, called a unit kernel, captures the more intricate structural properties of the function, such as smoothness or periodicity ([Hennig et al., 2022](#)).

Each kernel has a number of parameters which specify the precise shape of the covariance kernel, such as σ_f^2 . These parameters are often called hyperparameters. In practice, hyperparameters are often difficult or even impossible to specify a priori. Hence, hyperparameters are usually estimated from data using maximum likelihood estimation or cross-validation ([Rasmussen and Williams, 2008](#); [Canelles](#)). This is discussed in Subsection 3.4.

First, the squared-exponential (SE) kernel $k_{SE} : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ is defined by:

$$k_{SE}(\mathbf{x}, \tilde{\mathbf{x}} \mid \sigma_f^2, l) = k_{SE}(\|\mathbf{x} - \tilde{\mathbf{x}}\|_2 \mid \sigma_f^2, l) := \sigma_f^2 \cdot \exp \left\{ -\frac{\|\mathbf{x} - \tilde{\mathbf{x}}\|_2^2}{2l^2} \right\} = \sigma_f^2 \cdot \exp \left\{ -\frac{(\mathbf{x} - \tilde{\mathbf{x}})^T (\mathbf{x} - \tilde{\mathbf{x}})}{2l^2} \right\}, \quad (56)$$

where $\|\cdot\|_2$ denotes the Euclidean norm. Observe that k_{SE} is isotropic, hence it is also called the (Gaussian) radial basis function kernel. Another synonym is the Gaussian kernel, since it is equal to the $\mathcal{N}(x; \tilde{x}, l)$ PDF when $\sigma_f^2 = 1/\sqrt{2\pi}l^2$ and $d = 1$. However, the term squared-exponential kernel is commonly used in the GP literature (Kanagawa et al., 2018). k_{SE} is the most widely used, i.e., the de-facto, kernel in Gaussian processes and also Bayesian quadrature.

The scalar $l > 0$ is the (characteristic) length-scale hyperparameter that determines the length-scale of the associated hypothesis space of functions. When l is small, the term $\exp\{-\|\mathbf{x} - \tilde{\mathbf{x}}\|_2^2 / (2l^2)\}$ decays quickly implying that only points extremely close to each other are considered similar. Consequently, the induced GP sample paths tend to vary rapidly. In contrast, when l increases, the decay of the kernel becomes slower making points farther apart considered to be similar. This enlarges the neighbourhood in which function values influence each other and leading to visibly smoother sample paths. In fact, k_{SE} is an infinitely differentiable function (Rasmussen and Williams, 2008). Hence, l does not influence the smoothness of k_{SE} but it controls the rate at which the smoothness changes.

Figure 1 illustrates the influence of l on k_{SE} . When $l = 0.5$, the kernel decays so rapidly that only very close points retain a high covariance. This is reflected in the heatmap by a narrow diagonal band, indicating that similarity drops sharply away from the diagonal. When l is increased to $l = 3$, the kernel decays more slowly and the diagonal widens accordingly. Points farther apart in the input domain maintain a stronger covariance, signalling a broader region of influence. Figure 2 shows some sample functions drawn from $\mathcal{GP}(0, k_{SE})$. For $l = 0.5$, the restriction of similarity only in a small neighbourhood enforced by the kernel produces rapidly fluctuating functions. When the length-scale is increased to $l = 3$, correlations persist over a larger range of inputs.

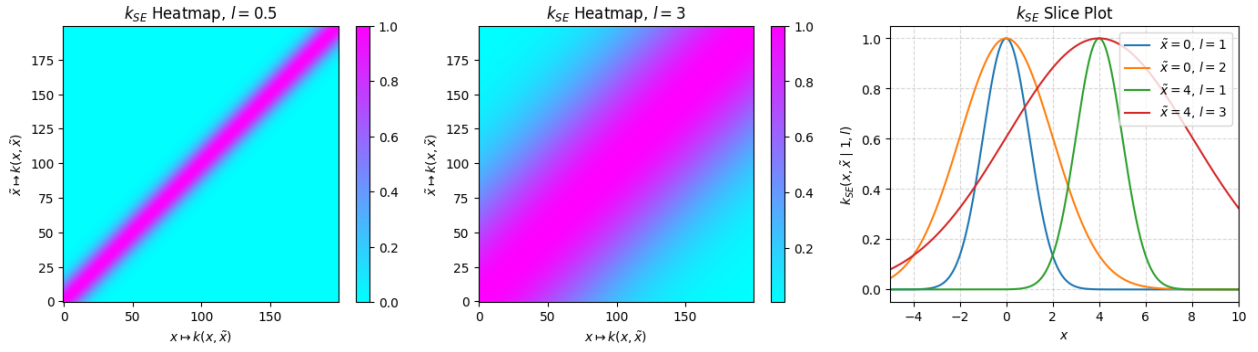


Figure 1: k_{SE} heatmaps and slice plot.

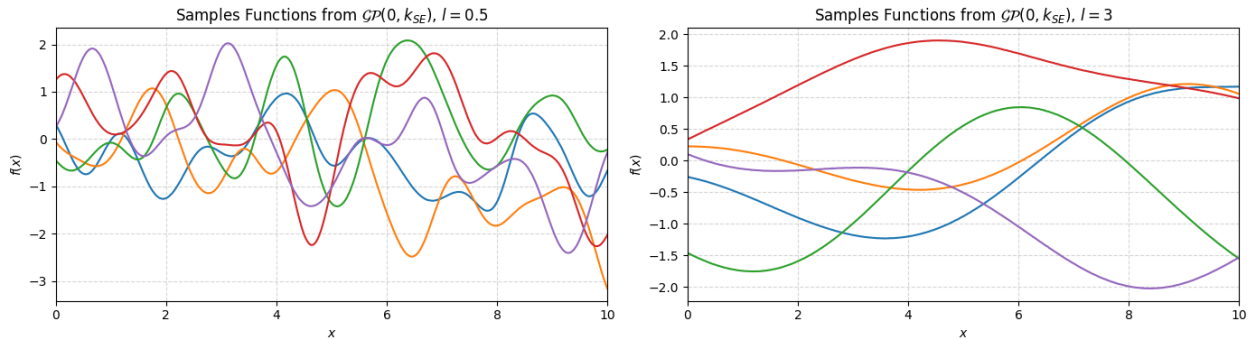


Figure 2: Sample functions drawn from $\mathcal{GP}(0, k_{SE})$, for $l = 0.5$ and $l = 3$.

To draw sample functions from $\mathcal{GP}(0, k_{SE})$ shown in Figure 2, a grid of $n = 1000$ points is used, i.e., $\mathbf{X} := \{x_1, \dots, x_n\} \subset \mathbb{X}$. Then, the corresponding function values $\mathbf{f}_{\mathbf{X}} = (f(x_1), \dots, f(x_n))^T$ are sampled from $\mathcal{N}_n(\mathbf{0}, \mathbf{K}_{\mathbf{X}\mathbf{X}} = [k_{SE}(x_i, x_j) \mid \sigma_f^2 = 1, l]_{i,j=1}^n)$. To ensure numerical stability and avoid singularity issues of

$\mathbf{K}_{\mathbf{X}\mathbf{X}}$, a small error term, called a jitter, of magnitude 10^{-6} is added to the diagonal of the Gram matrix $\mathbf{K}_{\mathbf{X}\mathbf{X}}$ (Kanagawa et al., 2018). And the multivariate Gaussian is generated using the Cholesky decomposition. Next, the points $(x_i, f(x_i))$, for $i \in \{1, \dots, n\}$, are plotted and connected by straight line segment, i.e., piecewise linear interpolation. In each plot of Figure 2, this procedure is repeated five times to illustrate different realizations from the GP prior.

Second, the Matérn kernel $k_M : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ constitutes a broad class of covariance kernels depending on two hyperparameters: the smoothness parameter $\alpha > 0$ and the length-scale $l > 0$. The general expression of k_M is complicated as it involves the modified Bessel function and the gamma function (Rasmussen and Williams, 2008). However, for the most useful and commonly-used values $\alpha \in \{1/2, 3/2, 5/2\}$, relatively simple closed-form expression are available:

$$\begin{aligned}
k_M\left(\mathbf{x}, \tilde{\mathbf{x}} \mid \sigma_f^2, \alpha = \frac{1}{2}, l\right) &= k_M\left(\|\mathbf{x} - \tilde{\mathbf{x}}\|_2 \mid \sigma_f^2, \alpha = \frac{1}{2}, l\right) := \exp\left\{-\frac{\|\mathbf{x} - \tilde{\mathbf{x}}\|_2}{l}\right\} \\
k_M\left(\mathbf{x}, \tilde{\mathbf{x}} \mid \sigma_f^2, \alpha = \frac{3}{2}, l\right) &= k_M\left(\|\mathbf{x} - \tilde{\mathbf{x}}\|_2 \mid \sigma_f^2, \alpha = \frac{3}{2}, l\right) \\
&:= \left(1 + \frac{\sqrt{3} \cdot \|\mathbf{x} - \tilde{\mathbf{x}}\|_2}{l}\right) \cdot \exp\left\{-\frac{\sqrt{3} \cdot \|\mathbf{x} - \tilde{\mathbf{x}}\|_2}{l}\right\} \\
k_M\left(\mathbf{x}, \tilde{\mathbf{x}} \mid \sigma_f^2, \alpha = \frac{5}{2}, l\right) &= k_M\left(\|\mathbf{x} - \tilde{\mathbf{x}}\|_2 \mid \sigma_f^2, \alpha = \frac{5}{2}, l\right) \\
&:= \left(1 + \frac{\sqrt{5} \cdot \|\mathbf{x} - \tilde{\mathbf{x}}\|_2}{l} + \frac{5 \cdot \|\mathbf{x} - \tilde{\mathbf{x}}\|_2^2}{3l^2}\right) \cdot \exp\left\{-\frac{\sqrt{5} \cdot \|\mathbf{x} - \tilde{\mathbf{x}}\|_2}{l}\right\}.
\end{aligned} \tag{57}$$

Figure 3 shows that the Matérn kernel with $\alpha = 3/2$ closely resembles the squared-exponential kernel, but it is slightly narrower around the mean. The length-scale hyperparameter l determines the size of the neighbourhood of similar points, while α governs the smoothness of the sample functions. As α increases, the functions drawn from the corresponding Gaussian process become smoother.

Figure 4 depicts samples from the Gaussian process $\mathcal{GP}(0, k_M)$ for $\alpha \in \{1/2, 3/2, 5/2\}$. For $\alpha = 1/2$, the corresponding Gaussian process is very rough, because it is mean-square continuous but not mean-square differentiable (Rasmussen and Williams, 2008). In fact, for $\alpha = 1/2$ and $d = 1$, the corresponding Gaussian process is called the Ornstein-Uhlenbeck (OU) process (Uhlenbeck and Ornstein, 1930). Moreover, the squared-exponential kernel is a limiting case of the Matérn kernel and it can be obtained by taking the limit $\alpha \rightarrow \infty$. In this sense, the squared-exponential kernel lies within the Matérn class, representing infinitely smooth sample functions.

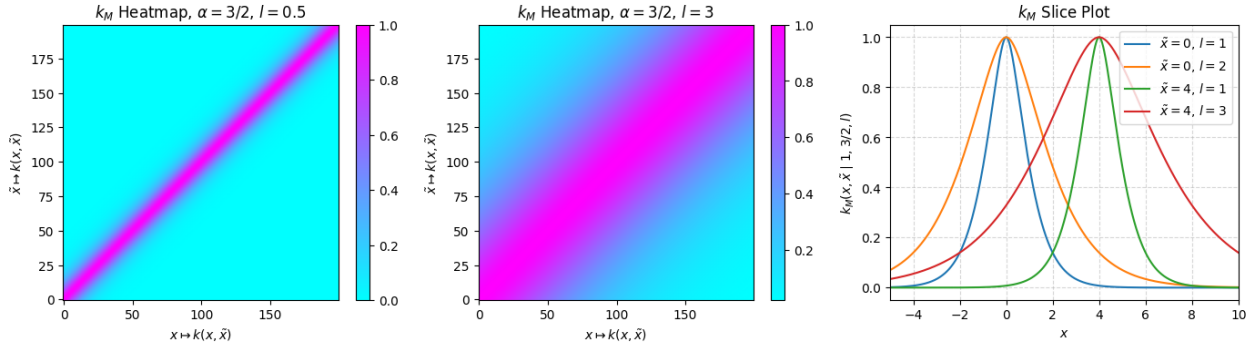


Figure 3: k_M heatmaps and slice plot.

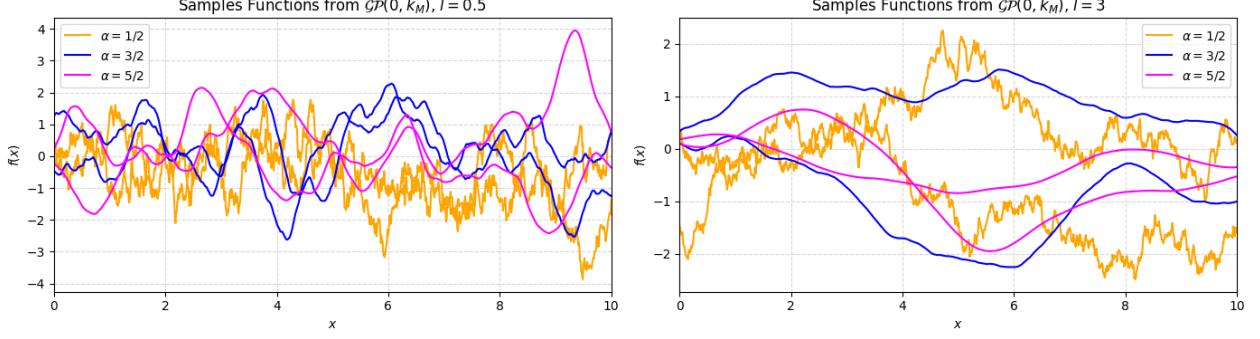


Figure 4: Sample functions drawn from $\mathcal{GP}(0, k_M)$, for $l = 0.5$ and $l = 3$.

Fourth, the periodic kernel $k_\pi : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ is defined by (MacKay, 1998):

$$k_\pi(\mathbf{x}, \tilde{\mathbf{x}} \mid \sigma_f^2, \boldsymbol{\lambda}, \mathbf{l}) := \sigma_f^2 \cdot \exp \left\{ -\frac{1}{2} \cdot \sum_{i=1}^d \frac{1}{l_i^2} \cdot \sin \left(\frac{\pi \cdot (x_i - \tilde{x}_i)}{\lambda_i} \right)^2 \right\}, \quad (58)$$

where $\boldsymbol{\lambda} \in \mathbb{R}_{>0}^d$ is the vector of period-length hyperparameters and $\mathbf{l} \in \mathbb{R}_{>0}^d$ is the vector of length-scale hyperparameters. This kernel is stationary, but not isotropic, since it depends on the periodic distance between inputs rather than on any norm of the difference vector such as $\|\mathbf{x} - \tilde{\mathbf{x}}\|_2$.

Figure 5 illustrates that similar values occur near the local extrema of the periodic function, reflecting the kernel’s periodic structure. Additionally, Figure 6 shows sample functions drawn from the Gaussian process prior $\mathcal{GP}(0, k_\pi)$ for different values of l , with a fixed period-length $\lambda = 2$. The frequency of the resulting sample functions is solely determined by λ , while l determines the smoothness.

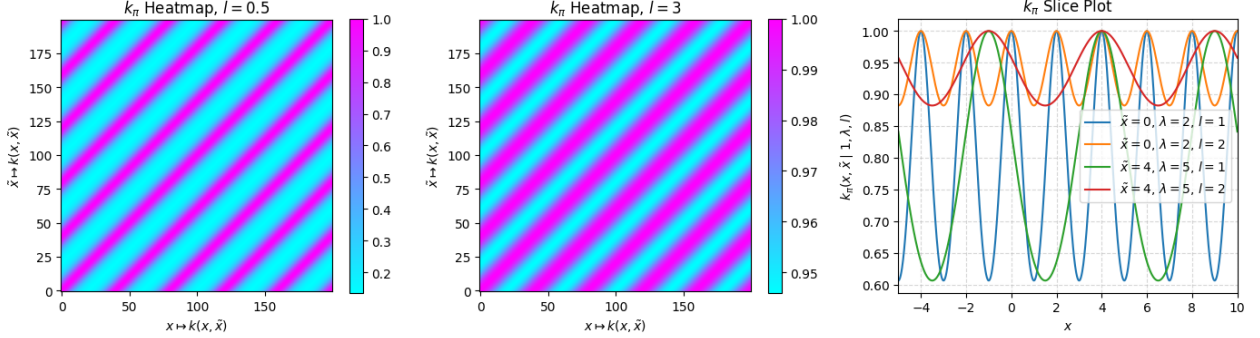


Figure 5: k_π heatmaps and slice plot.

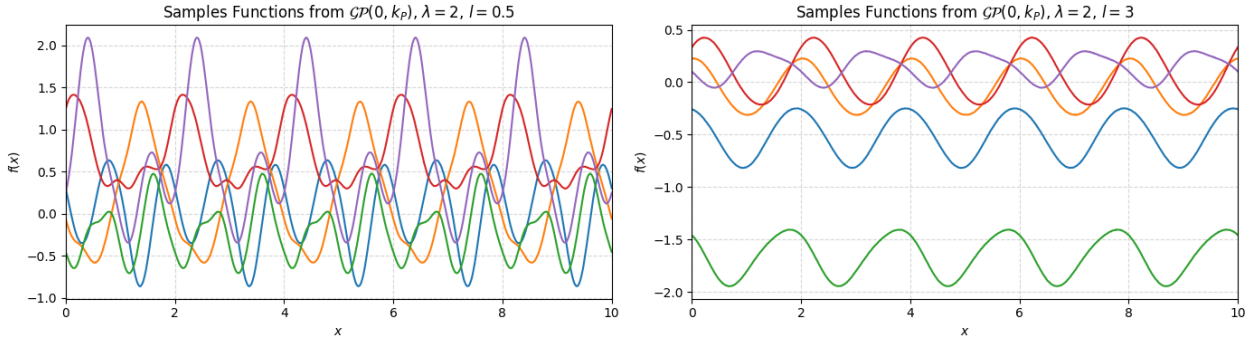


Figure 6: Sample functions drawn from $\mathcal{GP}(0, k_\pi)$, for $l = 0.5$ and $l = 3$ with $\lambda = 2$.

Finally, by Proposition 2.5, we can construct new kernels by combining existing ones through addition or multiplication. Figure 7 illustrates the sum of a periodic kernel k_π and a Matérn kernel k_M with smoothness parameter $\alpha = 3/2$. The left plot displays two corresponding kernel functions showing that the bell-shaped structure of the Matérn kernel is preserved, but some periodic oscillating behaviour is added. The right plot shows three sample paths drawn from the Gaussian process $\mathcal{GP}(0, k_\pi + k_M)$ with length-scale hyperparameter $l = 3$.

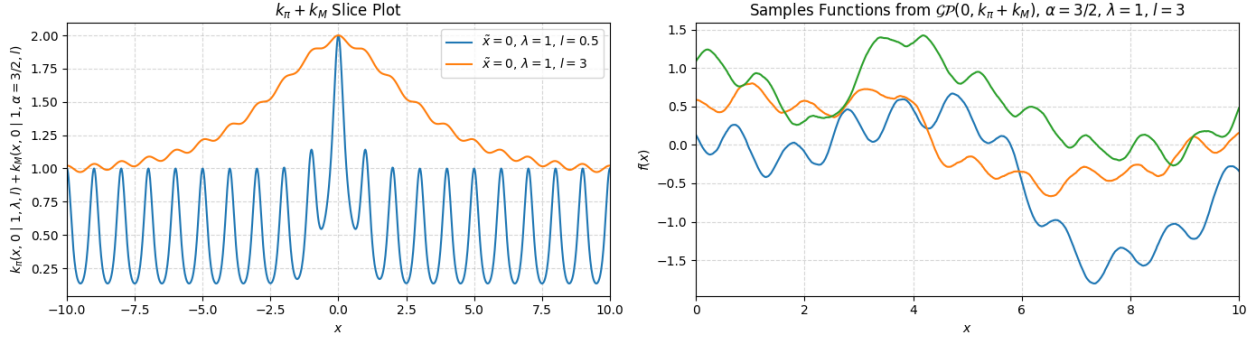


Figure 7: $k_\pi + k_M$ slice plot and sample functions drawn from $\mathcal{GP}(0, k_\pi + k_M)$.

Other examples of kernels can be found in, for instance, [Rasmussen and Williams \(2008\)](#), [Kanagawa et al. \(2018\)](#), [Ong et al. \(2002\)](#), and [Duvenaud \(2014\)](#).

In summary, the choice of covariance kernels determines the structural properties of the sample functions drawn from a particular Gaussian process. These properties include smoothness, scale and periodicity, all of which are determined by the choice of covariance kernel and the corresponding hyperparameters.

2.3.6 Posterior Conditioning (GP Regression)

Gaussian process regression, also known as Kriging or Wiener-Kolmogorov prediction, is a Bayesian non-parametric method for regression. Let $f : \mathbb{X} \rightarrow \mathbb{R}$ be a function. Suppose we observe a collection of datapoints $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n = \{\mathbf{X}, \mathbf{y}\}$ such that:

$$y_i = f(\mathbf{x}_i) + \epsilon_i, \quad i \in \{1, \dots, n\}, \quad \mathbf{x}_i \in \mathbb{X}, \quad (59)$$

where ϵ_i is a zero-mean random variable representing the noise. The aim of regression is to estimate the unknown regression function f based on the data $\mathcal{D}$. The specific Gaussian process regression approach puts a prior Gaussian process on the function f , with some mean m and kernel k reflecting the prior knowledge or belief about f . Consequently, the Gaussian process prior updates to some posterior distribution given the observed data.

However, for computer simulations, the observations, i.e., function evaluations of $\mathbf{x}_i$ at f , appear noise-free. Therefore, we assume $\epsilon_i \equiv 0$, which implies that $y_i = f(\mathbf{x}_i)$, $\forall i \in \{1, \dots, n\}$, and regression becomes equivalent to interpolation. Consequently, the distribution of $y_i | f(\mathbf{x}_i)$ is the Dirac measure $\delta_{y_i}(f(\mathbf{x}_i))$ implying that likelihood function of f becomes degenerate. Hence, it is not possible to apply Bayes's rule to derive the posterior distribution of f . Fortunately, Bayes's rule is not needed to derive the posterior distribution of f .

Theorem 2.5. (Posterior Gaussian Process)

Assume that $f \sim \mathcal{GP}(m, k)$ with mean function $m : \mathbb{X} \rightarrow \mathbb{R}$ and covariance kernel $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$. Let $\mathbf{X} := \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \in \mathbb{X}^n$ and $\mathbf{f}_\mathbf{X} = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))^T \in \mathbb{R}^n$ denote the observations. Furthermore, assume that the kernel matrix $\mathbf{K}_{\mathbf{X}\mathbf{X}} = [k(\mathbf{x}_i, \mathbf{x}_j)]_{i,j=1}^n \in \mathbb{R}^{n \times n}$ is invertible. Then, the posterior distribution of f , i.e., the conditional distribution of f given $\mathcal{D} := \{\mathbf{X}, \mathbf{y}\}$, is also a Gaussian process:

$$f | \mathcal{D} \sim \mathcal{GP}(m_{f,\circlearrowleft}, k_{f,\circlearrowleft}), \quad (60)$$

where the posterior mean $m_{f,\curvearrowright} : \mathbb{X} \rightarrow \mathbb{R}$ and the posterior covariance kernel $k_{f,\curvearrowright} : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ are defined by:

$$\begin{aligned} m_{f,\curvearrowright}(\mathbf{x}) &:= m(\mathbf{x}) + \mathbf{k}_{\mathbf{x}\mathbf{X}} \mathbf{K}_{\mathbf{X}\mathbf{X}}^{-1} [\mathbf{f}_{\mathbf{X}} - \mathbf{m}_{\mathbf{X}}] \\ k_{f,\curvearrowright}(\mathbf{x}, \tilde{\mathbf{x}}) &:= k(\mathbf{x}, \tilde{\mathbf{x}}) - \mathbf{k}_{\mathbf{x}\mathbf{X}} \mathbf{K}_{\mathbf{X}\mathbf{X}}^{-1} \mathbf{k}_{\mathbf{X}\tilde{\mathbf{x}}} \end{aligned} \quad (61)$$

with $\mathbf{m}_{\mathbf{X}} := (m(\mathbf{x}_1), \dots, m(\mathbf{x}_n))^T \in \mathbb{R}^n$ and $\mathbf{k}_{\mathbf{X}\mathbf{x}} = \mathbf{k}_{\mathbf{x}\mathbf{X}}^T = (k(\mathbf{x}_1, \tilde{\mathbf{x}}), \dots, k(\mathbf{x}_n, \tilde{\mathbf{x}}))^T \in \mathbb{R}^n$.

A proof is provided in (Kanagawa et al., 2018). The argument relies on applying Theorem 2.4.

Figure 8 illustrates the Gaussian process regression mechanism. Subfigure 8a shows six observations, while Subfigure 8b presents sample functions drawn from the prior Gaussian process with a squared-exponential kernel, using hyperparameters selected by trial-and-error. Finally, Subfigure 8c displays four sample functions from the posterior Gaussian process along with the posterior mean. It is evident from the plot that the posterior sample function intersect the observed data points. However, the suitability of this model for f is uncertain, since the kernel hyperparameters in the Gaussian process prior are selected manually. In practice, hyperparameters are commonly estimated by maximising the (log) likelihood of the data., as discussed in Section 3.4.

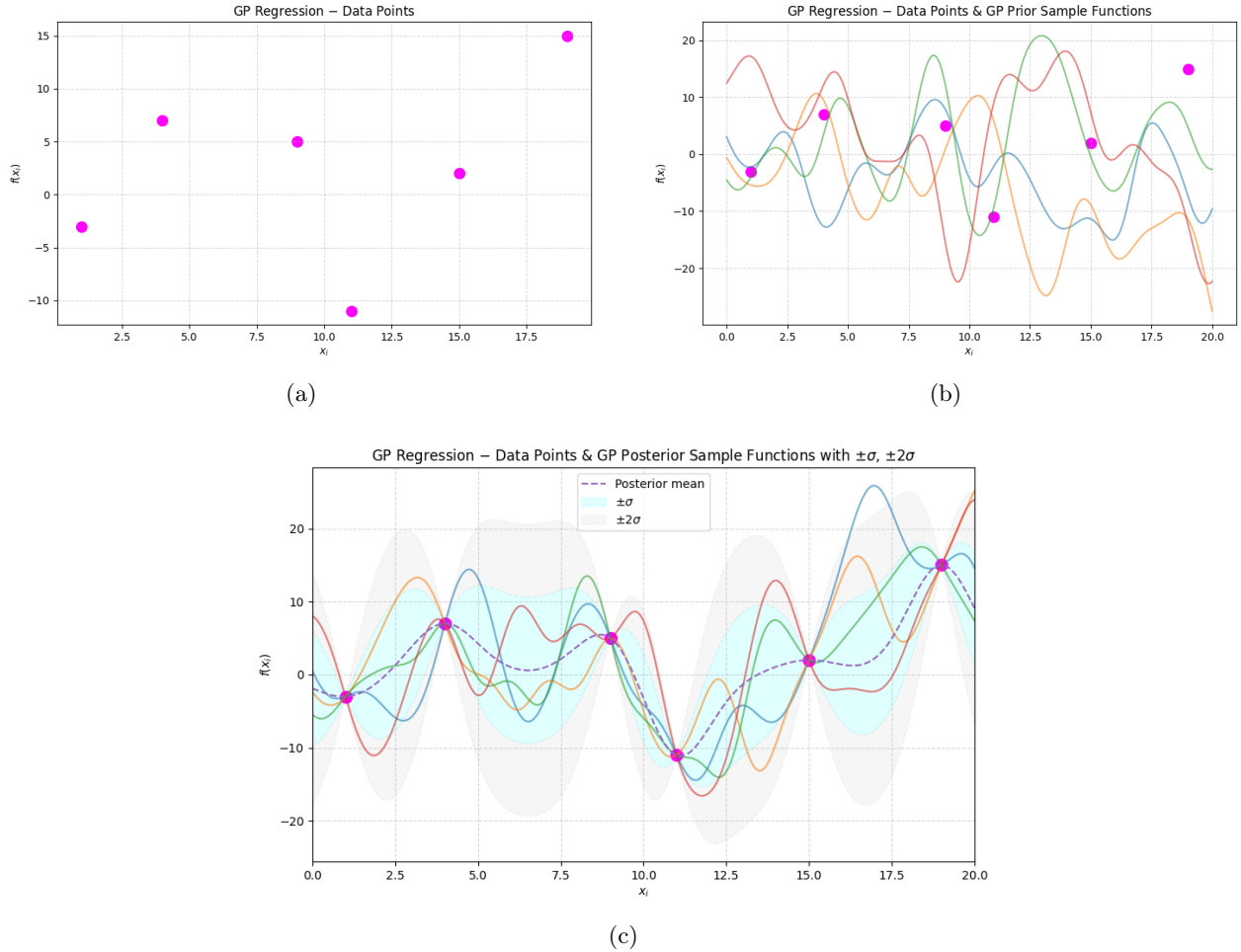


Figure 8: A Gaussian process regression using a squared-exponential kernel k_{SE} and hyperparameters selected by trial-and-error. Subfigure 8a displays the six observed data points, while Subfigure 8b displays four sample functions drawn from the Gaussian process prior. Subfigure 8c presents four sample function from the corresponding posterior Gaussian process.

2.3.7 Closedness under Bounded Linear Operators

A pivotal property of GPs, particularly in the context of Bayesian quadrature, is that they are closed under bounded linear operators.

Theorem 2.6.

Let $\mathbb{H}_1$ and $\mathbb{H}_2$ be separable Hilbert spaces. Let $f := f(\cdot, \omega) : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow \mathbb{H}_1$ be a $\mathbb{H}_1$ -valued Gaussian random variable, with mean $m := \mathbb{E}[f] \in \mathbb{H}_1$ and covariance operator $\Sigma : \mathbb{H}_1 \rightarrow \mathbb{H}_1$ defined by the property $\text{Cov}(\langle f, \mathbf{u}_1 \rangle_{\mathbb{H}_1}, \langle f, \mathbf{v}_1 \rangle_{\mathbb{H}_1}) = \text{Cov}(\mathbf{u}_1, \mathbf{v}_1) = \langle \Sigma \mathbf{u}_1, \mathbf{v}_1 \rangle_{\mathbb{H}_1}$. In addition, assume that $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$ be a bounded linear operator.

Then, $T[f]$ is a $\mathbb{H}_2$ -valued Gaussian random variable with the following mean and covariance operator:

$$\begin{aligned} \mathbb{E}[T[f]] &= T[\mathbb{E}[f]] = T[m]; \\ \Sigma_T &:= T\Sigma T^*, \end{aligned} \tag{62}$$

where $T^* : \mathbb{H}_2 \rightarrow \mathbb{H}_1$ is the adjoint operator of T .

A proof is provided in Appendix 6.8.3. This theorem has a pivotal corollary which states that a Gaussian process is closed under any linear bounded operator between Hilbert spaces.

Corollary 2.3. (Closedness of Gaussian Process under Bounded Linear Operators)

Let $\mathbb{H}_1$ and $\mathbb{H}_2$ be separable Hilbert spaces, and let $T : \mathbb{H}_1 \rightarrow \mathbb{H}_2$ be a bounded linear operator. Assume that $f \sim \mathcal{GP}(m, k)$, and let K be the covariance operator on $\mathbb{H}_1$ corresponding to k .

Then:

$$T[f] \sim \mathcal{GP}(T[m], TkT^*), \tag{63}$$

with $T[\mathbf{f}_{\mathbf{X}}] \sim \mathcal{N}_n(T[\mathbf{m}_{\mathbf{X}}], TKT^*)$, for some $\mathbf{X} \subseteq \mathbb{R}^d$.

This is a special case of Theorem 2.6 with $\mathbb{H}_1 \subseteq \mathbb{R}^{\mathbb{X}}$. Matsumoto and Sullivan (2024) extend this result to unbounded linear operators under some technical conditions. Furthermore, $\mathbb{H}_1$ is often a reproducing kernel Hilbert space, L^p , or Sobolev space (Kanagawa et al., 2018; Ghoghgh et al., 2021; Muandet et al., 2017).

2.4 Bayesian Quadrature

In this subsection, the objective of the probabilistic integration technique Bayesian quadrature (BQ) is introduced and motivated. This is followed by an explanation of the mechanism underlying BQ, accompanied by applications of a special case of BQ.

2.4.1 Motivation

Integrals play an important role in mathematics. However, integrals do not always have an analytical closed-form solution. Therefore, in practice, intractable integrals are approximated by numerical integration techniques. Numerical integration is sometimes referred to as quadrature. Suppose that we are interested in approximating the following integral:

$$I = T[f] := \int_{\mathbb{X}} f(\mathbf{x}) \, d\nu(\mathbf{x}), \tag{64}$$

where $f : \mathbb{X} \subseteq \mathbb{R}^d \rightarrow \mathbb{R}$ and ν is a measure on $\mathbb{X}$.

Most quadrature techniques can be classified into two classes: classical numerical integration, (such as the trapezoidal rule and Gaussian quadrature) and stochastic numerical integration (such as Monte Carlo (MC) integration). Both approaches yield a point estimate. For instance, when ν is the Lebesgue measure or a probability measure $\mathbb{P}$, MC integration estimates the integral, rewritten as an expectation, by the sample mean following the (weak) Law of Large Numbers:

$$\hat{I}_{MC} := \frac{1}{n_{MC}} \sum_{i=1}^{n_{MC}} f(\mathbf{x}_i) \xrightarrow[n \rightarrow \infty]{\mathbb{P}} I = \int_{\mathbb{X}} f(\mathbf{x}) \, d\mathbb{P}(\mathbf{x}) = \int_{\mathbb{X}} f(\mathbf{x}) \cdot p(\mathbf{x}) \, d\mathbf{x} = \mathbb{E}[f(\mathbf{X})], \tag{65}$$

where n_{MC} denotes the number of samples, and $\mathbf{x}_i \sim \mathbb{P}$ with some proposal PDF p . Note that MC integration requires the existence of the Radon-Nikodym derivative (Theorem 6.1), since the penultimate equality must hold. When ν is the Lebesgue measure, or when sampling from p is difficult, the importance sampling reweighting trick $[f(\mathbf{x})/p(\mathbf{x})] \cdot p(\mathbf{x})$ or $[f(\mathbf{x}) \cdot p(\mathbf{x})/q(\mathbf{x})] \cdot q(\mathbf{x})$ can be applied.

In contrast to MC integration, classical quadratures suffer from the curse of dimensionality, i.e., the number of function evaluations required to attain a certain accuracy grows exponentially with the dimension d . The convergence rate of MC integration is $\mathcal{O}(n_{MC}^{-1/2})$, independent of the dimension d . Hence, MC integration does not suffer from the curse of dimensionality, and is therefore preferable to classical quadratures for approximating high-dimensional integrals. However, MC has the worst possible convergence rate, under any reasonable model of the integrand f , in every dimension d (Hennig et al., 2022).

In addition, O’Hagan (1987) argues that MC integration is statistically unsound. First, it requires sampling from a proposal PDF p , which can be chosen almost arbitrarily due to the reweighting trick, and it conveys no information about f or I . Second, MC integration ignores all spatial information contained in $\{\mathbf{x}_1, \dots, \mathbf{x}_{n_{MC}}\}$, only the function evaluations $f(\mathbf{x}_i)$ are used. For example, when $\mathbf{x}_i$ and $\mathbf{x}_{i \neq j}$ are very close, the function evaluation $f(\mathbf{x}_i) \approx f(\mathbf{x}_j)$ effectively receives double weight. Thus, MC integration wastes most of the available information about f and I . The aim of BQ is to incorporate more (prior) information about the integral I and the integrand f when estimating I .

The intuition behind BQ is to model the integrand of interest f by a GP, such that the integral of the posterior mean function $m_{f, \circlearrowleft}$ of the posterior GP serves as an estimate of the integral itself. Suppose we want to estimate the following one-dimensional Lebesgue-integral:

$$I := \int_0^5 f(x) \, dx := \int_0^5 e^{\sin(x^2) \cdot \cos(x)} \, dx. \quad (66)$$

Subfigure 9a displays the integrand f with $n = 6$ observations. The integral in equation (66) does not have a known closed-form solution. However, we approximate it by the trapezoid rule, and treat this numerical value as the true integral. Subfigure 9b shows sample functions drawn from the GP prior of f , using a squared-exponential kernel. The corresponding kernel hyperparameters are estimated via maximum likelihood, as discussed in section 3. The left plot in Subfigure 9c shows one sample function drawn from the posterior GP, and the right plot shows the corresponding integral value of this function along with the integral value of the posterior mean. In addition, the true integral value is also included for comparison. All integrals are approximated by the trapezoid rule. Subfigures 9d and 9e show the same plots but with more posterior sample functions and their corresponding integral values. Observe that the distribution of these integral values looks approximately Gaussian. This is because Gaussian processes are closed under linear operations such as integration.

One factor contributing to the discrepancy between the BQ estimate and the true integral is the limited number of observations. Figures 18 and 19 in Appendix 6.1.1 repeat the same procedure as in Figure 9, but with a larger number of observations. This results in a more accurate estimate and in a posterior distribution that centers more closely around the true integral value.

2.4.2 Mechanism

The aim of BQ is to estimate the integral I in equation (64) while incorporating prior information or beliefs about the integrand f and the integral I . Analogous to Section 2.3, the integrand $f : \mathbb{X} \rightarrow \mathbb{R}$ is modelled as a GP with some mean function $m : \mathbb{X} \rightarrow \mathbb{R}$ and covariance kernel $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$, representing the prior belief. In practice, m is frequently set to zero. Additionally, we evaluate the integrand f at n points $\mathbf{X} := \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. The observations are collected in $\mathcal{D} = \{\mathbf{X}, \mathbf{y}\}$, where $\mathbf{y} := \mathbf{f}_\mathbf{X} = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_n))^T \in \mathbb{R}^n$, since we assume zero observation error.

Observe that the integral of interest I in equation (64) is an integration operator $T : L^p(\mathbb{X}, \nu) \rightarrow \mathbb{R}$ similar to equation (9). Hence, Corollary 2.3 enables us to push the integral through the GP:

$$T[f] \sim \mathcal{GP}(T[m], TkT^*), \quad (67)$$

with $T[\mathbf{f}_\mathbf{X}] \sim \mathcal{N}_1(T[\mathbf{m}_\mathbf{X}], T\mathbf{K}_{\mathbf{X}\mathbf{X}}T^*)$. Combining this with $\mathbf{f}_\mathbf{X} \sim \mathcal{N}_n(\mathbf{m}_\mathbf{X}, \mathbf{K}_{\mathbf{X}\mathbf{X}})$ results in the following joint distribution:

$$\begin{bmatrix} T[f] \\ \mathbf{f}_\mathbf{X} \end{bmatrix} \sim \mathcal{N}_{n+1} \left(\begin{bmatrix} T[\mathbf{m}_\mathbf{X}] \\ \mathbf{m}_\mathbf{X} \end{bmatrix}, \begin{bmatrix} T\mathbf{K}_{\mathbf{X}\mathbf{X}}T^* & \boldsymbol{\kappa}_\mathbf{X}^T \\ \boldsymbol{\kappa}_\mathbf{X} & \mathbf{K}_{\mathbf{X}\mathbf{X}} \end{bmatrix} \right), \quad (68)$$

where the off-diagonal are determined as follows by using Theorem 2.1 and Fubini's Theorem 2.2:

$$\begin{aligned}
\text{Cov}(T[f], f(\mathbf{x}_i)) &= \mathbb{E}[(T[f] - \mathbb{E}[T[f]]) \cdot (f(\mathbf{x}_i) - \mathbb{E}[f(\mathbf{x}_i)])] \\
&= \mathbb{E}\left[\int_{\mathbb{X}} f(\mathbf{x}) - m(\mathbf{x}) \, d\nu(\mathbf{x}) \cdot (f(\mathbf{x}_i) - \mathbb{E}[f(\mathbf{x}_i)])\right] \\
&= \int_{\mathbb{X}} \mathbb{E}[f(\mathbf{x}) - m(\mathbf{x}) \cdot (f(\mathbf{x}_i) - \mathbb{E}[f(\mathbf{x}_i)])] \, d\nu(\mathbf{x}) \\
&= \int_{\mathbb{X}} \text{Cov}(f(\mathbf{x}), f(\mathbf{x}_i)) \, d\nu(\mathbf{x}) = \int_{\mathbb{X}} k(\mathbf{x}, \mathbf{x}_i) \, d\nu(\mathbf{x}) = \kappa(\mathbf{x}_i).
\end{aligned} \tag{69}$$

Consequently, $\boldsymbol{\kappa}_{\mathbf{X}} := (k(\mathbf{x}_1), \dots, k(\mathbf{x}_n))^T \in \mathbb{R}^n$. When ν is a probability measure, $\boldsymbol{\kappa}_{\mathbf{X}}$ is called the kernel mean embedding of ν in the reproducing kernel Hilbert space of k . Additionally, $T\mathbf{K}_{\mathbf{X}\mathbf{X}}T^*$ can be rewritten as:

$$T\mathbf{K}_{\mathbf{X}\mathbf{X}}T^* = \int_{\mathbb{X}} \int_{\mathbb{X}} k(\mathbf{x}, \tilde{\mathbf{x}}) \, d\nu(\mathbf{x}) \, d\nu(\tilde{\mathbf{x}}) =: k_{I,0} \in \mathbb{R}^{>0}. \tag{70}$$

Therefore, by Theorem 2.4, the posterior distribution of the integral $I = T[f]$ is:

$$p(I \mid \mathbf{f}_{\mathbf{X}}) = p(I \mid \mathcal{D}) = \mathcal{N}(I; m_{I,\curvearrowright}, k_{I,\curvearrowright}), \tag{71}$$

where the posterior integral mean $m_{I,\curvearrowright} : \mathbb{X} \rightarrow \mathbb{R}$ and the posterior integral variance $k_{I,\curvearrowright} : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ are defined as:

$$\begin{aligned}
m_{I,\curvearrowright} &= T[m] + \boldsymbol{\kappa}_{\mathbf{X}}^T \mathbf{K}_{\mathbf{X}\mathbf{X}}^{-1} [\mathbf{f}_{\mathbf{X}} - \mathbf{m}_{\mathbf{X}}] \\
&= \int_{\mathbb{X}} m(\mathbf{x}) + \mathbf{k}_{\mathbf{x}\mathbf{X}} \mathbf{K}_{\mathbf{X}\mathbf{X}}^{-1} [\mathbf{f}_{\mathbf{X}} - \mathbf{m}_{\mathbf{X}}] \, d\nu(\mathbf{x}),
\end{aligned} \tag{72}$$

and

$$\begin{aligned}
k_{I,\curvearrowright} &= k_{I,0} - \boldsymbol{\kappa}_{\mathbf{X}}^T \mathbf{K}_{\mathbf{X}\mathbf{X}}^{-1} \boldsymbol{\kappa}_{\mathbf{X}} \\
&= \int_{\mathbb{X}} \int_{\mathbb{X}} k(\mathbf{x}, \tilde{\mathbf{x}}) - \mathbf{k}_{\mathbf{x}\mathbf{X}} \mathbf{K}_{\mathbf{X}\mathbf{X}}^{-1} \mathbf{k}_{\mathbf{X}\tilde{\mathbf{x}}} \, d\nu(\mathbf{x}).
\end{aligned} \tag{73}$$

The estimate of I is the posterior integral mean, because it is the maximum a posteriori (MAP) estimator. Observe that the estimate only depends on the integrand f via the n observations, i.e., $\mathbf{f}_{\mathbf{X}}$. The posterior integral variance is independent of the integrand f and solely depends on the kernel k . In addition, observe that $m_{I,\curvearrowright}$ and $k_{I,\curvearrowright}$ are the integrals of the mean $m_{f,\curvearrowright}$ and covariance kernel $k_{f,\curvearrowright}$ of the posterior GP on f in equation (60).

BQ can be viewed as a probabilistic generalisation of classical and stochastic numerical integration techniques, because the trapezoidal rule is BQ with Brownian motion, defined in equation (48), as the kernel of the prior GP. Furthermore, the MC estimate is the limit of $m_{I,\curvearrowright}$ as $c \rightarrow \infty$ under the $\mathcal{GP}(0, \mathbb{I}\{\mathbf{x} - \tilde{\mathbf{x}}\} + c^{-1})$ prior (Hennig et al., 2022).

If the integrals $T[m]$, $\kappa(\mathbf{x}_i)$ and $k_{I,0}$ have an analytical closed-form solution, then $m_{I,\curvearrowright}$ and $k_{I,\curvearrowright}$ also have a closed-form expression. Otherwise, these three integrals need to be approximated. When the kernel is a squared-exponential kernel and ν a Gaussian measure, i.e., $\nu(\mathbf{x}) = \mathbb{P}(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu, \sigma^2)$, the three integrals have the following simplified form for a constant prior mean $m \equiv c \in \mathbb{R}$ in one-dimension:

$$\begin{aligned}
T[m] &= \int_{\mathbb{X}} c \, d\nu(x) = c; \\
\kappa(x_i) &= \sigma_f^2 \cdot \sqrt{\frac{l^2}{l^2 + \sigma^2}} \cdot e^{-(x_i - \mu)^2 / 2(l^2 + \sigma^2)}; \\
k_{I,0} &= \sigma_f^2 \cdot \sqrt{\frac{l^2}{l^2 + 2\sigma^2}}.
\end{aligned} \tag{74}$$

This is first modern formulation of BQ, called Bayes-Hermite quadrature (BHQ) (O'Hagan, 1991).

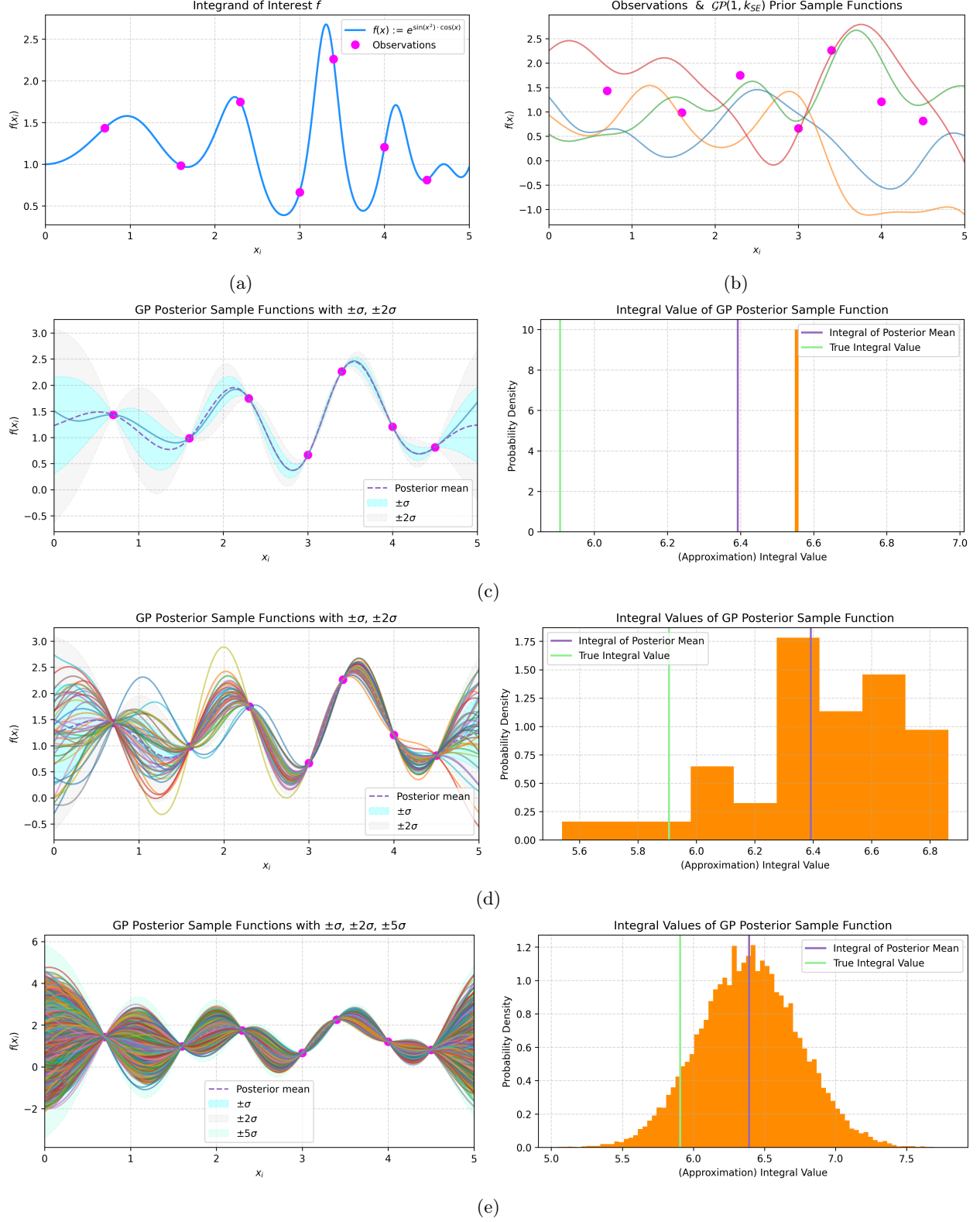


Figure 9: The distribution of the integrals of sample functions drawn from the posterior GP, based on $n = 10$ observations. Subfigure 9a shows the integrand. Subfigure 9b displays four sample functions drawn from the prior. Subfigures 9c, 9d, and 9e show the distributions of the integral computed from 1, 42, and 23,000 posterior sample functions, respectively. The last two histograms are created using the Freedman-Diaconis rule for optimal bin width.

Figure 10 illustrates the approximation of I by BQ. The initial estimate is not particularly accurate, as the absolute error is approximately 0.725. The posterior distribution of BQ is also shown, and its standard deviation is approximately 1. To examine how the estimate and the corresponding uncertainty change when additional information is provided, Figure 20 displays the effect of adding extra observations, either manually or at random. However, such approaches do not necessarily improve the estimate: although the absolute error decreases, the posterior standard deviation often increases significantly in both situations. Adaptive sampling offers a principled approach of selecting the next evaluation point according to some acquisition function, as discussed in Subsection 3.6. Moreover, the choice of kernel also plays a crucial role in the performance of the BQ estimator, as it encodes prior assumptions about the structural properties of the integrand f . However, there is no evidence that the squared-exponential kernel is the best choice of kernel among some family of kernels. This will be examined in Subsection 3.5.

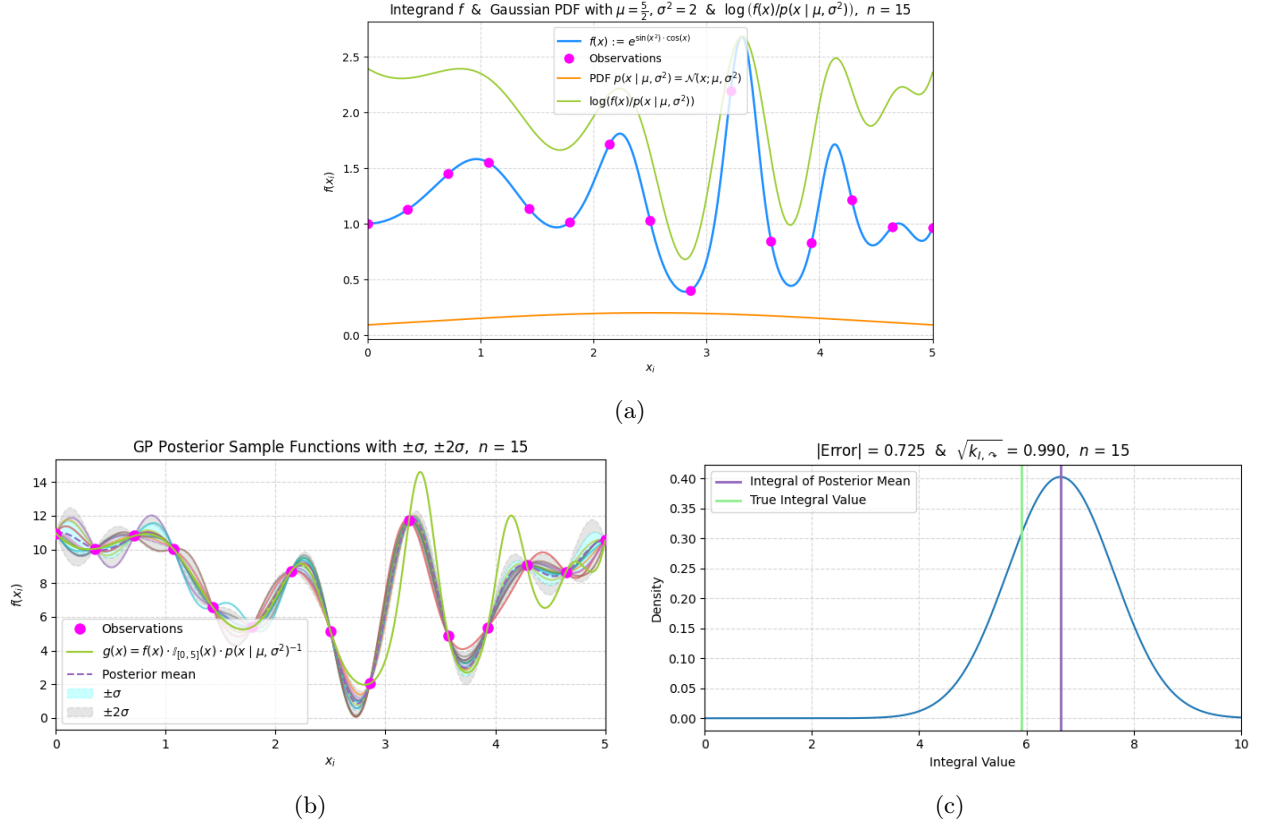


Figure 10: This figure illustrates Bayes-Hermite quadrature with a zero prior mean. Subfigure 10a displays the integrand of interest together with 14 observations, the Gaussian prior distribution, and the log-scaled transformed integrand. Subfigure 10b shows four sample functions drawn from the posterior Gaussian process. Subfigure 10c presents the posterior distribution of the integral, together with the true integral value.

3 Simulation Study

In this section, the setup of the comparative simulation study is described. In particular, we outline the computational setup, the methods included in the comparison, the three applications considered, the estimation procedure for the kernel and its hyperparameters, and the procedure for selecting new evaluation points. The three applications consist of a one-dimensional integral, the marginal likelihood of a real-data example, and a synthetic trimodal marginal likelihood.

3.1 Computational Setup

All simulations in this thesis were implemented using `Python 3.13`. Every figure and all GP and BQ simulations were produced from scratch, except for the optimisation algorithm used in the maximum likelihood estimation, for which the `SciPy` package was used. `NumPy` was used for numerical linear algebra operations, such as the Cholesky decomposition of the kernel matrix, and all plots were generated using the `Matplotlib` package. Furthermore, in each script a random seed was set to ensure reproducibility and fairness of comparison.

All code used to generate the figures and tables in this thesis is included in the accompanying document and is also available at the following GitHub repository: [GitHub link](#).

3.2 Methods Compared

In this subsection, we present an overview of the methods considered in the comparison. MC integration is compared with six variants of BQ. We consider BQ with a fixed grid and BQ with an uncertainty sampling scheme. For both approaches, the following three scenarios are examined: estimating the kernel and hyperparameters once, re-estimating hyperparameters only, and re-estimating both at each iteration.

First, MC integration is performed according to equation (65). The evaluation points are samples drawn from a proposal distribution, taken to be either a uniform or Gaussian distribution. MC integration has a convergence rate of order $\mathcal{O}(n^{-1/2})$, independent of the dimension d . Second, fixed-grid BQ refers to BQ applied to an equidistant grid or to a grid consisting of samples from the proposal distribution. Fixed-grid BQ is also referred to as vanilla BQ. Third, BQ with uncertainty sampling is considered as an adaptive sampling scheme, as discussed in Subsection 3.6. Additionally, the kernel and corresponding hyperparameters are estimated via maximum likelihood, as described in Subsections 3.4 and 3.5, respectively.

The seven methods are compared for different sample sizes on a grid of values evenly spaced on a $\log_{10}$ -scale. MC integration and fixed-grid BQ are computed in a non-nested manner, meaning that they are recomputed from scratch for each larger sample size. The accuracy is measured by the absolute error, i.e., the absolute value between the estimate and the true integral value, and the relative absolute error, defined as the absolute error divided by the absolute value of the true integral. Furthermore, in the marginal likelihood settings, a (log) scaling is applied because these values can be extremely small leading to machine precision or floating-point issues. Given a sequence $(s_n)_{n \in \mathbb{N}}$ of positive values, e.g., marginal likelihood values, we apply the following (log) scaling:

$$s_n = e^{\log(s_n)} = e^{L_0} e^{\log(s_n) - L_0}, \quad (75)$$

where $L_0 := \max_n \{\log(s_n)\}$. Since $\max_n \{e^{\log(s_n) - L_0}\} = 1$, the largest value of s_n is standardized to one, up to the scaling factor e^{L_0} .

Furthermore, in the synthetic trimodal likelihood example, in addition to the standard grid obtained by drawing evaluation points from the Gaussian prior, the corresponding marginal likelihood is also estimated using BQ with an alternative grid design. This design places 30 percent of the evaluation points in a small neighbourhood around one of the three modes.

3.3 Models & Integrals

In this subsection, the numerical integration problems used to evaluate and compare the performance of Monte Carlo integration and Bayesian quadrature. Each test problem defines an integrand f and an integration domain $\mathbb{D}$.

3.3.1 One-Dimensional Integral

The first test problem is the integral of the smooth function $f : \mathbb{X} \subset \mathbb{R} \rightarrow \mathbb{R}$ on $\mathbb{D} := [0, 5]$ as defined in equation (66). Subfigure 9a displays the graph of f on $\mathbb{D}$. No analytical closed-form solution is available for this integral. Therefore, it is approximated by the trapezoidal rule and this is treated as the true integral value.

3.3.2 Bayesian Linear Regression Model

The second test problem arises from a Bayesian linear regression model with a Gaussian prior and a Gaussian likelihood function. Let $\mathbf{y} \in \mathbb{R}^n$ denote the n observed data points, let $\mathbf{X} \in \mathbb{R}^{n \times d}$ denote the design matrix, and let $\boldsymbol{\beta} \in \mathbb{R}^d$ denote the regression coefficients. The standard Bayesian linear regression model is defined as:

$$\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}_1) \quad \& \quad \boldsymbol{\beta} \sim \mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}_2), \quad (76)$$

where $\mathbf{X}\boldsymbol{\beta} \in \mathbb{R}^n$, $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{n \times n}$, $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\boldsymbol{\Sigma}_2 \in \mathbb{R}^{d \times d}$.

We consider two datasets. First, we use a salary data containing 30 observations of two variables: salary and number of years of experience (Veerakumar, 2020). Second, we analyse a dataset on the profits of startup companies in the USA, consisting of 50 observations of five variables: research and development expenditure, administration cost, marketing expenditure, a state categorical variable, profit (Tiwari, 2019). For comparability, only the first 30 observations are used, and the categorical variable is omitted. Both datasets are standardized using the z-score standardization. Consequently, we assume that $\boldsymbol{\Sigma}_1 = \sigma^2 I$ and $\boldsymbol{\Sigma}_2 = \tau^2 I$, and we set $\sigma^2 = \tau^2 = 1$. Additionally, we assume $\boldsymbol{\mu} = \mathbf{0}$.

By writing the model as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, with $\boldsymbol{\epsilon} \sim \mathcal{N}_n(\mathbf{0}, \boldsymbol{\Sigma}_1)$, and applying Corollary 2.2, the marginal distribution of $\mathbf{y}$ is given by:

$$\mathbf{y} \mid \mathbf{X} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\mu}, \boldsymbol{\Sigma}_1 + \mathbf{X}\boldsymbol{\Sigma}_2\mathbf{X}^T) = \mathbb{P}_{\mathbf{y}}. \quad (77)$$

Equivalently, $p(\mathbf{y} \mid \mathbf{X})$ can also be obtain by marginalising $\mathbf{y}$ over $\boldsymbol{\beta}$:

$$p(\mathbf{y} \mid \mathbf{X}) = \int_{\mathbb{D} := \mathbb{R}^d} p(\mathbf{y}, \boldsymbol{\beta} \mid \mathbf{X}) \, d\boldsymbol{\beta} = \int_{\mathbb{R}^d} p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}) \cdot p(\boldsymbol{\beta}) \, d\boldsymbol{\beta}. \quad (78)$$

The scalar quantity $p(\mathbf{y} \mid \mathbf{X})$ is also referred to as the marginal likelihood or model evidence. Intuitively, $p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta})$ denotes the likelihood of the parameter $\boldsymbol{\beta}$, whereas $p(\mathbf{y} \mid \mathbf{X})$ represents the likelihood of the model itself. Note that in the salary dataset, $\boldsymbol{\beta} := (\beta_0, \beta_1)^T \in \mathbb{R}^{d=2}$ where β_0 denotes the intercept parameter. In the startup dataset, $\boldsymbol{\beta} \in \mathbb{R}^{d=4}$.

Figure 21 in Appendix 6.1.2 displays a three-dimensional plot of the likelihood function $p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta})$ for the salary dataset over a two-dimensional grid of $(\beta_0, \beta_1)^T \in \mathbb{R}^2$. Figure 22 in Appendix 6.1.2 shows a three-dimensional plot of the scaled likelihood function, the scaling is discussed in Subsection 3.2. Furthermore, Figure 23 in Appendix 6.1.2 presents a three-dimensional plot of the scaled $\boldsymbol{\beta}$ prior and the integrand $p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}) \cdot p(\boldsymbol{\beta})$. All four graphs are unimodal. Hence, given the Gaussian prior, the integrand is relatively simple.

BQ is applied on the marginal likelihood integral in equation (78) in its probability-measure form:

$$p(\mathbf{y} \mid \mathbf{X}) = \int_{\mathbb{R}^2} f(\boldsymbol{\beta}) \, d\boldsymbol{\beta} = \int_{\mathbb{R}^d} p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}) \, d\mathbb{P}_{\boldsymbol{\beta}}(\boldsymbol{\beta}), \quad (79)$$

where $\mathbb{P}_{\boldsymbol{\beta}}$ denotes the Gaussian prior distribution. The grid of evaluation points is obtained by drawing random samples from this prior distribution.

3.3.3 Gaussian Mixture Model as Likelihood

The third test problem is based on a Gaussian mixture model, which results in a more challenging likelihood function due to its multimodal structure. The aim is still to estimate a marginal likelihood, but without using real-data:

$$\int_{\mathbb{D} := \mathbb{R}^2} f(\boldsymbol{\beta}) \, d\boldsymbol{\beta} = \int_{\mathbb{R}^2} p(\mathbf{y} \mid \boldsymbol{\beta}) \cdot p(\boldsymbol{\beta}) \, d\boldsymbol{\beta} = \int_{\mathbb{R}^2} L(\boldsymbol{\beta}) \cdot p(\boldsymbol{\beta}) \, d\boldsymbol{\beta}. \quad (80)$$

A Gaussian mixture model (GMM) is a probability distribution whose PDF is a weighted linear combination of $K \in \mathbb{N}$ Gaussian PDFs (Bishop, 2006):

$$p(\mathbf{u}) = \sum_{k=1}^K w_k \cdot \mathcal{N}(\mathbf{u}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad (81)$$

where the mixing weights w_k satisfy $0 \leq w_k \leq 1$ and $\sum_{k=1}^K w_k = 1$. Each Gaussian PDF has its own mean vector $\boldsymbol{\mu}_k$ and covariance matrix $\boldsymbol{\Sigma}_k$. We assume $\boldsymbol{\mu}_k = \mathbf{0}$, $\forall k \in \{1, 2, 3\}$, and $\boldsymbol{\Sigma}_k = \sigma_k^2 I$ for some $\sigma_k^2 \in \mathbb{R}$ and $\forall k \in \{1, 2, 3\}$.

We assume that the likelihood function $p(\mathbf{y} | \boldsymbol{\beta}) = L(\boldsymbol{\beta})$ is a GMM with $K = 3$. Note that it is independent of $\mathbf{y}$. As before, we assume $\boldsymbol{\beta} \sim \mathcal{N}_2(\mathbf{0}, \tau^2 I)$. Figure 24 in Appendix 6.1.2 displays the scaled prior $p(\boldsymbol{\beta})$, scaled likelihood $L(\boldsymbol{\beta})$ and scaled integrand $f(\boldsymbol{\beta})$. Under this Gaussian prior, the integrand also trimodal, mirroring to the likelihood.

Similar to above, BQ is applied on the marginal likelihood integral in equation (80) in its probability-measure form:

$$\int_{\mathbb{R}^2} f(\boldsymbol{\beta}) \, d\boldsymbol{\beta} = \int_{\mathbb{R}^2} L(\boldsymbol{\beta}) \, d\mathbb{P}_{\boldsymbol{\beta}}(\boldsymbol{\beta}), \quad (82)$$

where $\mathbb{P}_{\boldsymbol{\beta}}$ denotes the Gaussian prior distribution. The grid of evaluation points is obtained by drawing random samples from this prior distribution.

Moreover, the integral can be computed analytically:

$$\begin{aligned} \int_{\mathbb{R}^2} f(\boldsymbol{\beta}) \, d\boldsymbol{\beta} &= \int_{\mathbb{R}^2} \left[\sum_{k=1}^K w_k \cdot \mathcal{N}_2(\boldsymbol{\beta}; \boldsymbol{\mu}_k, \sigma_k^2 I) \right] \cdot \mathcal{N}_2(\boldsymbol{\beta}; \mathbf{0}, \tau^2 I) \, d\boldsymbol{\beta} \\ &= \sum_{k=1}^K w_k \int_{\mathbb{R}^2} \mathcal{N}_2(\boldsymbol{\beta}; \boldsymbol{\mu}_k, \sigma_k^2 I) \cdot \mathcal{N}_2(\boldsymbol{\beta}; \mathbf{0}, \tau^2 I) \, d\boldsymbol{\beta} \\ &= \sum_{k=1}^K w_k \mathcal{N}_2(\boldsymbol{\beta}; \boldsymbol{\mu}_k, (\sigma_k^2 + \tau^2) I). \end{aligned} \quad (83)$$

3.4 Hyperparameter Estimation

In this subsection, we illustrate how the kernel hyperparameters can be estimated using maximum likelihood (ML) rather than being selected manually. Manual selection is generally impractical, especially in higher dimensions. Estimation by maximum likelihood is the standard approach in GP modelling.

Suppose $f \sim \mathcal{GP}(m, k)$. Let $\boldsymbol{\theta}$ denote the vector of kernel k hyperparameters, and let $\mathbf{y} = \mathbf{f}_{\mathbf{X}} + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 I)$. The marginal likelihood of $\mathbf{y} \in \mathbb{R}^n$ is given by:

$$p(\mathbf{y} | \mathbf{X}, \boldsymbol{\theta}) = \int_{\mathbb{R}^n} p(\mathbf{y}, \mathbf{f}_{\mathbf{X}} | \mathbf{X}, \boldsymbol{\theta}) \, d\mathbf{f}_{\mathbf{X}} = \int_{\mathbb{R}^n} p(\mathbf{y} | \mathbf{X}, \mathbf{f}_{\mathbf{X}}, \boldsymbol{\theta}) \cdot p(\mathbf{f}_{\mathbf{X}} | \mathbf{X}, \boldsymbol{\theta}) \, d\mathbf{f}_{\mathbf{X}} \quad (84)$$

where $p(\mathbf{y} | \mathbf{X}, \mathbf{f}_{\mathbf{X}}, \boldsymbol{\theta})$ is the likelihood of $\mathbf{y}$, i.e., the probability of the observed data given a specific $\mathbf{f}_{\mathbf{X}}$ from the GP. Importantly, $p(\mathbf{y} | \mathbf{X}, \boldsymbol{\theta})$ is also the likelihood of the hyperparameters $\boldsymbol{\theta}$. So, $p(\mathbf{y} | \mathbf{X}, \boldsymbol{\theta})$ is the marginal likelihood of the observations, but also the likelihood of the hyperparameters. Moreover, since we assume noise-free observations, i.e., $\mathbf{y} = \mathbf{f}_{\mathbf{X}}$, the likelihood of interest is $p(\mathbf{y} | \mathbf{X}, \boldsymbol{\theta}) = p(\mathbf{f}_{\mathbf{X}} | \mathbf{X}, \boldsymbol{\theta})$. Consequently, the log (marginal) likelihood of the hyperparameters is:

$$\log(p(\mathbf{f}_{\mathbf{X}} | \mathbf{X}, \boldsymbol{\theta})) = -\frac{n}{2} \cdot \log(2\pi) - \frac{1}{2} \cdot \log(\det(\mathbf{K}_{\mathbf{XX}})) - \frac{1}{2} \cdot (\mathbf{y} - \mathbf{m}_{\mathbf{X}})^T \mathbf{K}_{\mathbf{XX}}^{-1} (\mathbf{y} - \mathbf{m}_{\mathbf{X}}), \quad (85)$$

where $\mathbf{K}_{\mathbf{XX}}$ is the only term that depends on the hyperparameters $\boldsymbol{\theta}$. Rasmussen and Williams (2008) present an explicit expression for the partial derivative of $\log(p(\mathbf{f}_{\mathbf{X}} | \mathbf{X}, \boldsymbol{\theta}))$ with respect to θ_i . In practice, however, equation (85) is usually optimised numerically. We follow this approach using the `scipy.optimize` function.

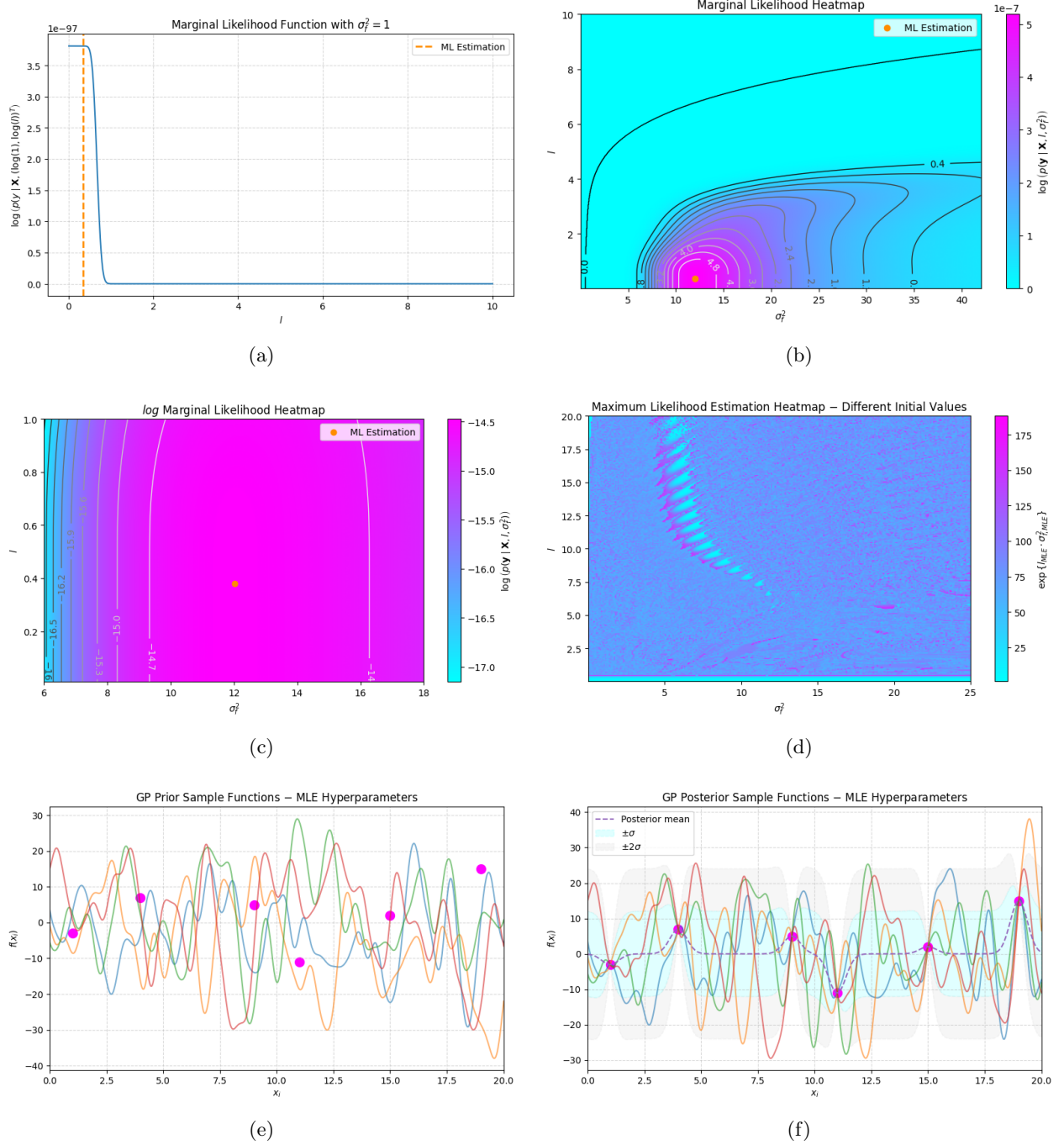


Figure 11: The hyperparameter estimation procedure for the one-dimensional integral example based on marginal likelihood optimisation. Subfigure 11a shows the likelihood as a function of the length-scale l , with σ_f^2 fixed to one. Subfigures 11b and 11c display heatmaps of the unscaled marginal likelihood and the log scaled marginal likelihood, respectively. Subfigure 11d presents a heatmap of the function $\exp\{l_{MLE} \cdot \sigma_{f,MLE}^2\}$ for different combinations of initial values used in the optimisation. Finally, Subfigures 11e and 11f show four sample functions drawn from the GP prior and posterior, respectively, using the estimated hyperparameters.

Figure 11 illustrates GP regression, analogous to Figure 8, but with hyperparameters estimated via maximum (log) likelihood. Subfigure 11a shows the likelihood as a function of the length-scale l , with σ_f^2 set to one. The marginal likelihood function is concentrated around the interval $[0, 1]$, indicating that the most likely values of l lie in this interval for the given dataset. Subfigures 11b and 11c display heatmaps of

the marginal likelihood and log marginal likelihood, respectively, with the maximum indicated by an orange dot. Subfigure 11d shows how different pairs of initial values used in the optimisation algorithm can yield different maximum likelihood estimates ($l_{MLE}, \sigma_{f,MLE}^2$). This can be due to multiple local maxima of the (marginal) likelihood function. Finally, Subfigures 11e and 11f present sample functions drawn from the GP prior and posterior, respectively, using the estimated hyperparameters.

However, performing maximum likelihood estimation on a dataset with six datapoints is not reliable and it is unlikely to yield accurate hyperparameter estimates. Figure 25 repeats the same procedure as in Figure 11, but with a larger number of observations.

3.5 Kernel Choice – Highest Maximum Likelihood

In this subsection, we select the best covariance kernel for the given collection of observations. The "best" kernel is defined as the kernel that maximises the log (marginal) likelihood among a predefined family of kernels. We choose a family of kernels containing the squared-exponential, rational quadratic, Matérn with $\alpha \in \{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}\}$, polynomial, periodic, and all pairwise sums of two distinct kernels from this collection (up to symmetries). In addition, the maximum likelihood estimated hyperparameters corresponding to the best kernel are used.

The best kernel depends heavily on the number of observations. For instance, with $n = 10$ observations, the best kernel is the sum of a rational quadratic and periodic kernel (see Figure 26). The squared-exponential kernel is one of worst choices of kernels in this situation. This improved kernel choice results in a significantly more accurate integral estimate with lower uncertainty compared to BHQ as shown in Figure 10. For visual clarity, the bar-plot excludes kernels with very low log (marginal) likelihoods, as these would otherwise distort the plot.

When the number of observations increases to $n = 15$ and $n = 20$ observations, the best kernel is the sum of a Matérn kernel with $\alpha = \frac{1}{2}$ and a periodic kernel (see Figures 12 and 27 in Appendix 6.1.4). Both the integral estimate and its posterior standard deviation improve significantly. At $n = 30$, the best kernel changes to the sum of a Matérn kernel with $\alpha = \frac{3}{2}$ and a periodic kernel (see Figure 28 in Appendix 6.1.4). Finally, for $n = 42$ observations, the best kernel becomes the sum of a squared-exponential and a rational quadratic kernel (see Figure 29 in Appendix 6.1.4). In both cases, the integral estimate and its posterior standard deviation continue improve drastically.

3.6 Adaptive Sampling Scheme

In this subsection, we discuss a principled approach of selecting the next evaluation point according to some acquisition function. As illustrated in Figure 20, adding observations does not necessarily lead to improved outcomes.

Suppose we have n observations. The aim of adaptive sampling is to select new evaluation points sequentially according to some criterion that depends on the observed function values $\mathbf{f}_{\mathbf{X}}$. Such a criterion is encoded via some acquisition function $A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^d$. Adaptive sampling is also referred to as active sampling or sequential Bayesian quadrature, while acquisition functions are sometimes called loss functions or utility functions.

The idea of adaptive sampling is to choose the next evaluation point $\mathbf{x}_{n+1}$ by optimising the acquisition function conditioned on the current observations $\mathcal{D}$ of the integrand f :

$$\mathbf{x}_{n+1} = \arg \max_{\mathbf{x}} A(\mathbf{x} \mid \mathcal{D}). \quad (86)$$

Two of the most important adaptive sampling schemes for BQ include uncertainty sampling and minimising the posterior integral variance. Other adaptive sampling schemes include minimising the posterior entropy of the integral, as discussed in [Gunter et al. \(2014\)](#) and [Gessner et al.](#)

3.6.1 Uncertainty Sampling

Uncertainty sampling selects the next evaluation point where the integrand f is most uncertain under the GP posterior of f . That is, the next point $\mathbf{x}_{n+1}$ is chosen to maximise the posterior variance of f :

$$\mathbf{x}_{n+1} = \arg \max_{\mathbf{x}} k_{f,\cdot}(\mathbf{x}, \mathbf{x}). \quad (87)$$

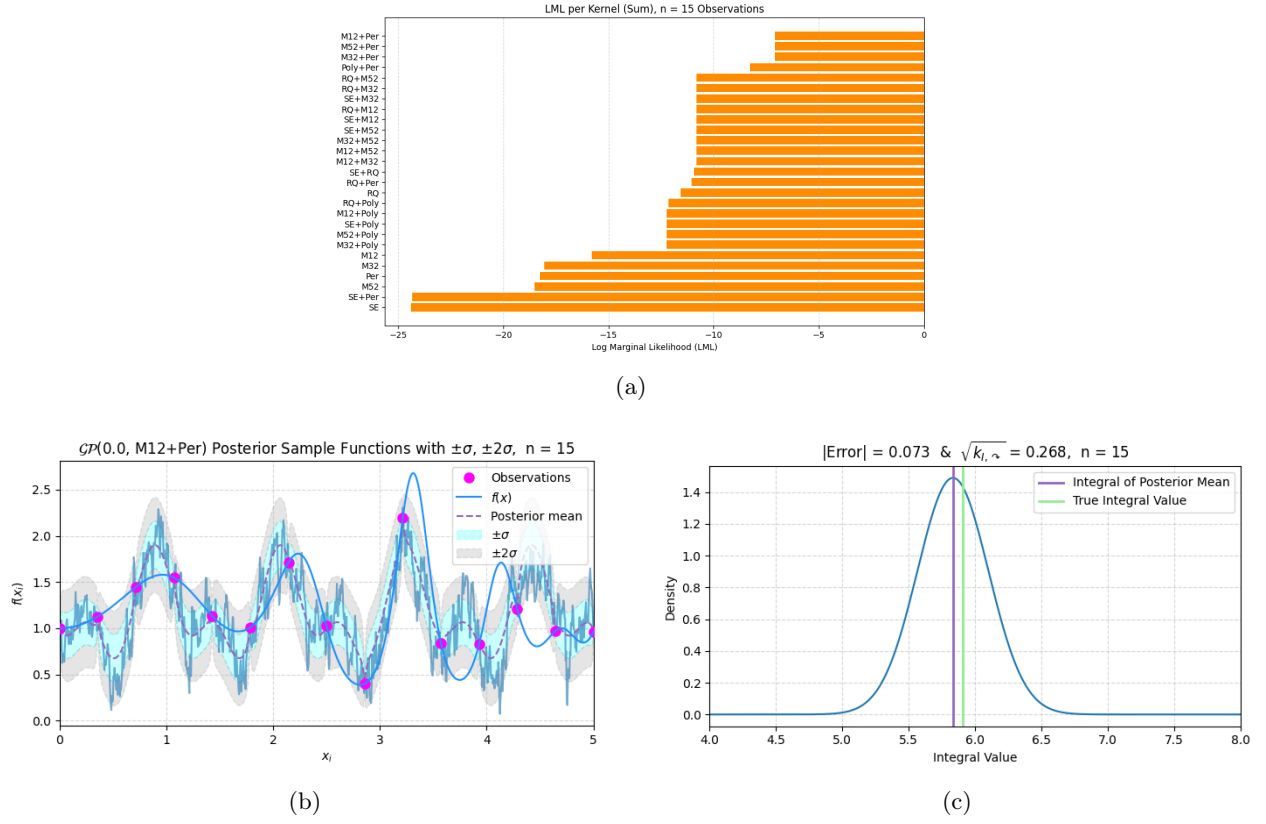


Figure 12: Subfigure 12a illustrates that the sum of the Matérn kernel with $\alpha = 1/2$ and periodic kernel has the highest log (marginal) likelihood for the one-dimensional integral example with $n = 15$ observations. Subfigure 12b shows the posterior Gaussian process with the ‘best’ kernel and hyperparameters for the one-dimensional integral example. Subfigure 12c shows the corresponding posterior distribution of the integral.

Note that for all of our considered kernels, $k(\mathbf{x}, \mathbf{x})$ is a constant σ_f^2 . Hence it is the same as finding the minimiser of $k_{\mathbf{x}\mathbf{x}}\mathbf{K}_{\mathbf{X}\mathbf{X}}^{-1}k_{\mathbf{x}\mathbf{x}}$. Only the polynomial kernel is non-stationary and $k_{f,\gamma}$ should be maximised.

Figure 14 illustrates the behaviour of uncertainty sampling when one and five additional evaluation points are added. When a single point is added, the absolute error increases substantially, whereas the posterior standard deviation decreases slightly. When five points are added, the absolute error is nearly as low as in the case without additional points, while the posterior standard deviation decreases as well. Figure 31 in Appendix 6.1.5 shows the results when 15 and 100 points are added using uncertainty sampling.

3.6.2 Minimising Posterior Integral Variance

Alternatively, we can choose the next evaluation point as the point that maximally reduces the posterior variance of the integral itself. That is, we choose the point $\mathbf{x}_{n+1}$ that reduces the uncertainty about the integral the most:

$$\mathbf{x}_{n+1} = \arg \max_{\mathbf{x}} \text{Var}(I \mid \mathcal{D} \cup \{\mathbf{x}, f(\mathbf{x})\}). \quad (88)$$

Figure 14 illustrates the behaviour of the posterior variance minimisation scheme when one evaluation point is added. The absolute error is significantly lower than in the case of adding a single point via uncertainty sampling, while the posterior standard deviation remains approximately the same. However, in contrast to uncertainty sampling, minimising the posterior integral variance as an adaptive sampling scheme is often computationally expensive. Figure 31 shows the results of this scheme when five and ten points are added. Although the posterior standard deviation improves, the absolute error increases. Therefore, we adopt uncertainty sampling as our adaptive sampling scheme of choice.

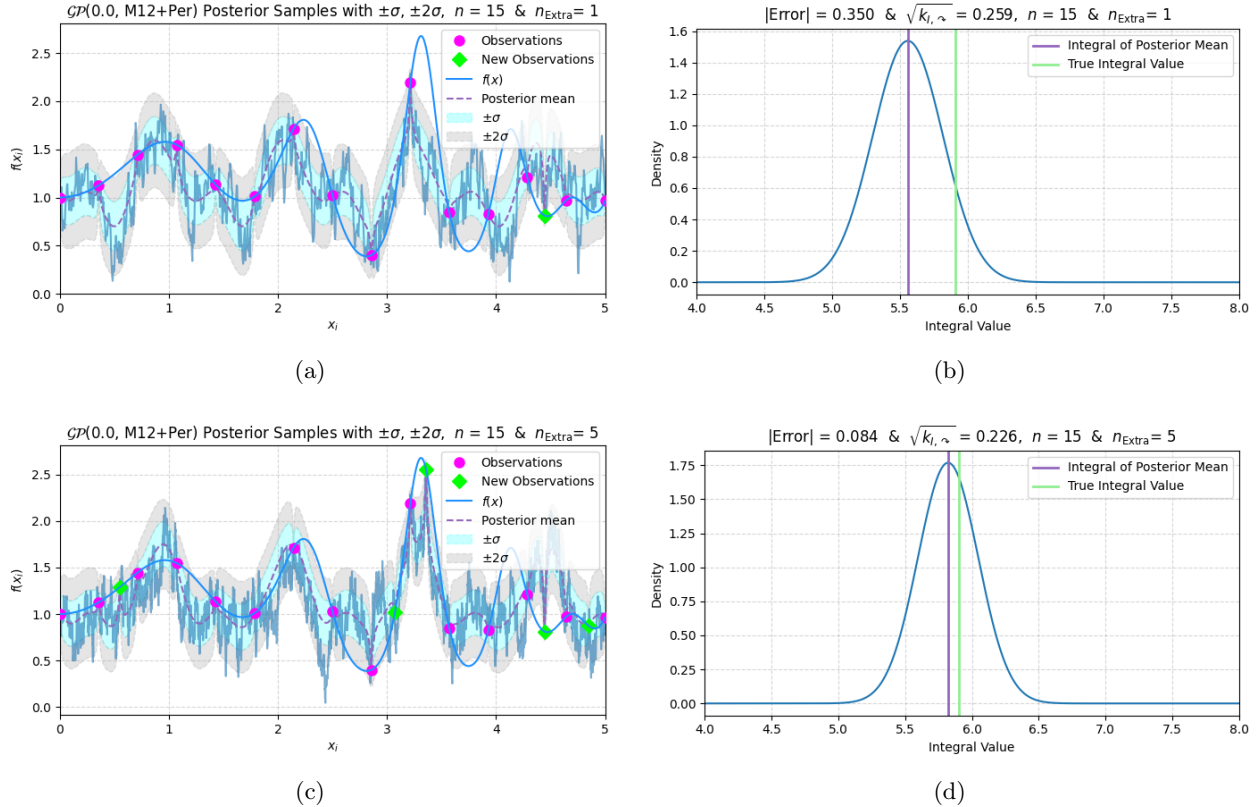


Figure 13: Subfigure 13a illustrates the uncertainty-sampling procedure for the one-dimensional integral with $n = 15$ observations. Subfigure 13b shows the corresponding posterior distribution of the integral. Subfigures 13c and 13d illustrate the same procedure, but with five evaluation points added iteratively instead of one.

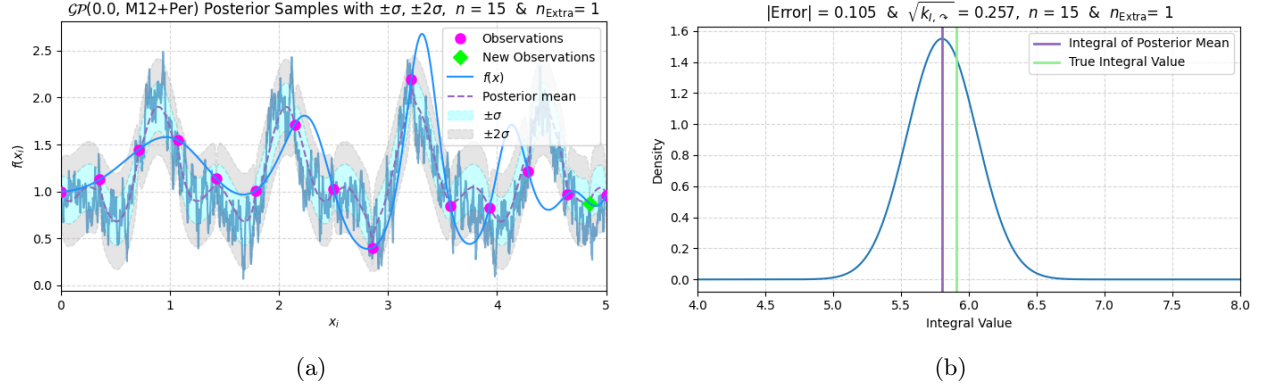


Figure 14: Subfigure 14a shows the adaptive scheme that minimises the posterior variance of the one-dimensional integral with $n = 15$ observations. Subfigure 14b shows the corresponding posterior distribution of the integral. This procedure is computationally expensive, and the resulting estimate deteriorates rather than improves.

4 Simulation Results

In this section, we discuss the simulation results for each of the three scenarios.

4.1 One-Dimensional Integral

Figure 32 shows the convergence of the estimates of the eight methods to the true value of the integral, which is approximately 5.9. All methods, except the two BQ methods that re-estimate only the hyperparameters, exhibit stable convergence behaviour. Re-estimating only the hyperparameters leads to highly deviant estimates when the number of samples is below 50. Figure 15 provides a more precise visualisations of the accuracy of the methods on a grid of 200 points between 1 and 1000. The MC integration estimates clearly satisfy the $\mathcal{O}(n^{-1/2})$ convergence rate. The MC estimate curves are the average of 100 estimates at each sample size, and the shaded region represents one posterior standard deviation.

All the BQ methods, except the vanilla BQ variant that re-estimates both the kernel and the hyperparameters, start from a grid of 15 evaluation points. This exception is made purely for illustrative comparison with MC integration at small sample sizes. Table 1 provides an overview of the maximum likelihood values for all kernels in the considered kernel family, together with their corresponding maximum-likelihood hyperparameter estimates based on 15 evaluation points. The sum of a Matérn kernel with $\alpha = 1/2$ and a periodic kernel yields the highest marginal likelihood. Moreover, the BQ methods form clear clusters: between 15 and 60 observations, the two methods that re-estimate both kernel and hyperparameters outperform the other BQ variants. BQ with uncertainty sampling and re-estimation of both kernel and hyperparameters is the most accurate method between 40 and 200 observations. When the number of observations reaches 700, all methods yield estimates of comparable accuracy. This is most likely due to the relatively small integration domain $\mathbb{D} = [0, 5]$, which becomes well explored after several hundred function evaluations. Note also that for sample sizes larger than 100, all BQ methods outperform MC integration.

However, MC integration is computationally less expensive in comparison to the BQ methods. Figure 33 displays the number of iterations each method performs within 0–300 seconds. The MC integration methods achieve far more iterations because the BQ methods are computationally intensive. This is due to repeated Cholesky decompositions and matrix multiplications, re-estimation procedures, and the uncertainty sampling calculations. Figures 34 and 35 show the absolute error between zero and 300 seconds, without and with the hyperparameter only re-estimation scenario, respectively. Due to randomness in the sampling of evaluation points, MC integration occasionally attains extremely accurate estimates for particular sample grids. Note that MC integration starts earlier than the BQ methods because determining the kernel and hyperparameters via maximum likelihood estimation requires additional computation time. Nevertheless, the BQ estimates are more stable and, on average, lead to more accurate results for this specific one-dimensional integral.

4.2 Marginal Likelihood Example – Real-World Data

This subsection is divided into two parts. The first concerns the estimation of a two-dimensional variant of the marginal likelihood in equation (78) using the salary dataset. The second addresses the estimation of a four-dimensional variant of the same marginal likelihood using the startup profit dataset. Both models include an intercept term.

4.2.1 Salary Data: $d = 2$

Figures 36 and 37 show the convergence of the estimates of the seven methods considered, using up to 100 observations, on an unscaled plot and on a $\log_{10}$ - $\log_{10}$ plot. In both cases, the MC integration estimates are more stable and more accurate. However, the true value of the integral is extremely small, on the order of 10^{-14} . Figure 16 displays the absolute error of the seven methods. This plot shows that uncertainty sampling BQ with re-estimation of the hyperparameters outperforms MC integration. Vanilla BQ with re-estimation of only the hyperparameters or of both the kernel and hyperparameters, as well as uncertainty sampling BQ, all exhibit improved accuracy compared with MC integration, although their estimate curves intersect the MC estimate curve at least eight times. In contrast, the BQ methods that

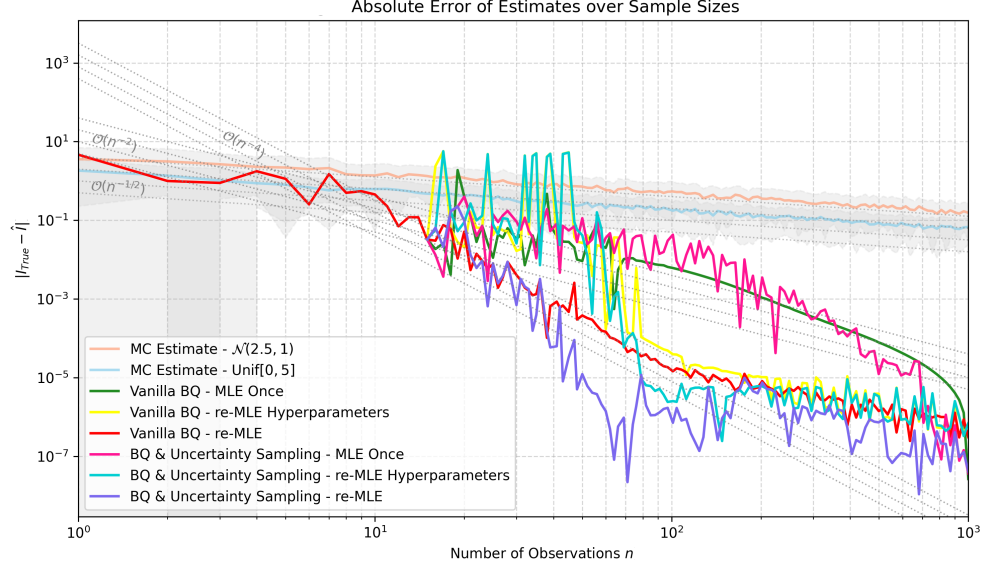


Figure 15: Comparison of the accuracy of six BQ methods and two MC methods for the one-dimensional integral example.

estimate the kernel and hyperparameters only once display highly uncertain and oscillatory behaviour. Figure 38 shows the corresponding relative absolute error.

Tables 2 and 4 display the hyperparameter values obtained from re-estimating the hyperparameters for both vanilla BQ and uncertainty sampling BQ. The best-performing kernel in this setting is the sum of a Matérn kernel with $\alpha = 1/2$ and a periodic kernel. In addition, Tables 3 and 5 show the hyperparameter values obtained in each iteration when both the kernel and its hyperparameters are re-estimated. Occasionally, the length-scale hyperparameters are estimated to be extremely large. Such values are highly implausible and likely indicate numerical instability or model misspecification.

Although certain BQ methods appear to outperform MC integration for specific sample sizes, their computational cost is substantially higher. Figure 39 illustrates that within 1000 seconds, the BQ methods reach grid sizes of only 30–40, despite starting from a grid size of 21. By contrast, MC integration reaches a grid size of 10,000 in the same time interval. Figure 40 shows the absolute error of the estimates as a function of computation time, up to 1000 seconds. Note that the BQ methods produce their first estimate only after 50–60 seconds, whereas MC integration produces its first estimate after approximately 0.002 seconds. The accuracy achieved by the BQ estimates is on the order of 10^{-14} , which is poor because it is comparable to the magnitude of the true value. In contrast, the MC estimates achieve significantly better accuracy after 1000 seconds.

4.2.2 Profit Startup Companies Data: $d = 4$

Figure 41 shows the convergence of the estimates of the seven methods for the four-dimensional marginal likelihood. Only the MC integration estimates exhibit clear convergence and oscillate around the true value. Vanilla BQ with re-estimation of both the kernel and hyperparameters, as well as uncertainty-sampling BQ with re-estimation of only the hyperparameters, produce estimates that remain clearly non-zero at 100 observations. All remaining BQ methods yield estimates that are effectively zero at this sample size.

4.3 Gaussian Mixture Model – Likelihood

Figures 42 and 43 show the convergence of the estimates of the seven methods, using up to 100 observations, on an unscaled plot and on a $\log_{10}$ - $\log_{10}$ plot. Again, the MC estimates appear more stable and more accurate than those of the BQ methods. Figure 17 displays the absolute error of the seven estimates. Both BQ methods that estimate the kernel and hyperparameters once or re-estimate them in

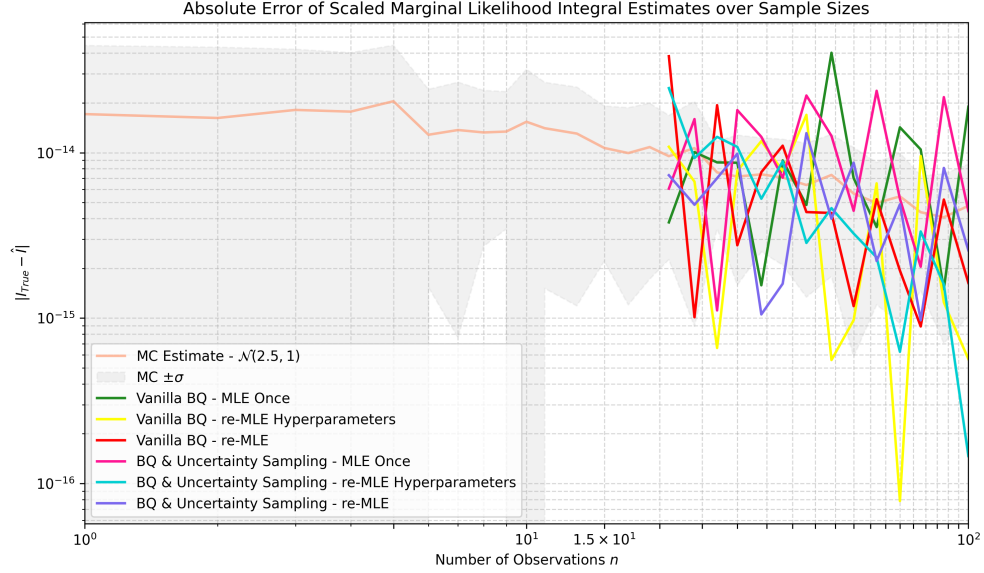


Figure 16: Comparison of the accuracy of six BQ methods and one MC method for the two-dimensional marginal likelihood example.

each iteration produce estimates whose absolute errors oscillate around those of MC integration. In contrast, BQ with re-estimation of both kernel and hyperparameters is more accurate than MC integration for most sample sizes.

Furthermore, both BQ methods that re-estimate only the hyperparameters cease to converge once the grid size exceeds approximately 30–40 points. This occurs because the maximum-likelihood optimisation is initialised at a default value of 1.0. When the optimisation fails (for example, due to a singular kernel matrix) the algorithm retains these initial hyperparameter values. This behaviour is clearly visible in Tables 7 and 9. As a consequence, the resulting integral estimate collapses to zero or near-zero, which is undefined on a $\log_{10}$ -scale, causing the corresponding line to disappear from the plot.

Tables 6 and 8 report the estimated hyperparameters when both the kernel and its hyperparameters are re-estimated. In two instances, a hyperparameter reaches the upper bound of 1000, which is imposed to prevent the optimisation algorithm from searching over an unbounded domain.

Figure 45 shows that within 1000 seconds, the BQ methods that re-estimate both kernel and hyperparameters reach a grid size of only 12. The other BQ methods reach grid sizes between 22 and 26, whereas MC integration achieves a grid size of 10,000. Similar to Subsection 4.2, Figure 46 demonstrates that the absolute errors of the BQ estimates are substantially larger than those of the MC estimates. Specifically, the BQ errors, which range between 10^{-2} and 10^{-3} , are larger than the magnitude of the true value ($\approx 6.5 \times 10^{-3}$).

Moreover, Figures 47 and 48 show the convergence of the estimates when using an adjusted grid design in which 30 percent of the evaluation points drawn from the Gaussian prior lie in a small neighbourhood around one of the three modes. Figures 49 and 50 present the corresponding absolute and relative absolute error plots, respectively. Uncertainty sampling BQ and both BQ methods that re-estimate only the hyperparameters are outperformed by MC integration when the number of observations exceeds approximately 60. Vanilla BQ with re-estimation of both kernel and hyperparameters, and uncertainty sampling BQ with hyperparameters estimated once, oscillate around the MC estimates but slightly outperform MC integration when the sample size exceeds 85. Uncertainty sampling BQ with re-estimation of both kernel and hyperparameters outperforms MC integration and all other BQ methods once the number of observations exceeds 60.

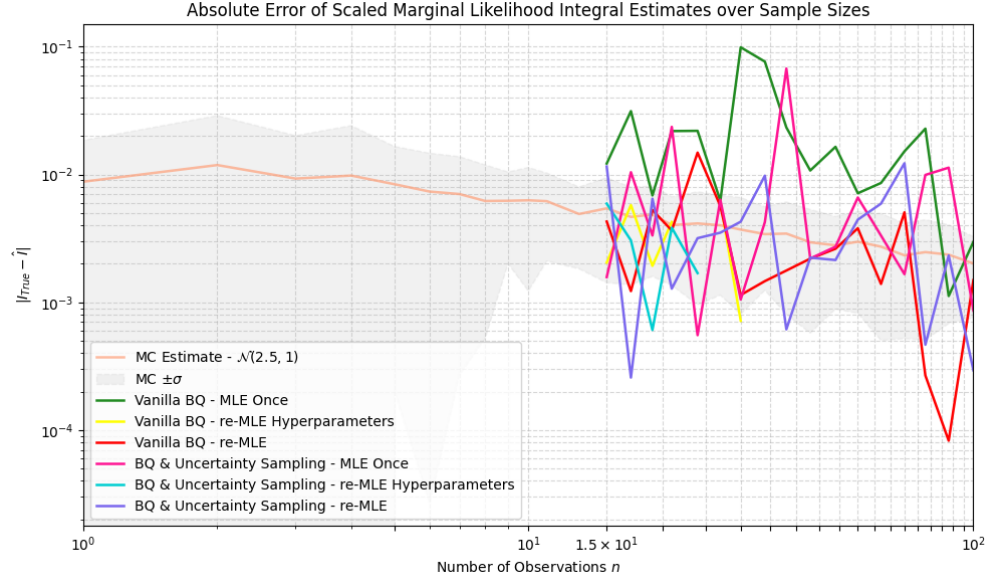


Figure 17: Comparison of the accuracy of six BQ methods and one MC method for the marginal likelihood with a synthetic trimodal likelihood example.

5 Conclusion & Discussion

5.1 Conclusion

This thesis has presented a detailed explanation of the Bayesian quadrature (BQ) mechanism by viewing Gaussian processes (GPs) as Gaussian measures on function spaces and exploiting their closedness property under bounded linear operators. Consequently, the integral of interest can be expressed as an integration operator applied to a GP, and the closedness property allows this operator to be pushed through the GP. This yields a posterior distribution for the integral, whose mean and covariance are obtained by applying the integration operator to the posterior mean function and covariance kernel of the GP.

In addition, a systematic comparison was conducted between Monte Carlo (MC) integration, fixed-grid (vanilla) Bayesian quadrature, and uncertainty sampling BQ across three scenarios: estimating the kernel and hyperparameters once, re-estimating only the hyperparameters, and re-estimating both the kernel and hyperparameters. The comparison was performed on three applications: a one-dimensional Lebesgue integral, and two marginal likelihood estimation problems involving real-world data and a synthetic trimodal likelihood. In the latter example, an alternative grid design was applied in which 30 percent of the evaluation points were placed near the modes. This led to a modest improvement for the BQ methods. However, BQ clearly outperformed MC integration only in the one-dimensional setting. In the higher-dimensional or more structurally complex scenarios, the advantages of BQ diminished, whereas MC integration remained more reliable. This behaviour is most likely due to the more complicated graph of the integrand, the higher dimensionality, and the increased difficulty of estimating kernels and hyperparameter.

Taking into account the trade-off between computational cost and estimation accuracy, either vanilla BQ or uncertainty sampling BQ with re-estimation of both kernel and hyperparameters appears to be the most effective standard BQ alternative to MC integration when integrand evaluations are relatively inexpensive. Re-estimating only the hyperparameters can lead to instability in their maximum likelihood estimates, resulting in poor integral approximations. Likewise, estimating the kernel and hyperparameters only once on an initial grid yields the least accurate results among the BQ variants considered.

Overall, these findings highlight that BQ is a powerful yet assumption-sensitive technique. Its performance depends on many factors, including kernel choice, hyperparameter estimation strategy, the suitability of prior assumptions, and the computational cost of evaluating the integrand. In contrast, MC integration is robust across all settings studied and demonstrates stable convergence behaviour regardless of integrand complexity. Moreover, BQ is computationally more expensive, which further favours the slower but uniform convergence rate of MC integration. Consequently, BQ is most effective as a specialised, tunable inference method for integrals when meaningful prior structural information is available and can be encoded through an appropriate kernel, especially when multi-dimensional or extremely small-valued integrals are considered.

Potential directions for future research include improving kernel family, kernel and hyperparameter selection. For example, by applying improved optimisation or numerical approximation algorithms, or by an alternative estimation procedure. Enhancing the robustness of BQ to constraints, such as non-negativity or high dimensionality, could also lead to improvements. Moreover, comparing BQ with other advanced techniques, such as thermodynamic integration, particularly in non-marginal likelihood applications, would provide further insight into BQ's role within the broader landscape of numerical integration techniques.

5.2 Discussion

The comparative results in this thesis illustrate that the performance of BQ is sensitive to the choice of kernel family, the estimation of kernels and hyperparameters, and the structural properties of the integrand. BQ performed well in the one-dimensional integral setting, where the integrand was smooth. However, the advantages of BQ diminished as the number of dimensions increased or the integrand became more complex. In contrast, MC integration remains robust across all settings studied and demonstrates stable convergence behaviour.

One of the main difficulties in BQ is the reliable estimation of kernels and their hyperparameters using maximum likelihood alone. In particular, when only the hyperparameters were re-estimated, the optimisation frequently failed, leading the algorithm to fall back to the default initial values. This behaviour was likely caused by (nearly) singular kernel matrices, violations of positive definiteness, or

instabilities in the optimisation routine. Although re-estimating both the kernel and hyperparameters improves performance, it also incurs a substantial computational cost. Overall, BQ methods (especially those involving repeated re-estimation) are significantly more expensive than MC integration due to repeated optimisation and large covariance-matrix decompositions and operations. This computational cost constraints limited the creation of plots on larger sample size and time grids. The restriction to personal computers also imposed practical limitations on the size and scope of the simulations.

Furthermore, the alternative grid design used in the synthetic trimodal likelihood example produced only minor improvements. As a result, MC integration provided more stable and more accurate estimates in both marginal likelihood applications.

Moreover, several critical assumptions and limitations must also be acknowledged. First, no reproducing kernel Hilbert spaces or Sobolev spaces were considered. Second, the initial grid sizes (15–25 points) were chosen without theoretical justification. Third, only 30 observations were used in the real-data marginal likelihood example. Fourth, the adaptive sampling procedure relied solely on uncertainty sampling, other acquisition functions were not explored. Fifth, in the marginal likelihood settings, the integrals needed for BQ were themselves approximated by MC integration, introducing an additional layer of uncertainty. Sixth, both BQ and MC drew evaluation points from the parameter prior, which leads to high variance for small sample sizes. Seventh, the posterior variance was not utilised in the comparison. Eighth, the only multi-dimensional integrals considered were marginal likelihoods with extremely small values, making the comparison numerically challenging.

Finally, when writing this thesis, I allocated too much time in trying to figure out how the BQ mechanism worked when not relying on important statements without proper proof. In particular, the closedness property of GPs under bounded linear operators and the theory of Bochner integrals. Furthermore, implementing all methods from scratch was also time-consuming and might have led to small coding bugs that could affect the numerical results.

6 Appendix

6.1 Some Figures

6.1.1 Subsection 2.4: Bayesian Quadrature

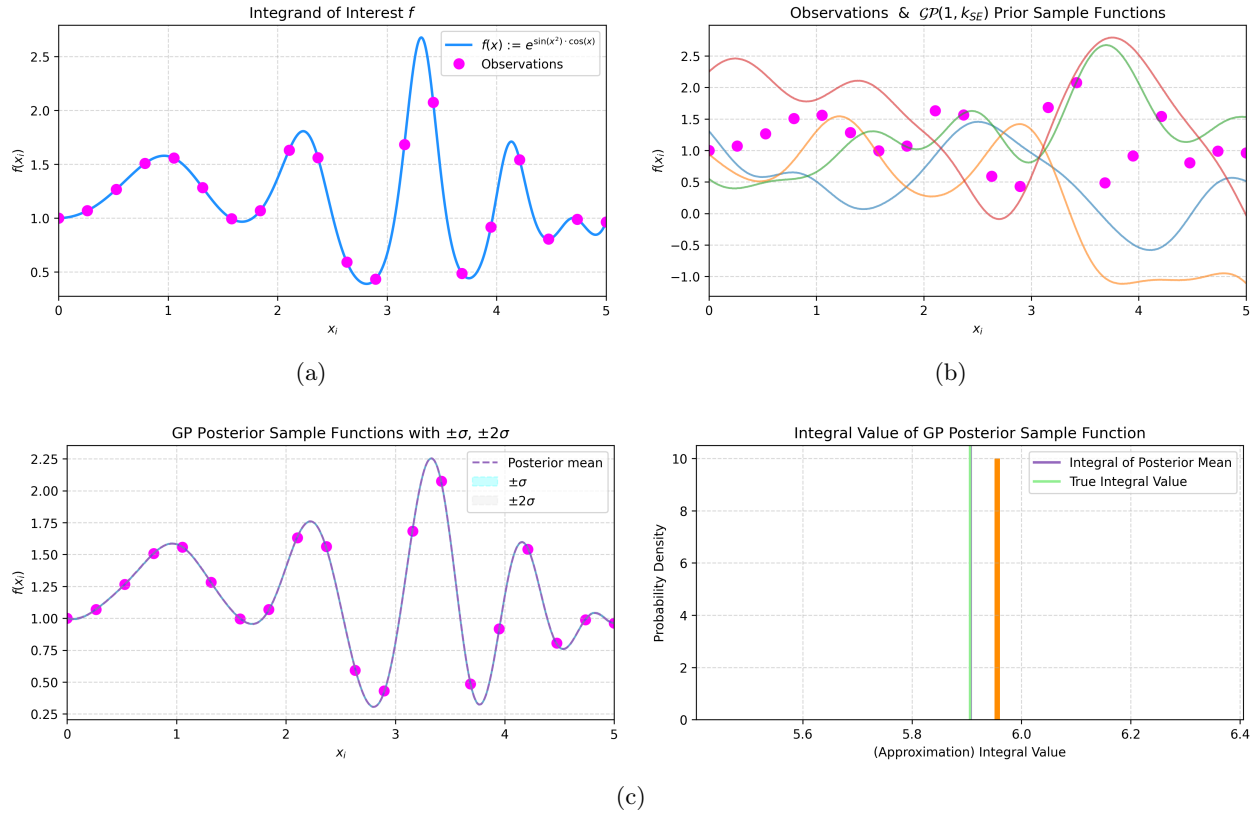
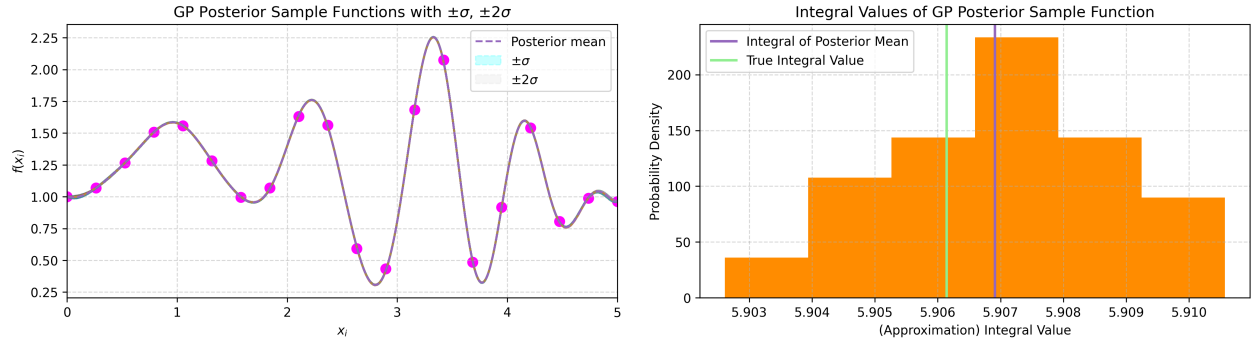
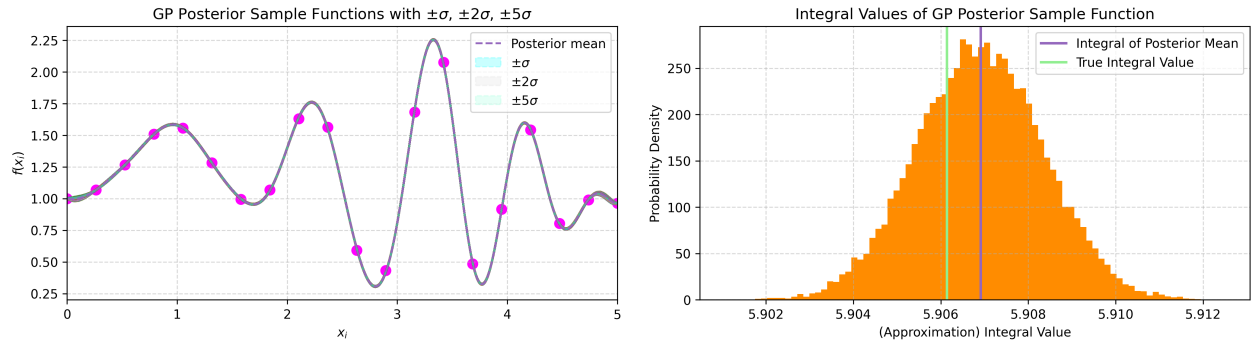


Figure 18: The distribution of the integrals of sample functions drawn from the posterior GP, based on $n = 20$ observations. Subfigure 18a shows the integrand. Subfigure 18b displays four sample functions drawn from the prior. Subfigure 18b shows the distributions of the integral computed from 1 posterior sample functions, respectively.



(a)



(b)

Figure 19: The distribution of the integrals of sample functions drawn from the posterior GP, based on $n = 20$ observations. Subfigures 19a and 19b show the distributions of the integral computed from 42 and 23,000 posterior sample functions, respectively. These histograms are created using the Freedman-Diaconis rule for optimal bin width.

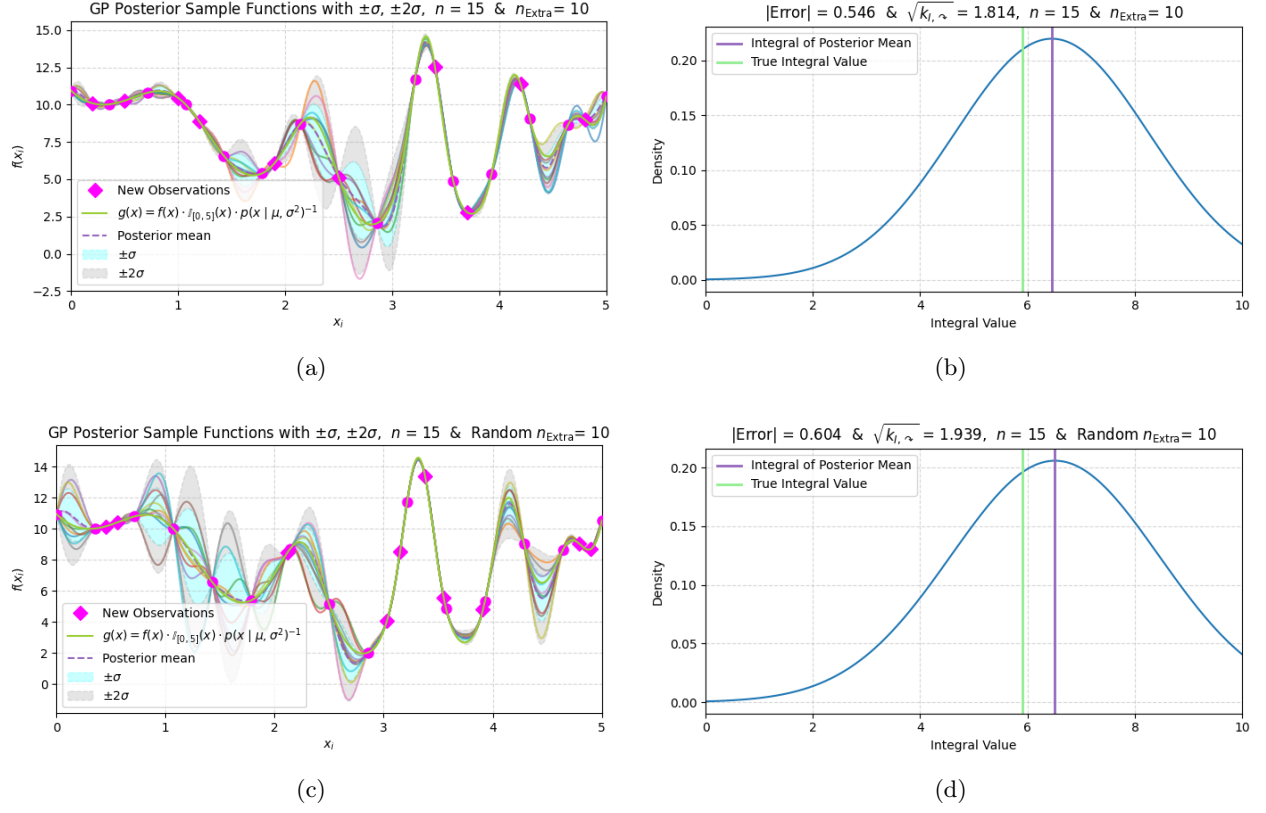


Figure 20: This figure illustrates Bayes-Hermite quadrature with a zero prior mean. Similar to Figure 10, but with extra observations.

6.1.2 Subsection 3.3: Models & Integrals

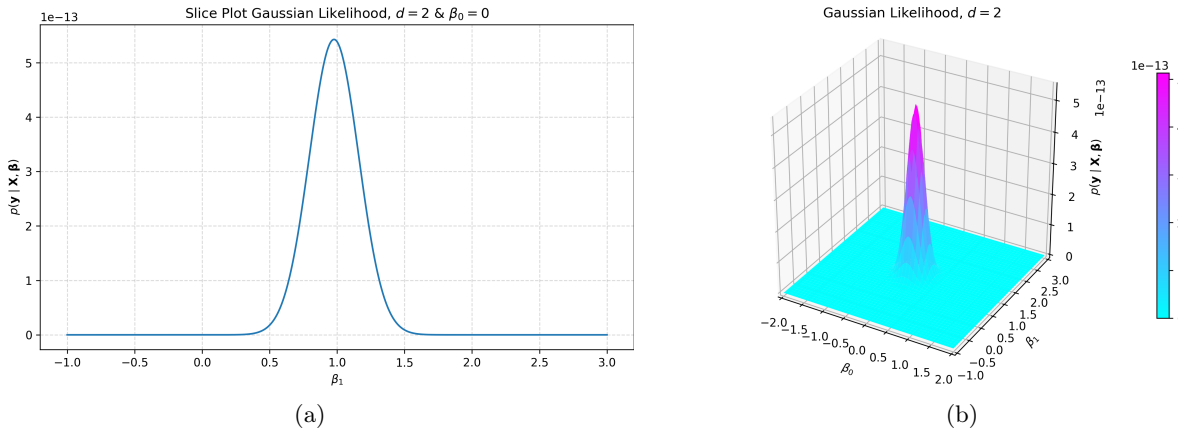


Figure 21: Unscaled likelihood plot for the two-dimensional marginal likelihood example.

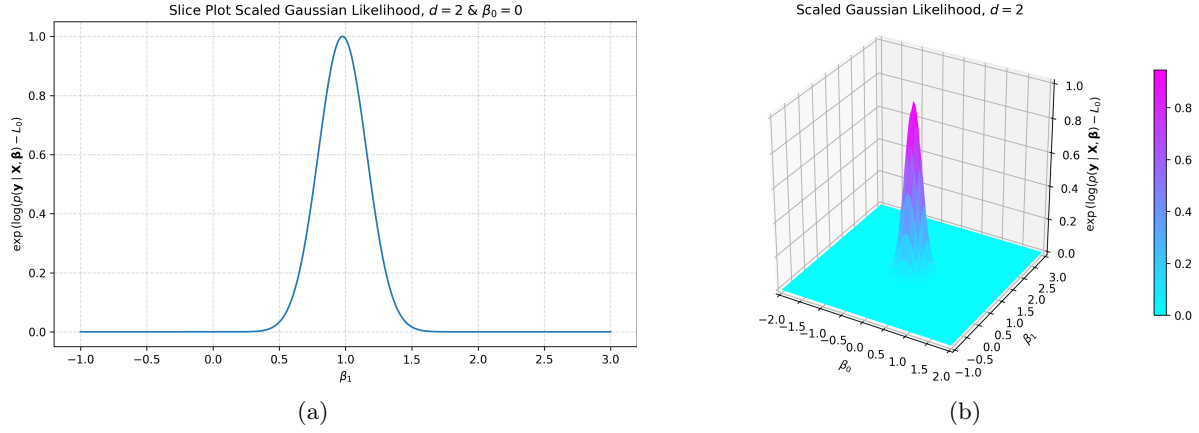


Figure 22: Scaled likelihood plot for the two-dimensional marginal likelihood example.

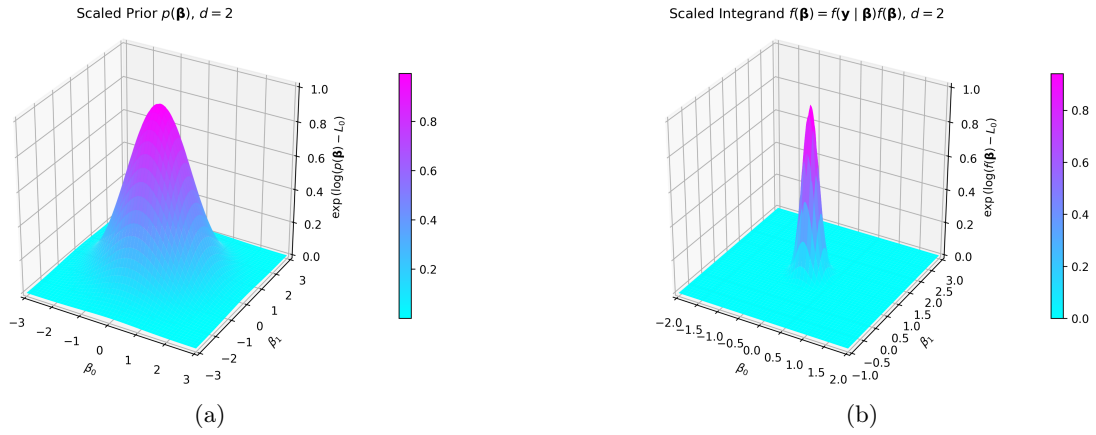


Figure 23: Scaled prior and integrand plots Unscaled likelihood plot for the two-dimensional marginal likelihood example.

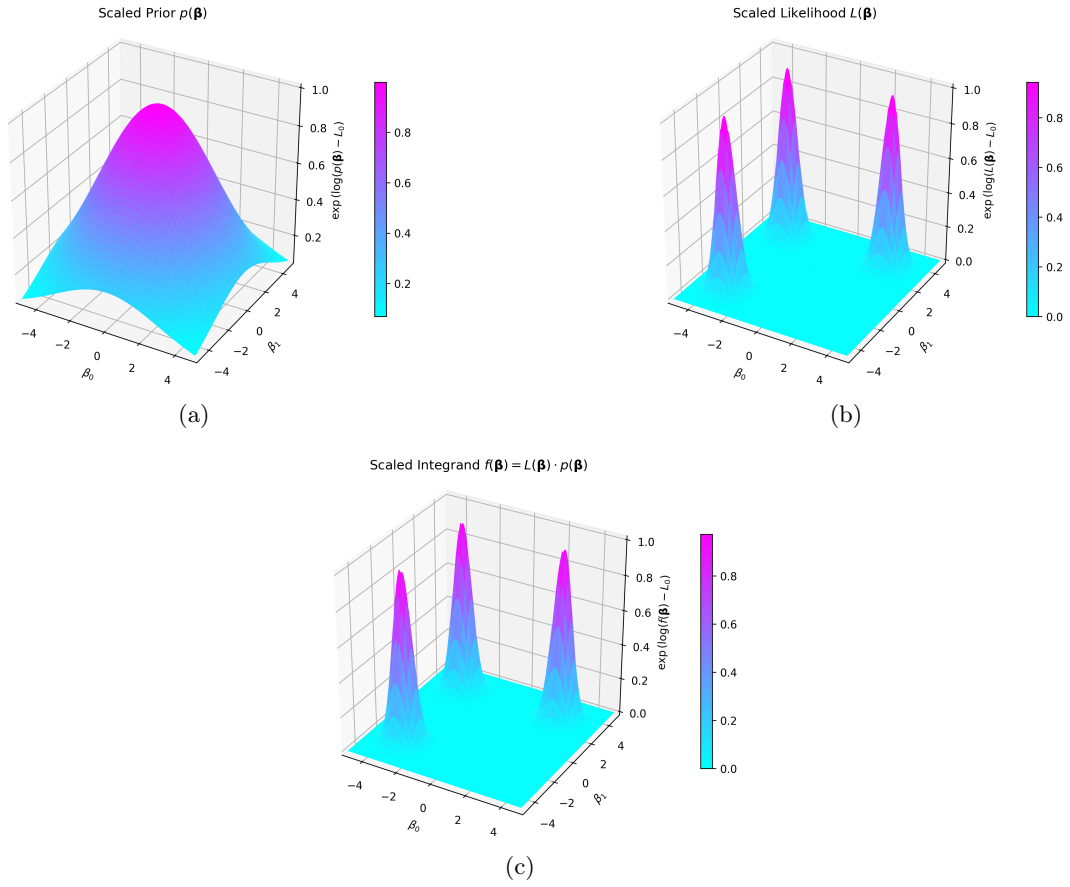


Figure 24: Plots of scaled prior, likelihood and integrand for the marginal likelihood example with synthetic trimodal likelihood.

6.1.3 Subsection 3.4: Hyperparameter Estimation

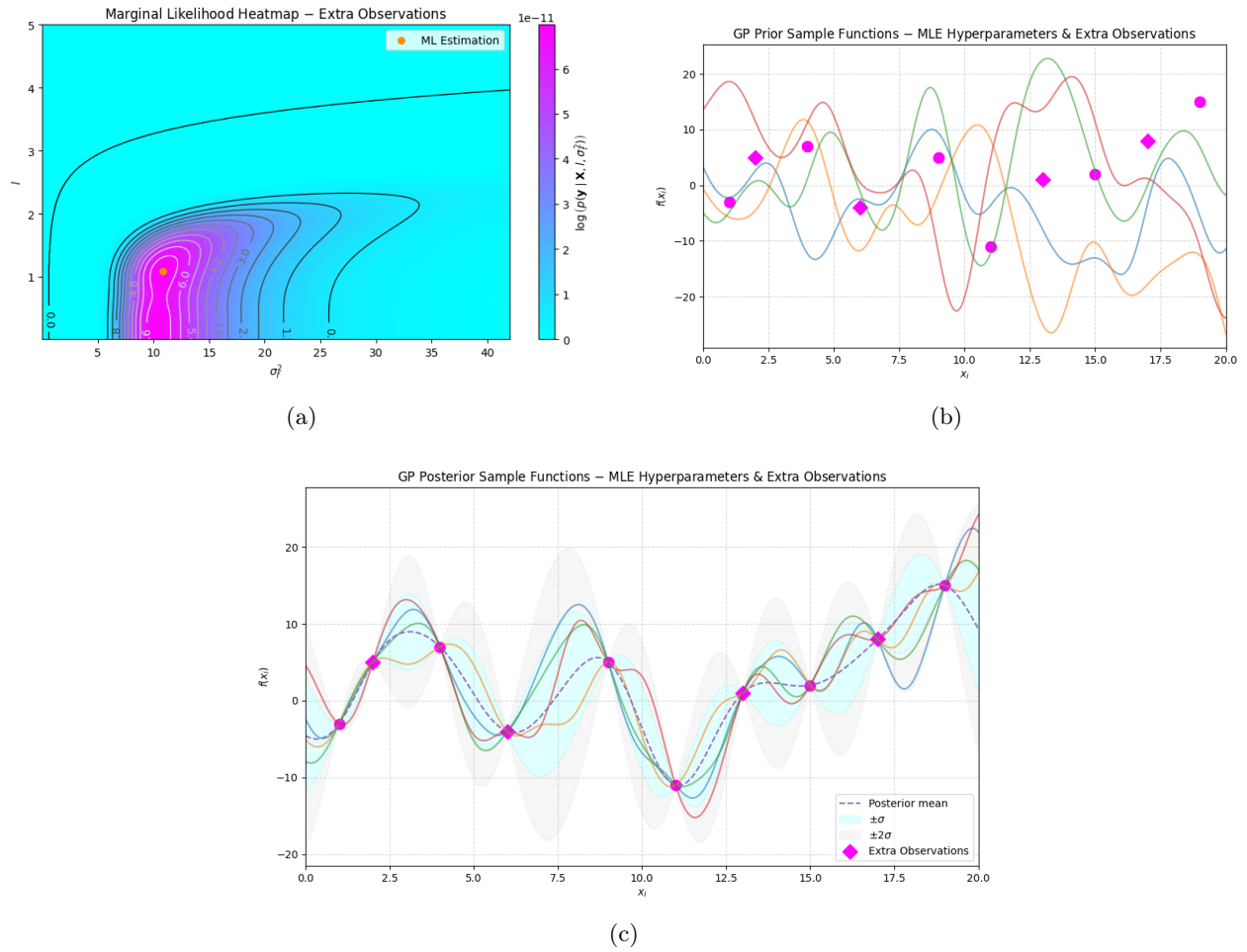


Figure 25: The hyperparameter estimation procedure for the one-dimensional integral example based on marginal likelihood optimisation. Similar to Figure 9, but with extra observations.

6.1.4 Subsection 3.5: Kernel Choice

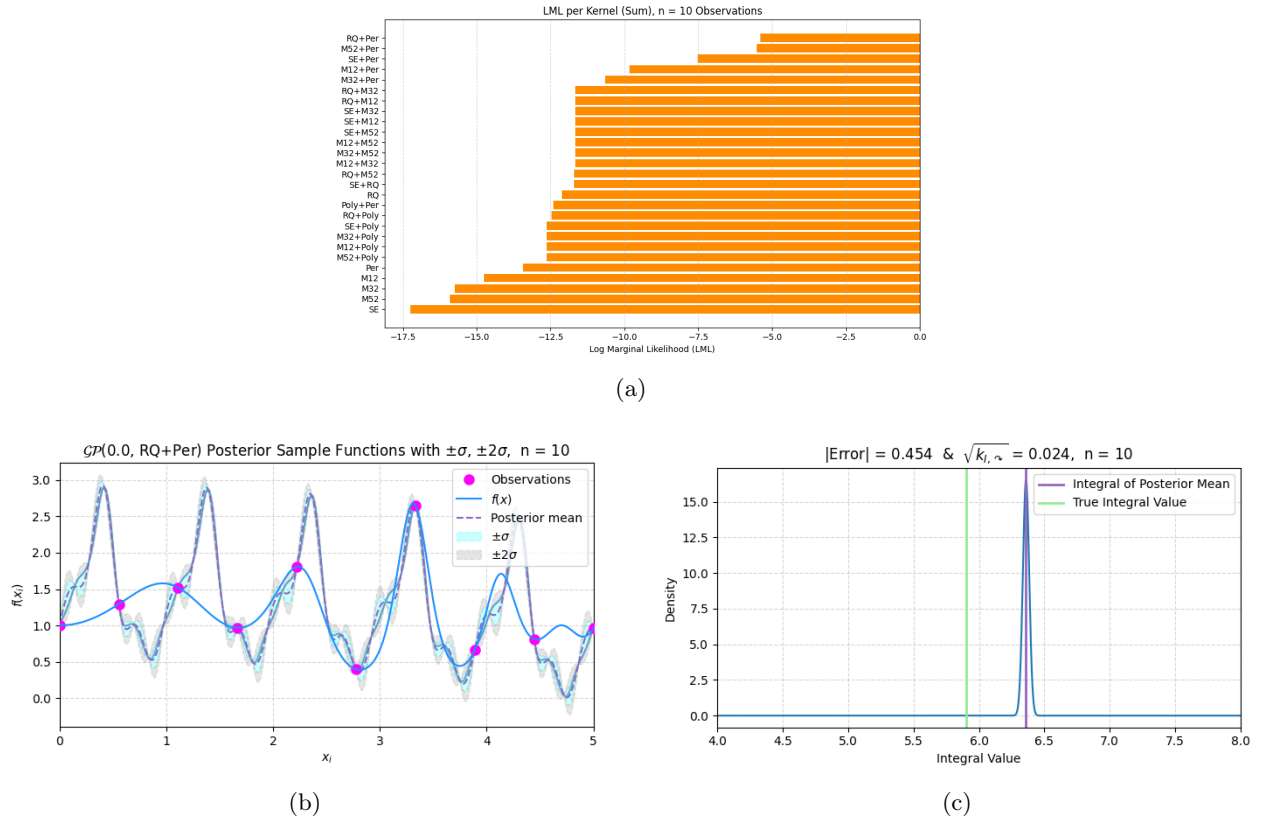


Figure 26: Bar-plot log (marginal) likelihood of the kernels and the corresponding posterior samples and posterior integral distribution, for the one-dimensional with $n = 10$ observations.

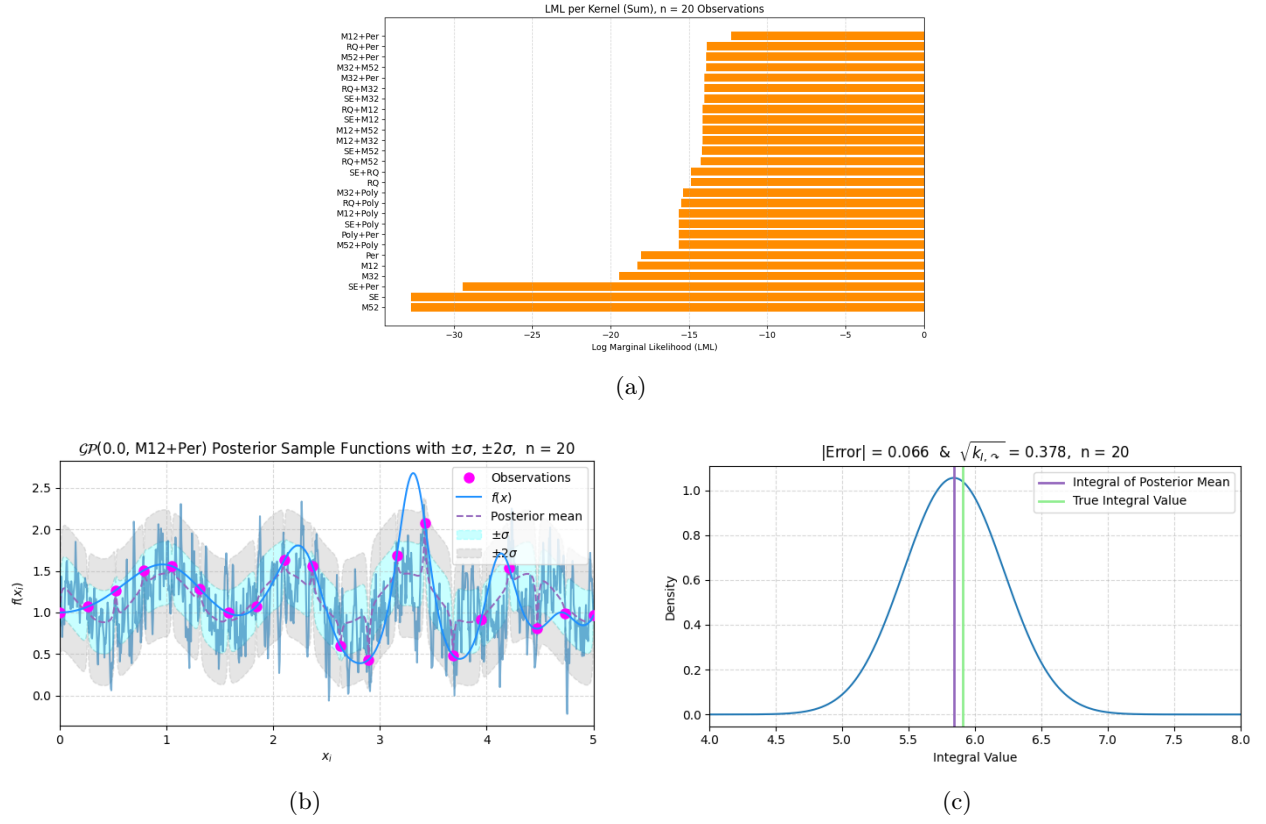


Figure 27: Bar-plot log (marginal) likelihood of the kernels and the corresponding posterior samples and posterior integral distribution, for the one-dimensional with $n = 20$ observations.

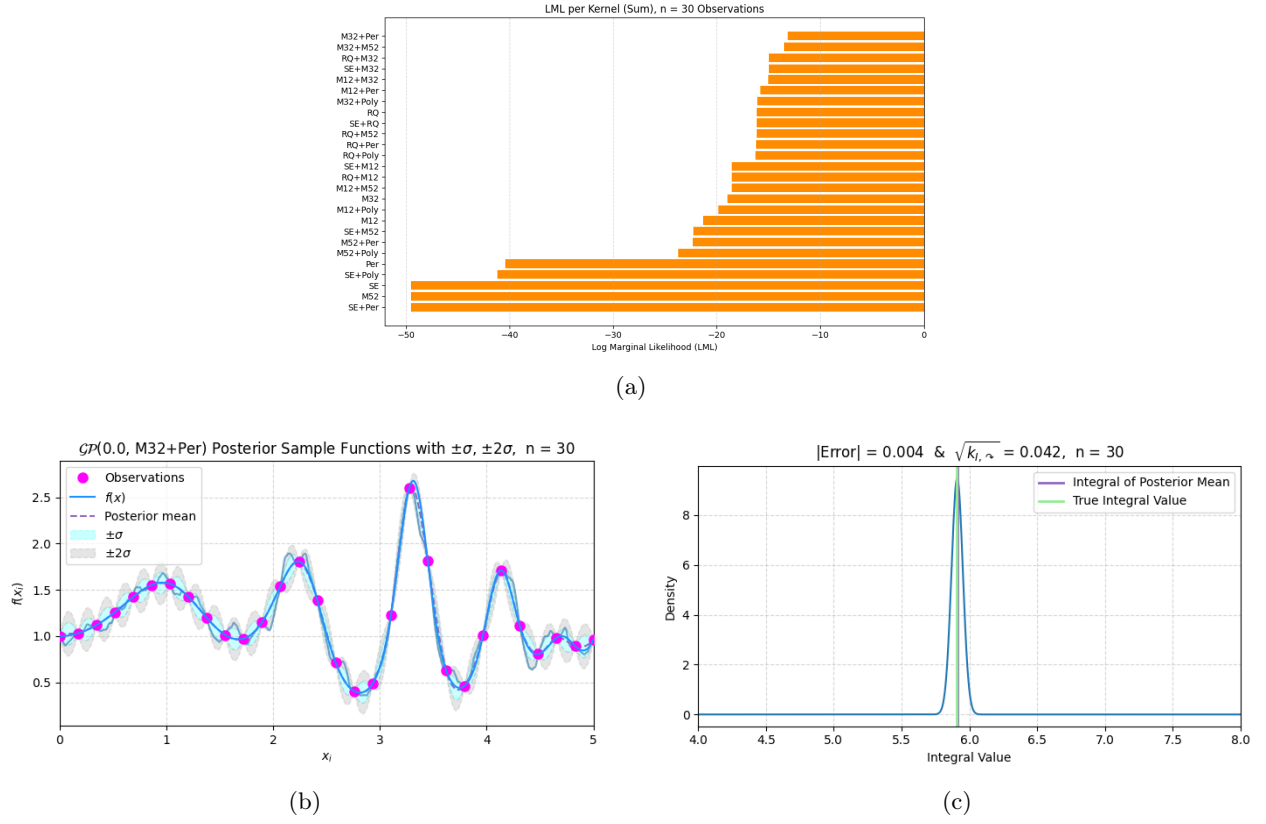


Figure 28: Bar-plot log (marginal) likelihood of the kernels and the corresponding posterior samples and posterior integral distribution, for the one-dimensional with $n = 30$ observations.

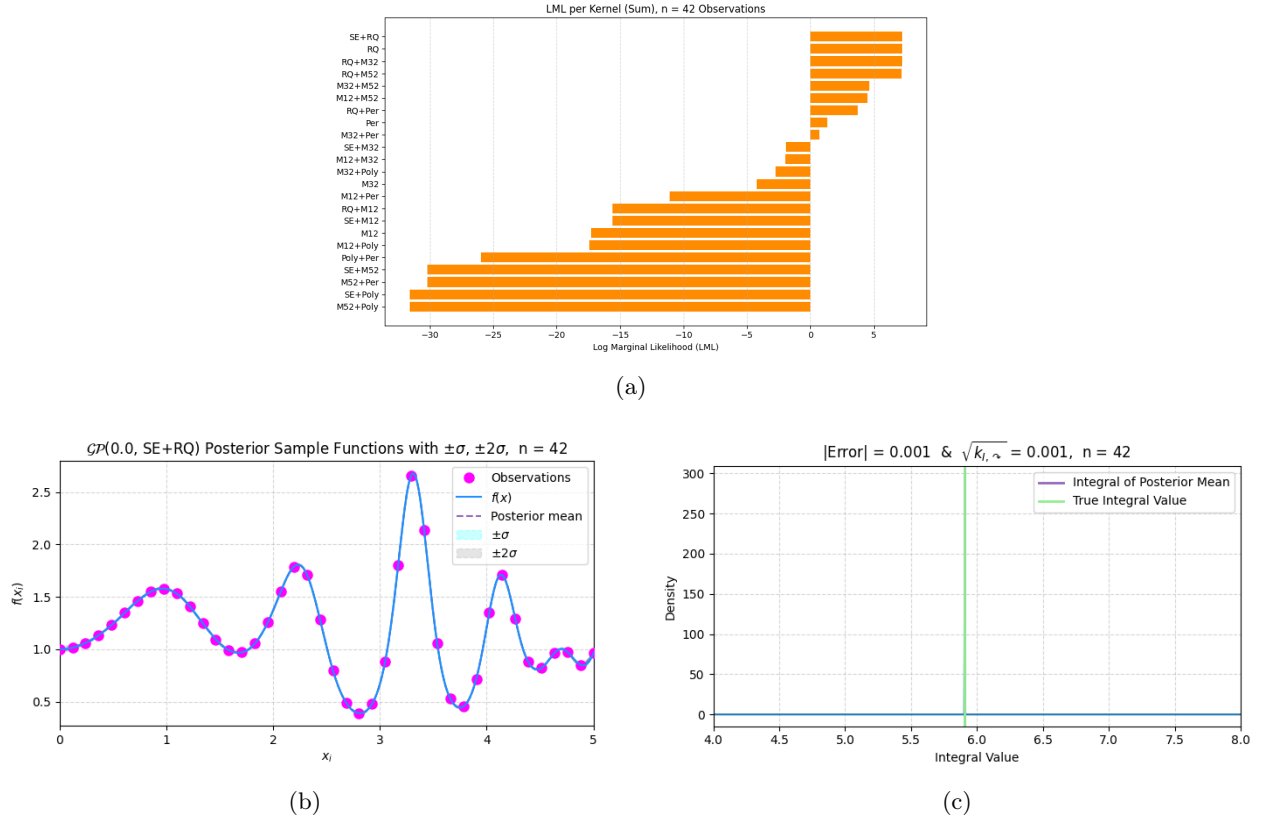
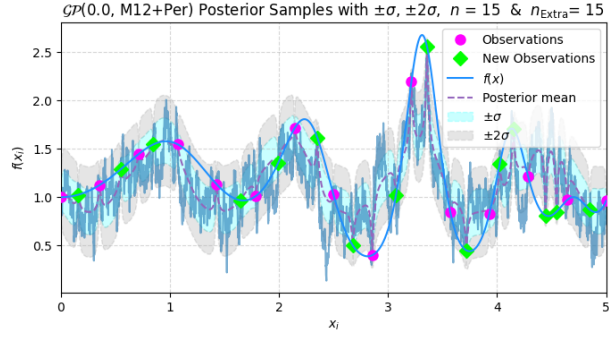
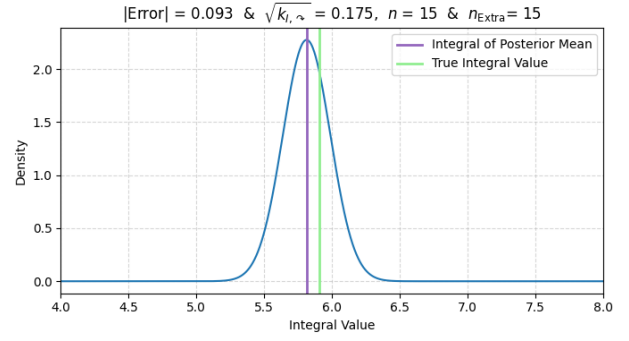


Figure 29: Bar-plot log (marginal) likelihood of the kernels and the corresponding posterior samples and posterior integral distribution, for the one-dimensional with $n = 42$ observations.

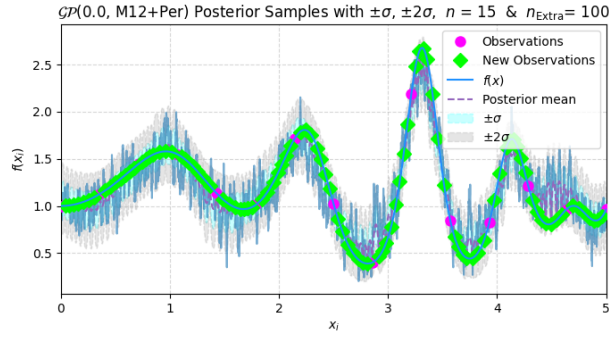
6.1.5 Subsection 3.6: Adaptive Sampling Scheme



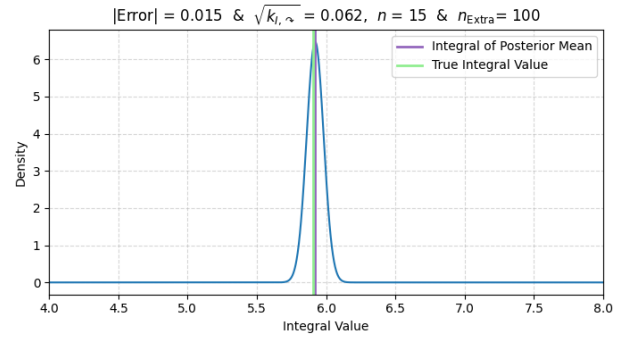
(a)



(b)



(c)



(d)

Figure 30: The uncertainty-sampling procedure for the one-dimensional integral with $n = 15$ observations.

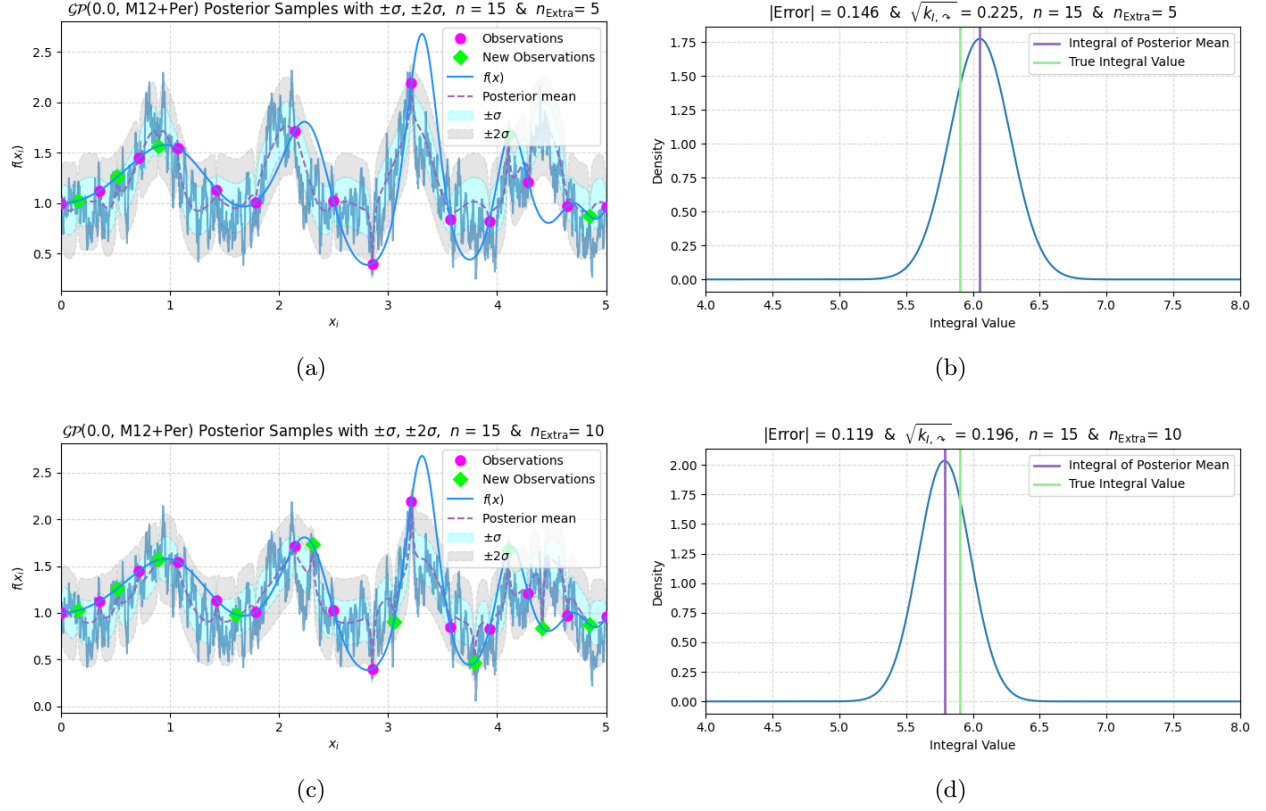


Figure 31: The adaptive scheme that minimises the posterior variance of the one-dimensional integral with $n = 15$ observations.

6.1.6 Subsection 4.1: Simulation Results – One-Dimensional Integral

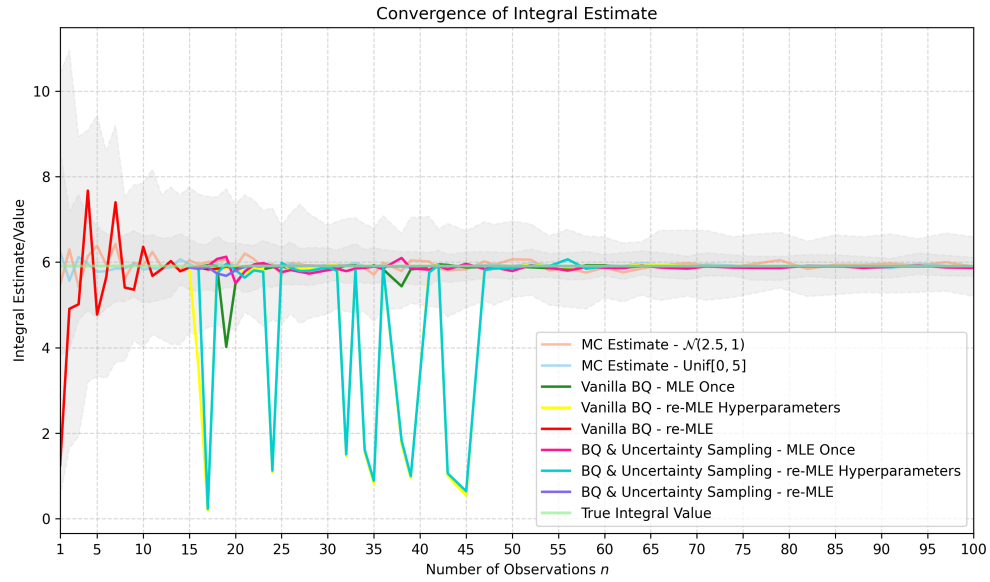


Figure 32: Comparison of the estimate of six Bayesian quadrature methods and two Monte Carlo methods for the one-dimensional integral example.

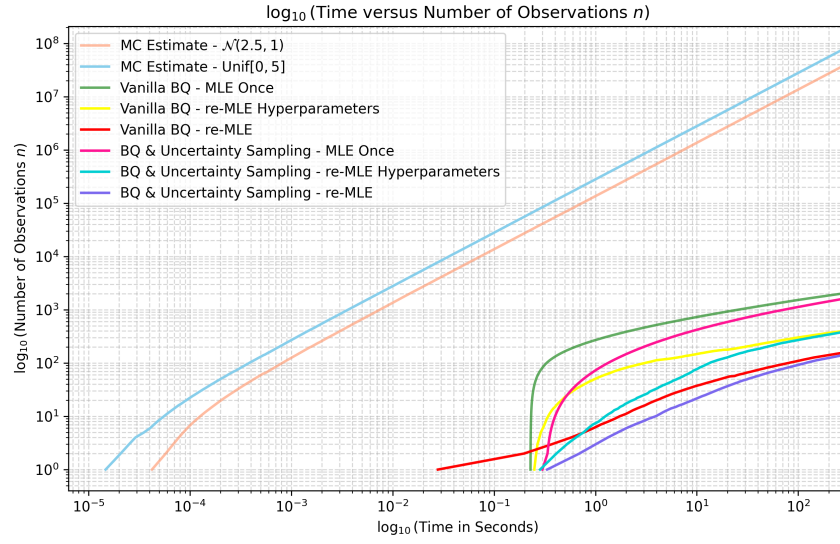


Figure 33: Comparison of the number of iterations over times of six Bayesian quadrature methods and two Monte Carlo methods for the one-dimensional integral example. Each iteration, the sample size is increased by one.

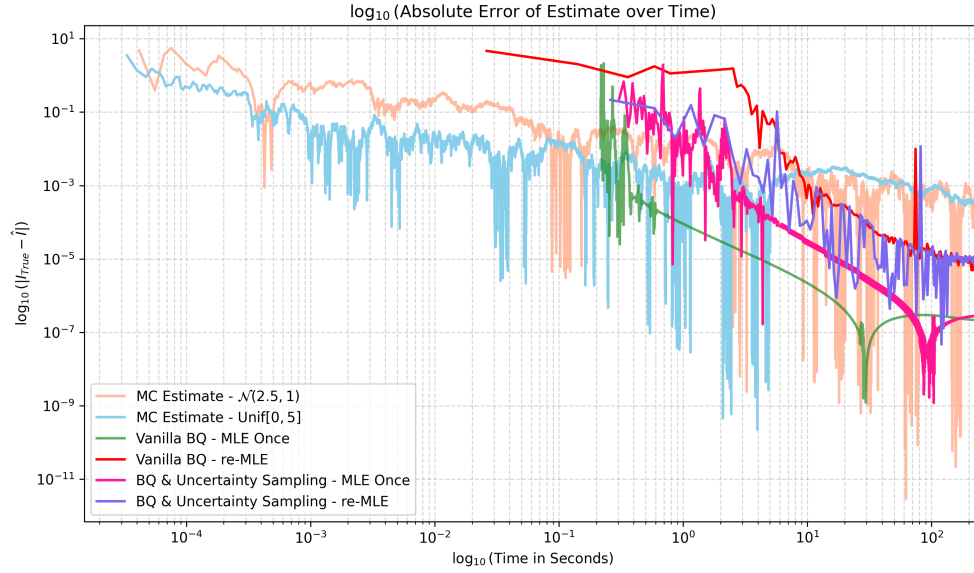


Figure 34: Comparison of the accuracy over times of four Bayesian quadrature methods and two Monte Carlo methods for the one-dimensional integral example.

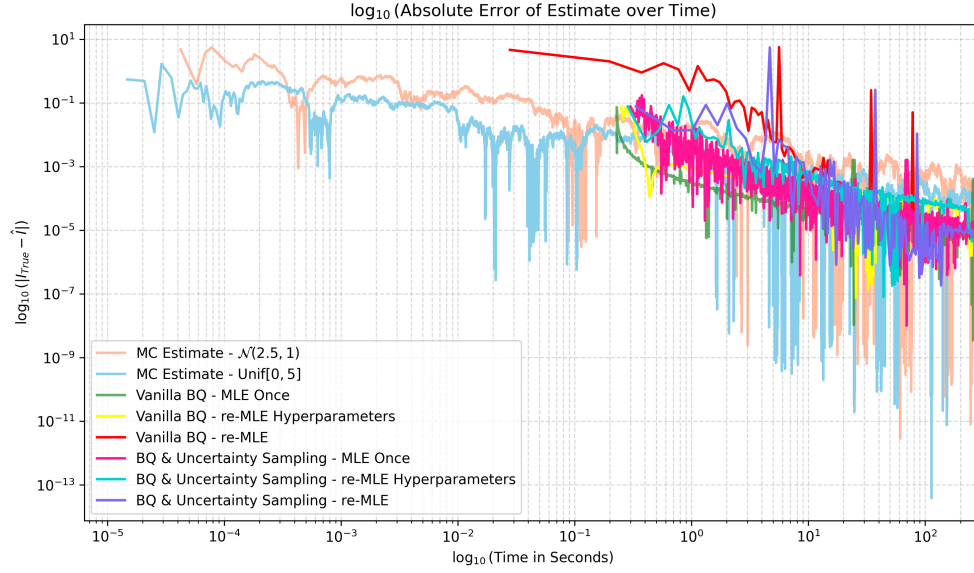


Figure 35: Comparison of the accuracy over times of six Bayesian quadrature methods and two Monte Carlo methods for the one-dimensional integral example.

6.1.7 Subsection 4.2: Simulation Results – Marginal Likelihood Example $d = 2$

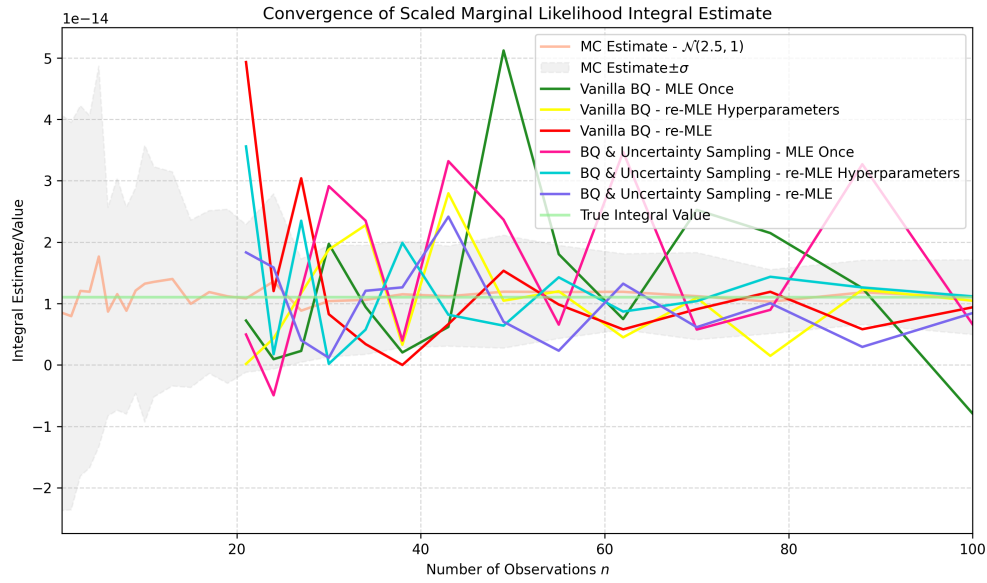


Figure 36: Comparison of the estimates of six BQ methods and one MC method for the two-dimensional marginal likelihood example.

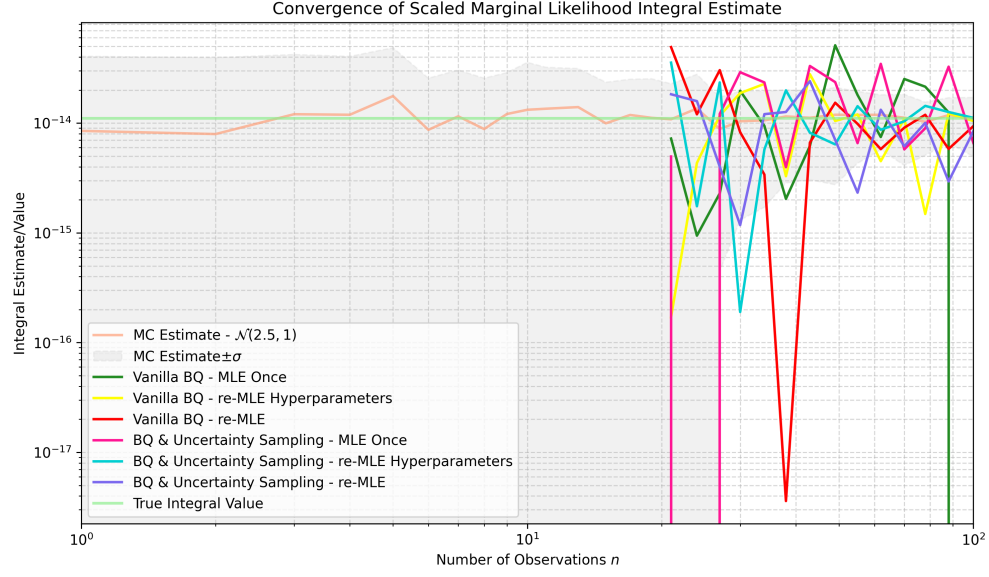


Figure 37: Comparison of the estimates of six BQ methods and one MC method for the two-dimensional marginal likelihood example on a $\log_{10}$ -scale.

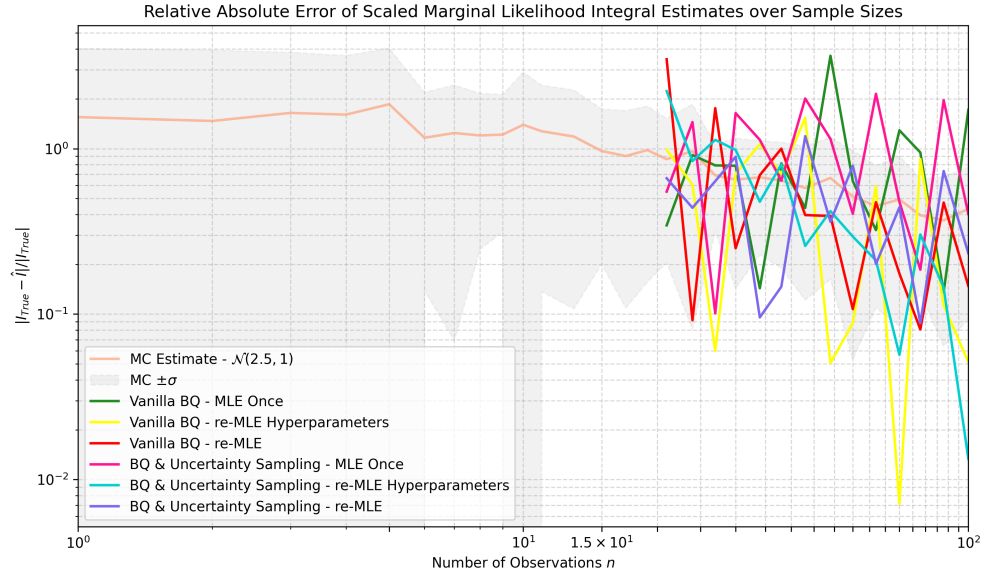


Figure 38: Comparison of the relative accuracy of six BQ methods and one MC method for the two-dimensional marginal likelihood example.

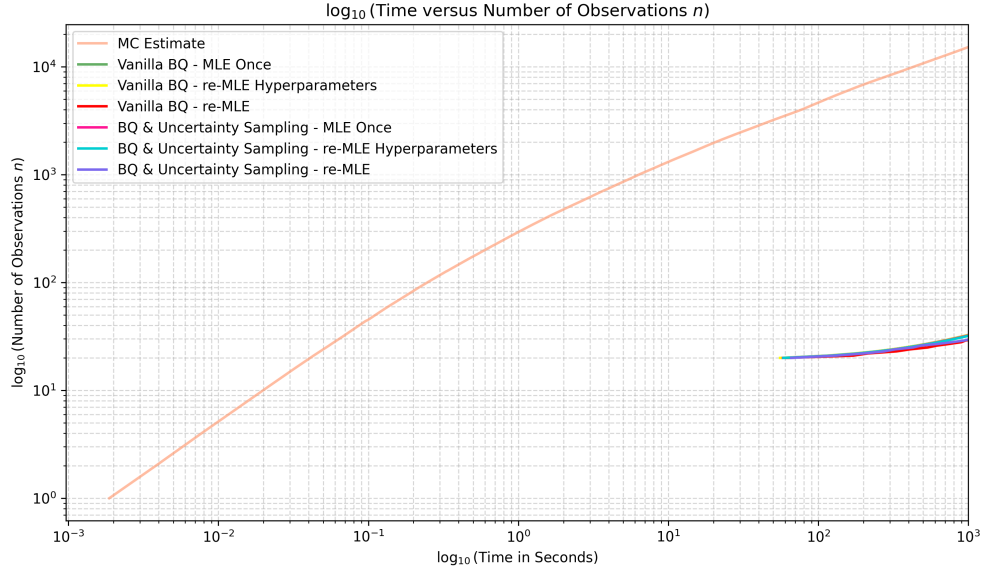


Figure 39: Comparison of the number of iterations over time of six BQ methods and one MC method for the two-dimensional marginal likelihood example. After each iteration, the sample size is increased by one.



Figure 40: Comparison of the accuracy over time of six BQ methods and one MC method for the two-dimensional marginal likelihood example.

6.1.8 Subsection 4.2: Simulation Results – Marginal Likelihood Example $d = 4$

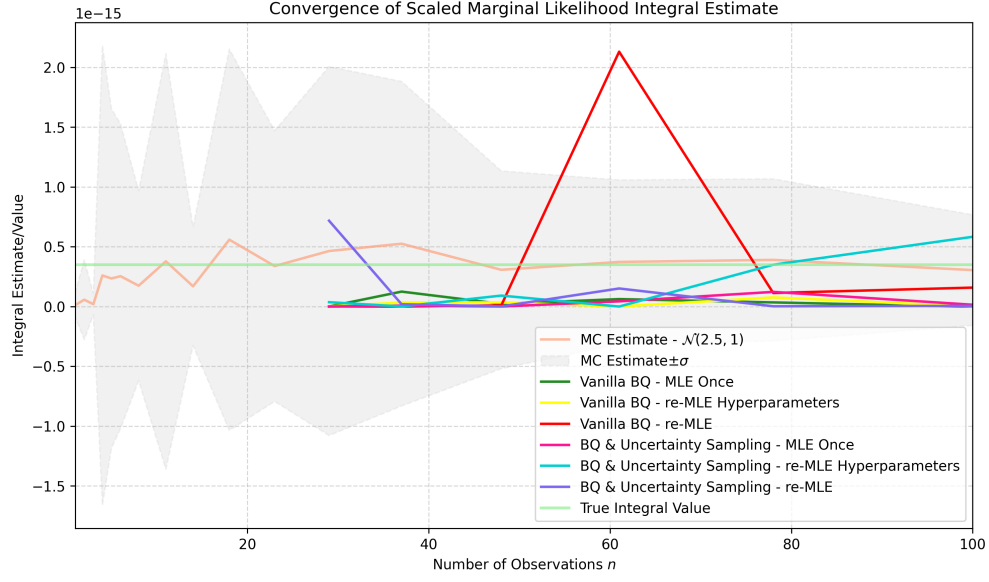


Figure 41: Comparison of the estimates of six BQ methods and one MC method for the four-dimensional marginal likelihood example.

6.1.9 Subsection 4.3: Simulation Results – Gaussian Mixture Model

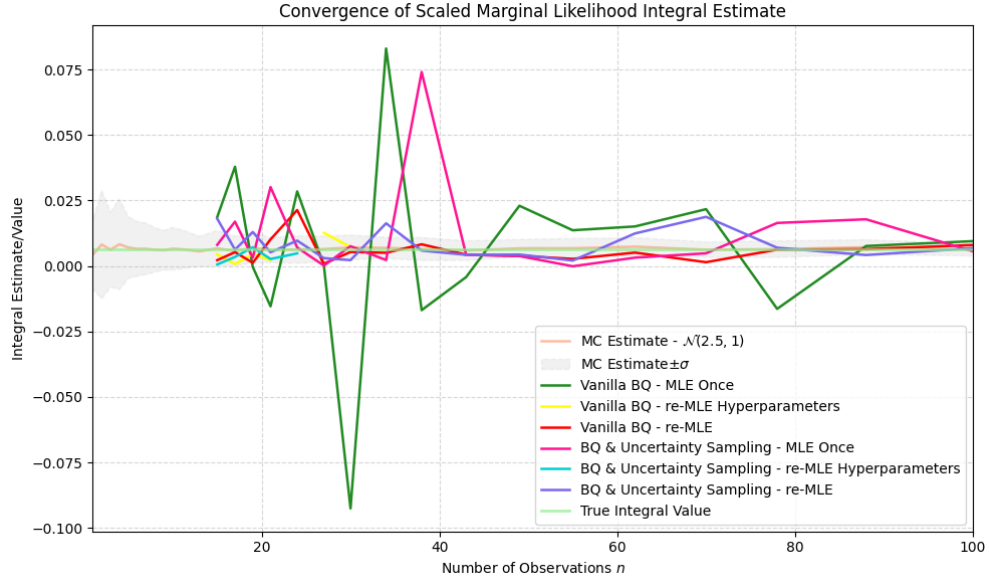


Figure 42: Comparison of the estimates of six BQ methods and one MC method for the two-dimensional marginal likelihood example with synthetic trimodal likelihood.

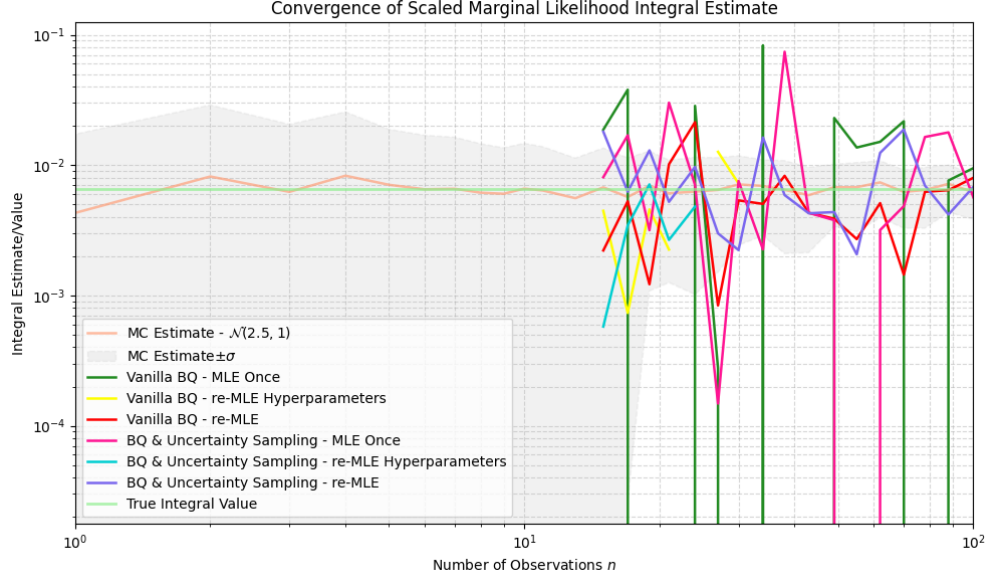


Figure 43: Comparison of the estimates on a $\log_{10}$ of six BQ methods and one MC method for the two-dimensional marginal likelihood example with synthetic trimodal likelihood.

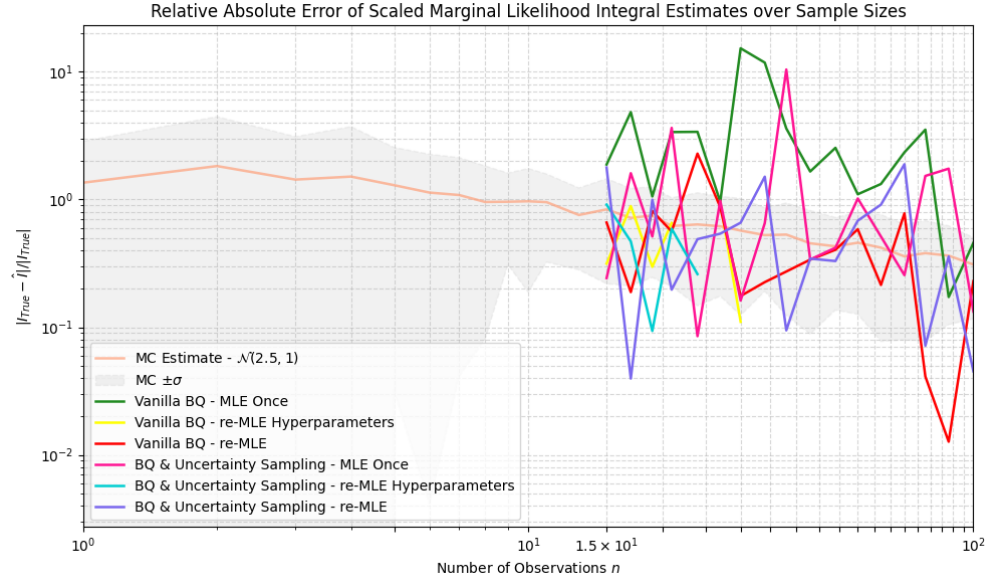


Figure 44: Comparison of the relative absolute error of six BQ methods and one MC method for the two-dimensional marginal likelihood example with synthetic trimodal likelihood.

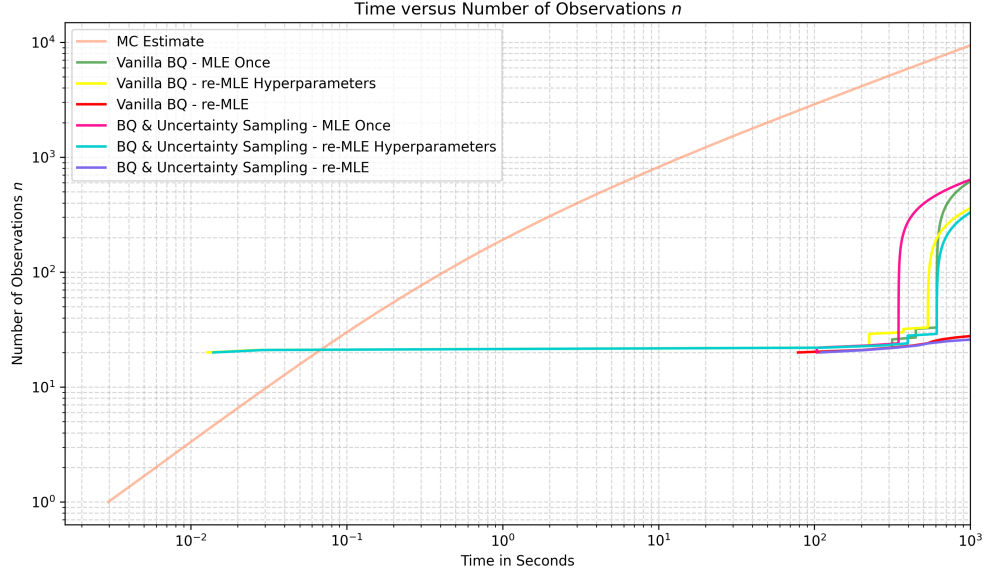


Figure 45: Comparison of the number of observations over time of six BQ methods and one MC method for the two-dimensional marginal likelihood example with synthetic trimodal likelihood.

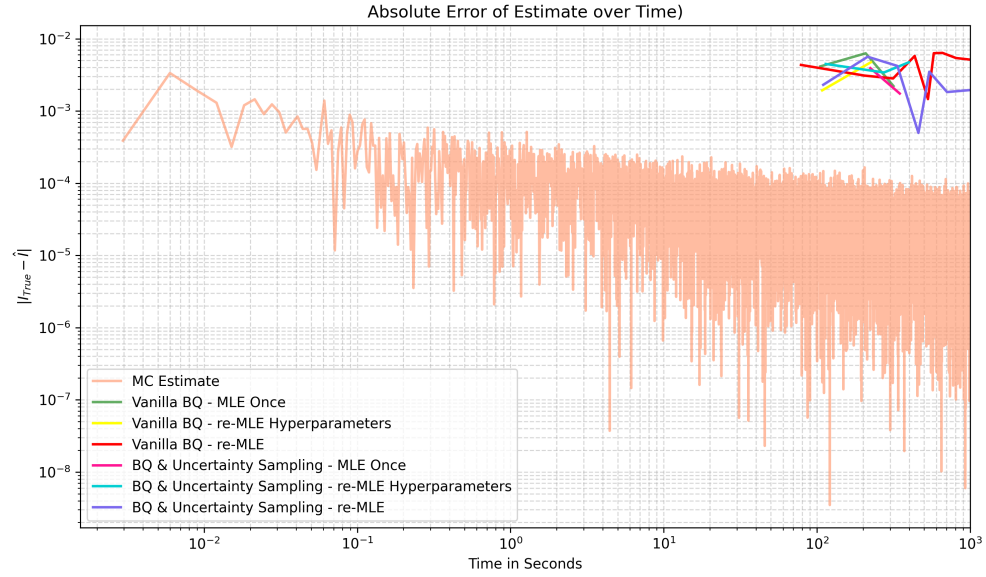


Figure 46: Comparison of the absolute error over time of six BQ methods and one MC method for the two-dimensional marginal likelihood example with synthetic trimodal likelihood.

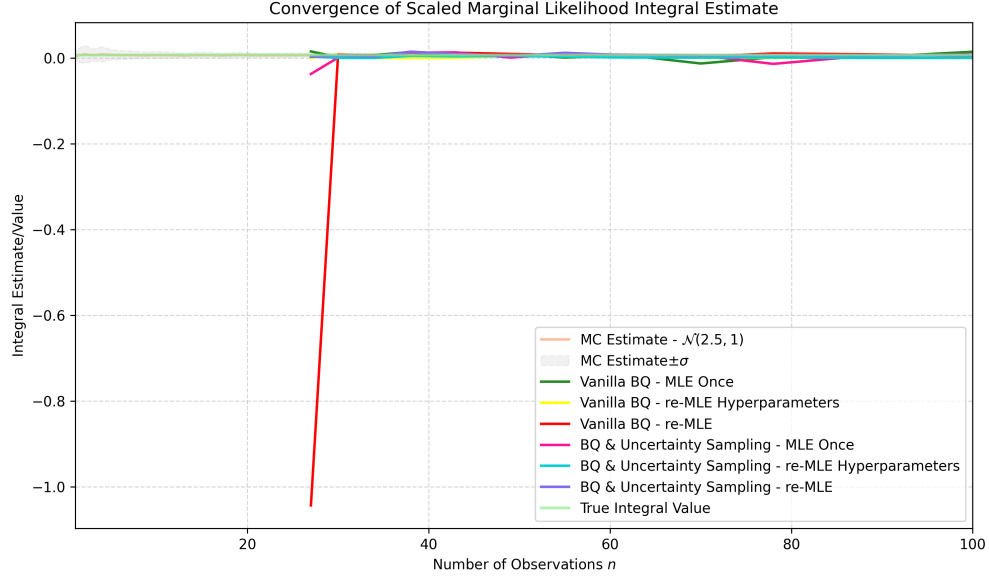


Figure 47: Comparison of the estimates of six BQ methods and one MC method for the two-dimensional marginal likelihood example with synthetic trimodal likelihood, with grid design.

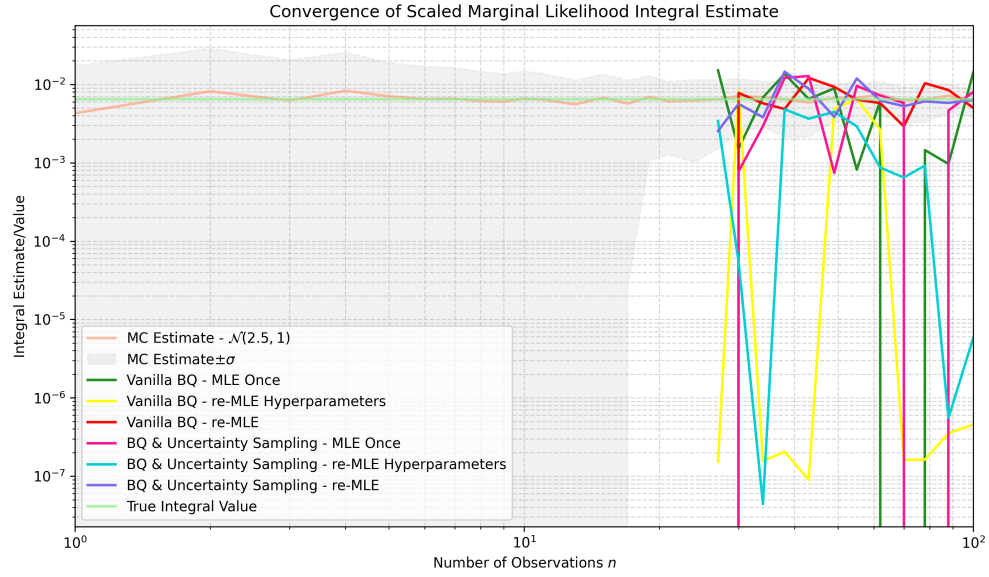


Figure 48: Comparison of the estimates on a $\log_{10}$ -scale of six BQ methods and one MC method for the two-dimensional marginal likelihood example with synthetic trimodal likelihood, with grid design.

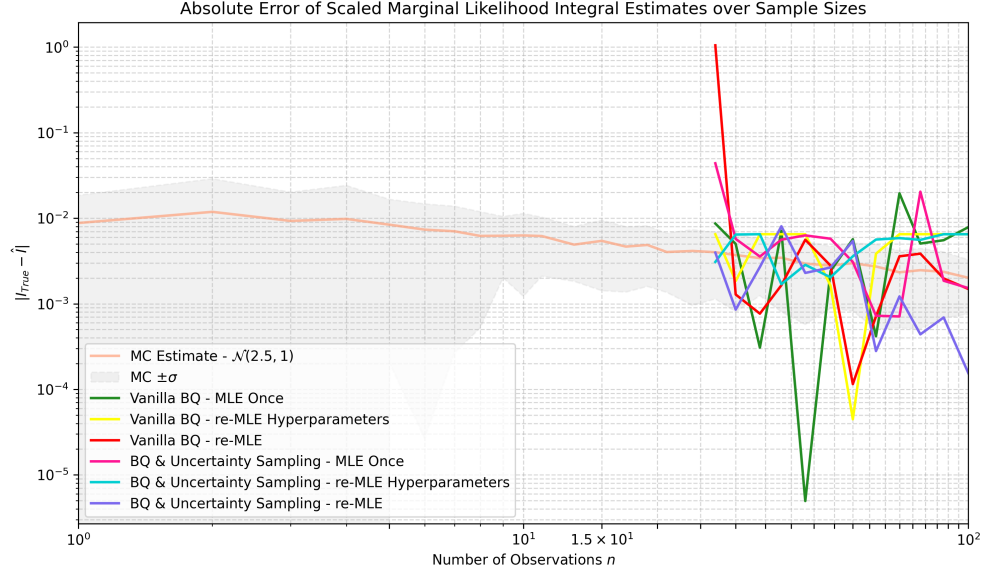


Figure 49: Comparison of the accuracy of six BQ methods and one MC method for the two-dimensional marginal likelihood example with synthetic trimodal likelihood, with grid design.

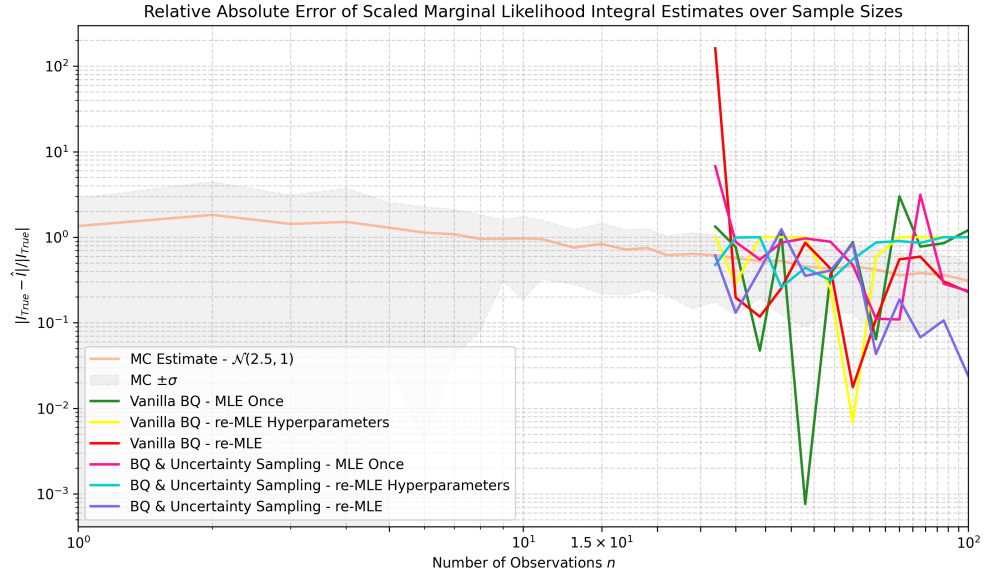


Figure 50: Comparison of the relative accuracy of six BQ methods and one MC method for the two-dimensional marginal likelihood example with synthetic trimodal likelihood, with grid design.

6.2 Tables

6.2.1 Subsection 4.1: Simulation Results – One-Dimensional Integral

Table 1: One-dimensional integral example. Hyperparameters estimated on $n = 15$ observed data points. All values are rounded to three decimal places.

Kernel	LML	MLE hyperparameters
M12+Per	-7.096	[0.021, 0.216, 1.176, 0.829, 1.290]
M52+Per	-7.096	[0.040, 0.216, 0.274, 0.829, 1.290]
RQ+Per	-10.710	[100.000, 0.010, 1.151, 0.010, 0.010, 0.184]
M32+Per	-10.833	[0.010, 0.426, 0.714, 100.000, 1.334]
SE+RQ	-10.838	[0.010, 0.426, 100.000, 0.010, 1.149]
RQ+M52	-10.838	[100.000, 0.010, 1.149, 0.010, 0.426]
RQ+M32	-10.838	[100.000, 0.231, 1.148, 0.010, 0.426]
RQ+M12	-10.838	[100.000, 0.442, 1.148, 0.019, 0.426]
M12+M52	-10.842	[0.019, 0.426, 100.000, 1.144]
M32+M52	-10.842	[0.032, 0.426, 100.000, 1.144]
M12+M32	-10.849	[0.021, 0.426, 100.000, 1.135]
RQ	-11.835	[0.010, 0.011, 1.245]
M32+Poly	-13.247	[0.016, 0.438, 9.947, 0.010]
M52+Poly	-13.247	[0.010, 0.438, 9.947, 0.010]
M12+Poly	-13.247	[0.010, 0.438, 9.947, 0.010]
RQ+Poly	-13.346	[0.010, 0.010, 1.331, 0.010, 0.010]
M12	-15.784	[1.858, 1.181]
SE+Per	-16.991	[0.010, 0.010, 0.010, 0.010, 1.787]
M32	-18.050	[0.544, 1.117]
M52	-24.409	[0.010, 1.232]
SE	-24.409	[0.010, 1.232]
Per	-29.957	[0.075, 0.075, 13.400]
Poly+Per	-2440514.294	[1.000, 1.000, 1.000, 1.000, 1.000]
SE+Poly	-20831688.610	[1.000, 1.000, 1.000, 1.000]
Poly	-119109462.854	[1.000, 1.000]

6.2.2 Subsection 4.2: Simulation Results – Marginal Likelihood Example $d = 2$

Table 2: Marginal likelihood ($d = 2$) example without grid design. Hyperparameters used by fixed-grid BQ with re-estimated hyperparameters. All values are rounded to three decimal places.

n	l_{M12}	σ_{M12}^2	λ_π	l_π	σ_π^2
21	0.003	0.216	0.095	0.016	0.001
24	0.925	0.293	1.792	0.544	0.007
27	0.166	0.240	0.668	1.170	0.002
30	0.386	0.234	2.949	998.528	0.001
34	0.299	0.199	1.027	1.634	0.001
38	3.459	0.028	1.533	0.250	0.042
43	0.721	0.022	1.003	0.309	0.038
49	0.282	0.185	0.845	2.658	0.003
55	0.134	0.204	0.663	0.361	0.002
62	0.026	0.117	507.510	0.003	0.007
70	228.383	0.001	663.917	0.001	0.020
78	0.102	0.149	0.676	1.749	0.001
88	0.126	0.174	0.061	12.370	0.001
100	0.114	0.113	0.001	0.001	0.001

Table 3: Marginal likelihood ($d = 2$) example without grid design. Hyperparameters used by fixed-grid BQ with re-estimated kernel and hyperparameters. All values are rounded to three decimal places.

n	l_{M12}	σ_{M12}^2	l_{SE}	σ_{SE}^2	l_{RQ}	α_{RQ}	σ_{RQ}^2	λ_π	l_π	σ_π^2
21								1.030	0.506	0.089
24	0.264	0.252						0.742	0.001	0.007
27			645.588	0.001				0.950	0.743	0.119
30					0.576	999.122	0.229			
34			1000.000	0.004				3.285	0.307	0.087
38	0.298	0.214						1.284	0.322	0.002
43	0.035	0.126	0.332	0.111						
49	0.145	0.172								
55			0.350	0.190				0.404	2.442	0.001
62			0.345	0.203				0.567	0.744	0.001
70	0.183	0.141								
78	0.134	0.161						0.448	0.476	0.001
88					0.321	577.931	0.079	1.106	0.187	0.008
100	0.001	0.026						3.682	0.206	0.017

Table 4: Marginal likelihood ($d = 2$) example without grid design. Hyperparameters used by BQ with uncertainty sampling and re-estimated hyperparameters. All values are rounded to three decimal places.

n	l_{M12}	σ_{M12}^2	λ_π	l_π	σ_π^2
21	0.262	0.001	1.637	0.480	0.071
24	0.286	0.001	0.812	0.223	0.047
27	411.006	0.001	0.829	0.408	0.056
30	0.210	0.197	0.900	1.048	0.007
34	0.154	0.236	0.188	1000.000	0.001
38	0.282	0.225	0.985	86.676	0.001
43	0.238	0.200	1.302	1.473	0.001
49	0.456	0.259	1.223	8.447	0.001
55	0.110	0.151	0.905	1.011	0.001
62	0.134	0.199	0.321	1000.000	0.001
70	0.148	0.159	1.855	0.489	0.001
78	0.056	0.004	2.134	0.243	0.022
88	0.107	0.158	0.224	944.231	0.001
100	0.154	0.134	984.757	999.921	0.001

Table 5: Marginal likelihood ($d = 2$) example without grid design. Hyperparameters used by BQ with uncertainty sampling and re-estimated kernel and hyperparameters. All values are rounded to three decimal places.

n	l_{M12}	σ_{M12}^2	l_{SE}	σ_{SE}^2	l_{RQ}	α_{RQ}	σ_{RQ}^2	c_{Poly}	σ_{Poly}^2	λ_π	l_π	σ_π^2
21			0.500	0.274								
24			5.066	0.001						1.372	0.616	0.092
27	1.479	0.118								0.971	0.452	0.054
30			0.506	0.055						0.621	0.448	0.080
34	0.174	0.206								0.499	1.672	0.007
38								0.115	0.001	1.733	0.558	0.108
43	0.274	0.053								0.050	0.308	0.032
49	0.339	0.188										
55	0.001	0.022								2.956	0.341	0.063
62	0.130	0.142										
70	1000.000	0.001								1.100	0.280	0.024
78	0.110	0.155										
88			0.001	0.009						2.393	0.240	0.034
100					0.257	12.427	0.158					

6.2.3 Subsection 4.3: Simulation Results – Gaussian Mixture Model

Table 6: Gaussian mixture model example without grid design. Hyperparameters used by fixed-grid BQ with re-estimated hyperparameters. All values are rounded to three decimal places.

n	l_{M32}	σ_{M32}^2	λ_π	l_π	σ_π^2
15	1.417	0.225	0.693	2.375	0.531
17	1.157	0.310	0.702	4.717	0.289
19	1.299	0.212	1.209	3.689	0.444
21	0.868	0.245	0.038	46.746	0.020
24	1.000	1.000	1.000	1.000	1.000
27	1.842	0.239	0.321	4.106	0.410
30	1.000	1.000	1.000	1.000	1.000
34	1.000	1.000	1.000	1.000	1.000
38	1.000	1.000	1.000	1.000	1.000
43	1.000	1.000	1.000	1.000	1.000
49	1.000	1.000	1.000	1.000	1.000
55	1.000	1.000	1.000	1.000	1.000
62	1.000	1.000	1.000	1.000	1.000
70	1.000	1.000	1.000	1.000	1.000
78	1.000	1.000	1.000	1.000	1.000
88	1.000	1.000	1.000	1.000	1.000
100	1.000	1.000	1.000	1.000	1.000

Table 7: Gaussian mixture model example without grid design. Hyperparameters used by fixed-grid BQ with re-estimated kernel and hyperparameters. All values are rounded to three decimal places.

n	l_{M12}	σ_{M12}^2	l_{SE}	σ_{SE}^2	l_{RQ}	α_{RQ}	σ_{RQ}^2	λ_π	l_π	σ_π^2
15	1.102	0.001						1.445	0.773	0.161
17								10.121	0.519	0.096
19								7.098	0.539	0.112
21			2.166	0.298				3.900	0.914	0.009
24			1.856	0.036				1.077	0.702	0.112
27			1.991	0.270						
30			1.532	0.243	0.002	0.468	0.002			
34	2.449	0.234						0.885	1.808	0.005
38	0.853	0.137	1.914	0.161						
43			0.804	0.180				0.840	0.273	0.008
49			1.153	0.189						
55								6.235	0.171	0.041
62					0.885	1000.000	0.115	3.397	0.244	0.018
70	0.378	0.136	213.615	0.025						
78	1.044	0.196						1.910	2.004	0.005
88	1.250	0.175	9.813	0.044						
100			0.981	0.241						

Table 8: Gaussian mixture model example without grid design. Hyperparameters used by BQ with uncertainty sampling and re-estimated hyperparameters. All values are rounded to three decimal places.

n	l_{M32}	σ_{M32}^2	λ_π	l_π	σ_π^2
15	1.549	0.201	0.286	1.734	0.628
17	0.069	0.241	0.806	884.137	0.001
19	1.615	0.211	0.235	2.678	0.468
21	2.141	0.251	0.271	2.585	0.450
24	1.568	0.346	0.477	4.400	0.358
27	1.000	1.000	1.000	1.000	1.000
30	1.127	0.802	0.904	1.112	0.929
34	1.000	1.000	1.000	1.000	1.000
38	1.005	0.994	1.006	1.006	0.996
43	1.000	1.000	1.000	1.000	1.000
49	1.000	1.000	1.000	1.000	1.000
55	1.000	1.000	1.000	1.000	1.000
62	1.000	1.000	1.000	1.000	1.000
70	1.000	1.000	1.000	1.000	1.000
78	1.000	1.000	1.000	1.000	1.000
88	1.000	1.000	1.000	1.000	1.000
100	1.000	1.000	1.000	1.000	1.000

Table 9: Gaussian mixture model example without grid design. Hyperparameters used by BQ with uncertainty sampling and re-estimated kernel and hyperparameters. All values are rounded to three decimal places.

n	l_{M12}	σ_{M12}^2	l_{M32}	σ_{M32}^2	l_{SE}	σ_{SE}^2	l_{RQ}	α_{RQ}	σ_{RQ}^2	λ_π	l_π	σ_π^2
15										0.994	0.701	0.231
17	14.361	0.455								1.281	5.404	0.002
19	1.388	0.127	0.887	0.199								
21	1000.000	0.001								1.046	0.494	0.075
24					1.541	0.273						
27					0.950	0.208						
30			0.014	0.001						1.155	0.537	0.115
34	0.360	0.019			2.166	0.385						
38					1.176	0.287						
43	2.295	0.180										
49										2.118	0.267	0.027
55					1.030	0.181						
62	3.603	0.374										
70	2.692	0.284								1.377	11.459	0.005
78	0.917	0.178								3.113	0.774	0.009
88							0.646	1.697	0.163	0.654	2.120	0.002
100							0.742	4.512	0.144	1.653	1.436	0.002

6.3 Probability & Measure Theory

This subsection provides a brief overview of probability and measure theory.

6.3.1 Probability Foundations

Let $(\Omega, \mathcal{E}, \mathbb{P})$ be a probability space. The sample space Ω denotes the set of all possible outcomes and the σ -algebra $\mathcal{E}$ is the collection of all events, i.e., a collection of measurable subsets of Ω . The probability measure is a map $\mathbb{P} : \mathcal{E} \rightarrow [0, 1]$ such that $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$, and $\mathbb{P}$ satisfies the countable additivity property $\mathbb{P}(\bigcup_{i \in \mathbb{N}} E_i) = \sum_{i \in \mathbb{N}} \mathbb{P}(E_i)$ for any countable collection of pairwise disjoint sets $E_i \in \mathcal{E}$, $i \in \mathbb{N}$. $\mathbb{P}$ assigns probabilities to events $E \in \mathcal{E}$ (Rosenthal, 2009).

6.3.1.1 Distribution

A real-valued random variable is a measurable map $X : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B})$, where $\mathcal{B} := \mathcal{B}(\mathbb{R})$ denotes the Borel σ -algebra. X is a measurable mapping because $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{E}$, for all $B \in \mathcal{B}$. The distribution of a random variable X is the push-forward measure $\mathbb{P}_X : (\mathbb{R}, \mathcal{B}) \rightarrow [0, 1]$ defined by $\mathbb{P}_X := \mathbb{P} \circ X^{-1}$. Hence, for any $B \in \mathcal{B}$:

$$\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in B\}) =: \mathbb{P}(X \in B). \quad (89)$$

Observe that $\mathbb{P}$ and $\mathbb{P}_X$ are measures on different spaces, because $\mathbb{P}$ is a measure on $(\Omega, \mathcal{E})$, while $\mathbb{P}_X$ is the push-forward measure of $\mathbb{P}$ that lives on $(\mathbb{R}, \mathcal{B})$. The cumulative distribution function (CDF) $F_X : \sigma(\mathcal{W}) \rightarrow [0, 1]$ of a random variable X is defined as $\mathbb{P}_X$ on the generating set $\mathcal{W} := \{(-\infty, x] : x \in \mathbb{R}\}$, i.e., $\mathcal{B} = \sigma(\mathcal{W})$:

$$F_X(x) := \mathbb{P}(X \leq x) = \mathbb{P}(X^{-1}((-\infty, x])) = \mathbb{P}_X((-\infty, x]). \quad (90)$$

6.3.1.2 Expectation

Moreover, let $g : (\mathbb{R}, \mathcal{B}) \rightarrow (\bar{\mathbb{R}}, \bar{\mathcal{B}})$ be any measurable map, where $\bar{\mathbb{R}} := [-\infty, \infty]$ denotes the extended real-line and $\bar{\mathcal{B}} := \mathcal{B}(\bar{\mathbb{R}})$ the Borel σ -algebra on $\bar{\mathbb{R}}$. Since the composition of measurable mappings is also a measurable mapping, the composition $g \circ X = g(X) : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow (\bar{\mathbb{R}}, \bar{\mathcal{B}})$ is also a random variable. By definition, in a measure space $(\bar{\Omega}, \bar{\mathcal{A}}, \nu)$, a measurable function $l : (\bar{\Omega}, \bar{\mathcal{A}}) \rightarrow (\bar{\mathbb{R}}, \bar{\mathcal{B}})$ is ν -integrable, if $\int_{\bar{\Omega}} |l| d\nu < \infty$. Hence, if $g \circ X$ is integrable, i.e., if $\int_{\Omega} |g(X)| d\mathbb{P} < \infty$, the expectation of the random variable $g(X)$ exists and is defined as follows (De Snoo and Winkler, 2018):

$$\mathbb{E}[g(X)] := \int_{\mathbb{R}} g(X(\omega)) d\mathbb{P}(\omega) = \int_{\mathbb{R}} g(X) d\mathbb{P}. \quad (91)$$

In addition, De Snoo and Winkler (2018) show that g is $\mathbb{P}_X$ -integrable if and only if $g \circ X$ is $\mathbb{P}$ -integrable, in which case the following holds:

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) d\mathbb{P}_X(x) = \int_{\mathbb{R}} g d\mathbb{P}_X. \quad (92)$$

6.3.1.3 Discrete Random Variables

A random variable X is discrete, if the distribution $\mathbb{P}_X$ is discrete, i.e. if it assigns all of its probability mass to countably many points. Therefore, X is discrete if it only attains countably many values. That is, there exists a countable set

$$\Lambda := \{x_i \in \mathbb{R} : \mathbb{P}_X(\{x_i\}) > 0, i \in \mathcal{I} \subset \mathbb{N}\} \quad (93)$$

such that $\mathbb{P}_X(\mathbb{R} \setminus \Lambda) = 0$, which implies that $\mathbb{P}_X(\mathbb{R}) = \mathbb{P}_X(\Lambda) = \sum_{i \in \mathcal{I}} \mathbb{P}_X(\{x_i\}) = 1$. The function $p_X : \Lambda \rightarrow \mathbb{R}$ defined by:

$$p_X(x_i) := \mathbb{P}_X(\{x_i\}) = \mathbb{P}(X = x_i) \quad (94)$$

is the probability mass function (PMF) of X .

For example, suppose $\mathbb{P}_X(\{c\}) = \mathbb{P}(X = c) = 1$ for some $c \in \mathbb{R}$ (Rosenthal, 2009). Then, the distribution of X is the point mass $\delta_c : (\mathbb{R}, \mathcal{B}) \rightarrow [0, \infty]$ defined by:

$$\delta_c(B) := \mathbb{I}_B(c) = \mathbb{I}\{c \in B\} = \begin{cases} 1, & c \in B \\ 0, & c \notin B \end{cases}, \quad B \in \mathcal{B}. \quad (95)$$

That is, $\mathbb{P}_X = \delta_c$, i.e., $X \sim \delta_c$. For any function $h : \mathbb{R} \rightarrow \mathbb{R}$, the expectation of $h(X)$ is:

$$\mathbb{E}[h(X)] = \int_{\Omega} h(X(\omega)) \, d\mathbb{P}(\omega) = \int_{\mathbb{R}} h(t) \, d\delta_c(t) = h(c). \quad (96)$$

The Dirac-delta function δ_x is defined by $\int_{\mathbb{R}} \delta_x(t)h(t) \, dt = h(x)$.

More generally, for a measure space $(\Omega, \mathcal{A})$ with $\omega \in \Omega$, the Dirac measure $\delta_{\omega} : \mathcal{A} \rightarrow [0, \infty]$ is defined by:

$$\delta_{\omega}(A) := \mathbb{I}_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}, \quad A \in \mathcal{A}. \quad (97)$$

Generalizing the above example yields an expression for the distribution measure of X :

$$\mathbb{P}_X = \sum_{i \in \mathcal{I}} p_X(x_i) \cdot \delta_{x_i}. \quad (98)$$

Therefore, for any measurable function $g : (\mathbb{R}, \mathcal{B}) \rightarrow (\bar{\mathbb{R}}, \bar{\mathcal{B}})$, the expectation of $g(X)$ is:

$$\begin{aligned} \mathbb{E}[g(X)] &= \int_{\mathbb{R}} g(x) \, d\mathbb{P}_X(x) = \int_{\mathbb{R}} g(x) \, d\left(\sum_{i \in \mathcal{I}} p_X(x_i) \cdot \delta_{x_i}\right)(x) \\ &= \sum_{i \in \mathcal{I}} p_X(x_i) \cdot \int_{\mathbb{R}} g(x) \, d\delta_{x_i}(x) \\ &= \sum_{i \in \mathcal{I}} p_X(x_i) \cdot g(x_i). \end{aligned} \quad (99)$$

6.3.1.4 Absolutely Continuous Random Variables

A random variable X is absolutely continuous, if the distribution $\mathbb{P}_X$ is absolutely continuous with respect to the Lebesgue measure λ , i.e., when:

$$\lambda(B) = 0 \implies \mathbb{P}_X(B) = 0, \quad B \in \mathcal{B}. \quad (100)$$

The above relation holds by the Radon-Nikodym derivative (De Snoo and Winkler, 2018).

Theorem 6.1. (Radon-Nikodym derivative)

Let ν_1 and ν_2 be σ -finite measures on a measurable space $(\tilde{\Omega}, \mathcal{A})$ such that ν_1 is absolutely continuous with respect to ν_2 . Then, there exists a ν_2 -measurable function $h : (\tilde{\Omega}, \mathcal{A}) \rightarrow [0, \infty]$ such that:

$$\nu_1(A) = \int_A h \, d\nu_2, \quad A \in \mathcal{A}. \quad (101)$$

The function h is uniquely determined up to sets of ν_2 -measure 0.

Furthermore, the function h is called the Radon-Nikodym derivative and it is usually denoted by:

$$h = \frac{d\nu_1}{d\nu_2}. \quad (102)$$

And for any measurable $g : (\tilde{\Omega}, \mathcal{A}) \rightarrow [0, \infty]$:

$$\int_{\tilde{\Omega}} g \, d\nu_1 = \int_{\tilde{\Omega}} gh \, d\nu_2. \quad (103)$$

We can translate Theorem 6.1 to our probability setting since $\mathbb{P}_X$ and λ are σ -finite measures on $(\mathbb{R}, \mathcal{B})$. Therefore, if the distribution measure $\mathbb{P}_X$ is absolutely continuous with respect to the Lebesgue measure λ , then there exists a measurable non-negative function $p_X : (\mathbb{R}, \mathcal{B}) \rightarrow [0, \infty)$ such that:

$$\mathbb{P}_X(B) = \int_B p_X(x) \, dx, \quad B \in \mathcal{B}. \quad (104)$$

The function $p_X = \frac{d}{d\lambda} \mathbb{P}_X$ is the probability density function (PDF) of X . Observe that no single point carries a positive probability: $\mathbb{P}_X(\{x\}) = 0, \forall x \in \mathbb{R}$. Probabilities are determined by the area under the density curve p_X . In addition, observe that p_X is not uniquely determined: if we change p_X in a set of Lebesgue measure zero, then the above property still holds. In other words, changing p_X on Lebesgue null-sets does not alter probabilities.

For any measurable function $g : (\mathbb{R}, \mathcal{B}) \rightarrow (\bar{\mathbb{R}}, \bar{\mathcal{B}})$ with $g(X)$ $\mathbb{P}$ -integrable, the expectation of the absolutely continuous random variable $g(X)$ is by equation (103):

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) \, d\mathbb{P}_X(x) = \int_{\mathbb{R}} g(x) \cdot p_X(x) \, dx. \quad (105)$$

Analogously to the fact that absolutely continuous random variables are absolutely continuous with respect to the Lebesgue measure λ , discrete random variables are absolutely continuous with respect to the counting measure $\# : \mathcal{P}(\Lambda) \rightarrow [0, \infty]$ on Λ , where $\mathcal{P}(\Lambda)$ is the power set of Λ . The counting measure is defined by:

$$\#(A) := \begin{cases} |A|, & \text{if } A \text{ is finite} \\ \infty, & \text{otherwise} \end{cases}. \quad (106)$$

Observe that $\#(\{x\}) = 1, \forall x \in \mathbb{R}$. Similar to equation (100), P_X from equation (98) is absolutely continuous with respect to $\#$, because the fact that $\#(A) = 0$ if and only if $A = \emptyset$ implies:

$$\#(A) = 0 \implies \mathbb{P}_X(A) = \mathbb{P}_X(\emptyset) = \sum_{i \in \mathcal{I}} p_X(x_i) \cdot \delta_{x_i}(\emptyset). \quad (107)$$

Hence, by Theorem 6.1, the PMF of a discrete random variable X equals the Radon-Nikodym derivative:

$$p_X(x) = \frac{d\mathbb{P}_X}{d\#} = \mathbb{P}(X = x). \quad (108)$$

Consequently, for any measurable function $g : (\mathbb{R}, \mathcal{B}) \rightarrow (\bar{\mathbb{R}}, \bar{\mathcal{B}})$:

$$\begin{aligned} \mathbb{E}[g(X)] &= \int_{\mathbb{R}} g(x) \, d\mathbb{P}_X(x) = \int_{\mathbb{R}} g(x) \cdot p_X(x) \, d\#(x) \\ &= \sum_{x \in \Lambda} p_X(x) \cdot g(x), \end{aligned} \quad (109)$$

where the last equality holds because $\int_{\tilde{\Omega}} h(\tilde{\omega}) \, d\#(\tilde{\omega}) = \sum_{\tilde{\omega} \in \tilde{\Omega}} h(\tilde{\omega})$, for a function $h : \tilde{\Omega} \rightarrow [0, \infty]$ and measure space $(\tilde{\Omega}, \mathcal{P}(\tilde{\Omega}), \#)$.

6.3.2 Random Vectors

Section 6.3.1 can be directly generalised to random vectors $\mathbf{X} : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow (\mathbb{R}^d, \mathcal{B}^d)$, where $\mathcal{B}^d := \mathcal{B}(\mathbb{R}^d)$ denotes the Borel σ -algebra for $d \in \mathbb{N}$. Hence, the distribution of a random vector $\mathbf{X} = (X_1, \dots, X_d)^T \in \mathbb{R}^d$ is the push-forward measure $\mathbb{P}_{\mathbf{X}} : (\mathbb{R}^d, \mathcal{B}^d) \rightarrow [0, 1]$ defined by $\mathbb{P}_{\mathbf{X}} := \mathbb{P} \circ \mathbf{X}^{-1}$. The generalisation of equation (89) is given by:

$$\mathbb{P}_{\mathbf{X}}(\mathbf{B}) = \mathbb{P}(\mathbf{X}^{-1}(\mathbf{B})) = \mathbb{P}(\{\omega \in \Omega : \mathbf{X}(\omega) \in \mathbf{B}\}) =: \mathbb{P}(\mathbf{X} \in \mathbf{B}), \quad \mathbf{B} \in \mathcal{B}^d. \quad (110)$$

The generalisation of equation (90) to the CDF of a random vector is:

$$F_{\mathbf{X}}(\mathbf{x}) := \mathbb{P}(\mathbf{X} \leq \mathbf{x}) = \mathbb{P}(X_1 \leq x_1, \dots, X_d \leq x_d) = \mathbb{P}_{\mathbf{X}}((-\infty, x_1] \times \dots \times (-\infty, x_d]), \quad (111)$$

for $\mathbf{x} = (x_1, \dots, x_d)^T \in \mathbb{R}^d$.

If $\mathbf{X} \in \mathbb{R}^d$ is a random vector and $g_i : (\mathbb{R}^d, \mathcal{B}^d) \rightarrow (\mathbb{R}, \mathcal{B})$ is a measurable mapping, then $g_i(\mathbf{X})$ is a random variable on $\mathbb{R}$ (Durrett, 2019). Analogously, if $\mathbf{g} = (g_1, \dots, g_k)^T : (\mathbb{R}^d, \mathcal{B}^d) \rightarrow (\mathbb{R}^k, \mathcal{B}^k)$, $k \in \mathbb{N}$, is a measurable mapping, then $\mathbf{g}(\mathbf{X})$ is a random variable. Hence, if $\mathbf{g} \circ \mathbf{X}$ is $\mathbb{P}$ -integrable, then the generalisation of equation (92) is:

$$\mathbb{E}[\mathbf{g}(\mathbf{X})] := \int_{\Omega} \mathbf{g}(\mathbf{X}(\omega)) \, d\mathbb{P}(\omega) = \int_{\mathbb{R}^d} \mathbf{g}(\mathbf{x}) \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \quad (112)$$

The expectation of a random vector is defined component-wise (van der Vaart, 1997):

$$\mathbb{E}[\mathbf{g}(\mathbf{X})] = \begin{bmatrix} \mathbb{E}[g_1(\mathbf{X})] \\ \vdots \\ \mathbb{E}[g_k(\mathbf{X})] \end{bmatrix} = \begin{bmatrix} \int_{\mathbb{R}^d} g_1(\mathbf{x}) \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \\ \vdots \\ \int_{\mathbb{R}^d} g_k(\mathbf{x}) \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \end{bmatrix}, \quad (113)$$

if each g_i is $\mathbb{P}_{\mathbf{X}}$ -integrable. If $k = d$ and g_i is the function evaluation, i.e., $g_i(\mathbf{x}) = x_i$, then equation (113) simplifies to:

$$\mathbb{E}[\mathbf{X}] = \begin{bmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_d] \end{bmatrix}. \quad (114)$$

The joint PDF of absolutely continuous random vectors can be obtained by generalising Theorem 6.1. If $\mathbb{P}_{\mathbf{X}}$ is absolutely continuous with respect to Lebesgue measure λ_d on $(\mathbb{R}^d, \mathcal{B}^d)$, then there exists a measurable non-negative function $p_{\mathbf{X}} : (\mathbb{R}^d, \mathcal{B}^d) \rightarrow [0, \infty)$ such that the generalisation of equation (104) is given by:

$$\mathbb{P}_{\mathbf{X}}(B) = \int_B p_{\mathbf{X}}(\mathbf{x}) \, d\mathbf{x}, \quad B \in \mathcal{B}^d. \quad (115)$$

$p_{\mathbf{X}}$ is called the joint PDF of $\mathbf{X}$. Hence, if $\mathbf{X}$ is absolutely continuous with respect to λ_d so that its PDF $p_{\mathbf{X}}$ exists, equation (112) can be written as

$$\mathbb{E}[g(\mathbf{X})] = \int_{\mathbb{R}^d} g(\mathbf{x}) \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) = \int_{\mathbb{R}^d} g(\mathbf{x}) \cdot p_{\mathbf{X}}(\mathbf{x}) \, d\mathbf{x}, \quad (116)$$

for any measurable $g : (\mathbb{R}^d, \mathcal{B}^d) \rightarrow (\bar{\mathbb{R}}, \bar{\mathcal{B}})$.

6.3.3 Marginal and Conditional Densities

Let $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)^T : (\Omega, \mathcal{E}, \mathbb{P}) \rightarrow (\mathbb{R}^d, \mathcal{B}^d) = (\mathbb{R}^l \times \mathbb{R}^q, \mathcal{B}^l \otimes \mathcal{B}^q)$ be a random vector, where $\mathbf{X}_1 \in \mathbb{R}^l$ and $\mathbf{X}_2 \in \mathbb{R}^q$ such that $l + q = d$ (De Snoo and Winkler, 2018). The push-forward measure $\mathbb{P}_{\mathbf{X}_1, \mathbf{X}_2} = \mathbb{P} \circ (\mathbf{X}_1, \mathbf{X}_2)^{-1}$ is the joint distribution of $\mathbf{X}$. If $\mathbb{P}_{\mathbf{X}_1, \mathbf{X}_2}$ is absolutely continuous with respect to λ_{l+q} , then the joint PDF exist $p_{\mathbf{X}} = p_{\mathbf{X}_1, \mathbf{X}_2}$ and its joint distribution is such that:

$$\mathbb{P}_{\mathbf{X}_1, \mathbf{X}_2}(B) = \int_B p_{\mathbf{X}_1, \mathbf{X}_2}(\mathbf{x}_1, \mathbf{x}_2) \, d\mathbf{x}_1 \, d\mathbf{x}_2, \quad B \in \mathcal{B}^l \otimes \mathcal{B}^q. \quad (117)$$

Let $\pi_1(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1$ and $\pi_2(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_2$ be projections. Whence, the marginal distributions are $\mathbb{P}_{\mathbf{X}_1} = \mathbb{P}_{\mathbf{X}_1, \mathbf{X}_2} \circ \pi_1^{-1}$ and $\mathbb{P}_{\mathbf{X}_2} = \mathbb{P}_{\mathbf{X}_1, \mathbf{X}_2} \circ \pi_2^{-1}$. In particular, the marginal distribution measure are given by:

$$\begin{aligned} \mathbb{P}_{\mathbf{X}_1}(B_1) &= \mathbb{P}(\mathbf{X}_1 \in B_1) = \mathbb{P}(\mathbf{X} \in B_1 \times \mathbb{R}^q), \quad B_1 \in \mathcal{B}^l; \\ \mathbb{P}_{\mathbf{X}_2}(B_2) &= \mathbb{P}(\mathbf{X}_2 \in B_2) = \mathbb{P}(\mathbf{X} \in \mathbb{R}^l \times B_2), \quad B_2 \in \mathcal{B}^q. \end{aligned} \quad (118)$$

Observe the equivalence between this and the Kolmogorov consistency condition C2). Next, since $\mathbb{P}_{\mathbf{X}_1}$ is absolutely continuous with respect to λ_l , the marginal PDF of $\mathbf{X}_1$ is given by:

$$p_{\mathbf{X}_1}(\mathbf{x}_1) = \int_{\mathbb{R}^q} p_{\mathbf{X}_1, \mathbf{X}_2}(\mathbf{x}_1, \mathbf{x}_2) \, d\mathbf{x}_2, \quad p_{\mathbf{X}_1} = \frac{d}{d\lambda_l} \mathbb{P}_{\mathbf{X}_1}, \quad (119)$$

where $\mathbf{x}_1 \in \mathbb{R}^l$. $p_{\mathbf{X}_2}$ is defined similarly.

Let $\mathbf{x}_2 \in \mathbb{R}^q$. If $\mathbb{P}(\mathbf{X}_2 = \mathbf{x}_2) > 0$, then the conditional probability distribution measure of $\mathbf{X}_1$ given that the event $\mathbf{X}_2 = \mathbf{x}_2$ has already occurred is defined as (Rosenthal, 2009):

$$\mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}(B) := \frac{\mathbb{P}_{\mathbf{X}_1, \mathbf{X}_2}(B \times \{\mathbf{x}_2\})}{\mathbb{P}_{\mathbf{X}_2}(\mathbf{x}_2)}, \quad B \in \mathcal{B}^l. \quad (120)$$

However, this is only true when $\mathbf{X}$ is a discrete random vector. The more general regular conditional probability measure $\mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}$ on $(\mathbb{R}^l, \mathcal{B}^l)$ is defined as:

$$\mathbb{P}_{\mathbf{X}_1, \mathbf{X}_2}(B_1 \times B_2) = \int_{B_2} \mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}(B_1) \, d\mathbb{P}_{\mathbf{X}_2}(\mathbf{x}_2), \quad B_1 \in \mathcal{B}^l, \quad B_2 \in \mathcal{B}^q. \quad (121)$$

If $p_{\mathbf{X}_2}(\mathbf{x}_2) > 0$, i.e., if $\mathbf{x}_2 \notin N := \{\mathbf{x}_2 \in \mathbb{R}^q : p_{\mathbf{X}_2}(\mathbf{x}_2) = 0\}$, and $\mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}$ is absolutely continuous with respect to λ_l , then the conditional PDF of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$ is given by:

$$p_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}(\mathbf{x}_1 | \mathbf{x}_2) = \frac{p_{\mathbf{X}_1, \mathbf{X}_2}(\mathbf{x}_1, \mathbf{x}_2)}{p_{\mathbf{X}_2}(\mathbf{x}_2)}, \quad p_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2} = \frac{d}{d\lambda_l} \mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}. \quad (122)$$

The set N has λ_q -measure zero and can be ignored almost everywhere. In particular, any integral over N with respect to the measure λ_q is zero. Equation (122) can be equivalently stated as:

$$p_{\mathbf{X}_1, \mathbf{X}_2}(\mathbf{x}_1, \mathbf{x}_2) = p_{\mathbf{X}_1|\mathbf{X}_2}(\mathbf{x}_1 | \mathbf{x}_2) \cdot p_{\mathbf{X}_2}(\mathbf{x}_2). \quad (123)$$

$p_{\mathbf{X}_1|\mathbf{X}_2}$ in equation (122) is a valid PDF because it is non-negative and integrates to one:

$$\int_{\mathbb{R}^p} p_{\mathbf{X}_1|\mathbf{X}_2}(\mathbf{x}_1, \mathbf{x}_2) \, d\mathbf{x}_1 = \frac{1}{p_{\mathbf{X}_2}(\mathbf{x}_2)} \int_{\mathbb{R}^p} p_{\mathbf{X}_1, \mathbf{X}_2}(\mathbf{x}_1, \mathbf{x}_2) \, d\mathbf{x}_1 = \frac{1}{p_{\mathbf{X}_2}(\mathbf{x}_2)} \cdot p_{\mathbf{X}_2}(\mathbf{x}_2) = 1. \quad (124)$$

The conditional probability measure can also be defined without assuming $p_{\mathbf{X}_2}(\mathbf{x}_2) > 0$, see Rosenthal (2009) and Durrett (2019). In addition, the conditional probability distribution measure can also be defined as:

$$\mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}(B_1) := \int_{B_1} p_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}(\mathbf{x}_1 | \mathbf{x}_2) \, d\mathbf{x}_1, \quad B_1 \in \mathcal{B}^l. \quad (125)$$

Moreover, let $g : \mathbb{R}^l \rightarrow \mathbb{R}$ be a measurable function. The conditional expectation of $g(\mathbf{X}_1)$ given $\mathbf{X}_2 = \mathbf{x}_2$ is given by:

$$\begin{aligned} \mathbb{E}[g(\mathbf{X}_1) | \mathbf{X}_2 = \mathbf{x}_2] &= \mathbb{E}[g(\mathbf{X}_1) \mathbb{I}_{\mathbf{X}_2=\mathbf{x}_2}] = \int_{\mathbb{R}^l} g(\mathbf{x}_1) \, d\mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}(\mathbf{x}_1) \\ &= \int_{\mathbb{R}^l} g(\mathbf{x}_1) p_{\mathbf{X}_1|\mathbf{X}_2}(\mathbf{x}_1 | \mathbf{x}_2) \, d(\mathbf{x}_1). \end{aligned} \quad (126)$$

Or, more generally:

$$\mathbb{E}[g(\mathbf{X}_1) \cdot \mathbb{I}_{\{\mathbf{X}_2 \in B_2\}}] = \int_{B_2} \mathbb{E}[g(\mathbf{X}_1) | \mathbf{X}_2 = \mathbf{x}_2] \, d\mathbb{P}_{\mathbf{X}_2}(\mathbf{x}_2), \quad B_2 \in \mathcal{B}^q. \quad (127)$$

Furthermore, the joint measure $\mathbb{P}_{\mathbf{X}_1, \mathbf{X}_2}$ can be expressed in terms of the conditional measure $\mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}$:

$$\mathbb{P}_{\mathbf{X}_1, \mathbf{X}_2}(B_1 \times B_2) = \int_{B_2} \mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}(B_1) \, d\mathbb{P}_{\mathbf{X}_2}(\mathbf{x}_2), \quad B_1 \in \mathcal{B}^l, \quad B_2 \in \mathcal{B}^q. \quad (128)$$

Analogously, the marginal measure $\mathbb{P}_{\mathbf{X}_1}$ can also be expressed in terms of the conditional measure $\mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}$:

$$\mathbb{P}_{\mathbf{X}_1}(B_1) = \int_{\mathbb{R}^q} \mathbb{P}_{\mathbf{X}_1|\mathbf{X}_2=\mathbf{x}_2}(B_1) \, d\mathbb{P}_{\mathbf{X}_2}(\mathbf{x}_2), \quad B_1 \in \mathcal{B}^l. \quad (129)$$

And when $p_{\mathbf{X}_1}$ exists:

$$p_{\mathbf{X}_1}(\mathbf{x}_1) = \int_{\mathbb{R}^q} p_{\mathbf{X}_1|\mathbf{X}_2}(\mathbf{x}_1 | \mathbf{x}_2) \cdot p_{\mathbf{X}_2}(\mathbf{x}_2) \, d\mathbf{x}_2. \quad (130)$$

Observe the difference between a marginal and a conditional probability measures. The former represents the probability of the occurrence a single event, independent of other events. While the latter represents the probability of the occurrence an event given that another event has already occurred.

6.3.4 Expectation and Covariance Properties

6.3.4.1 Linearity of Expectation

Expectation is a linear operator and closed under affine mappings.

Proposition 6.1.

Let X be a random variable, and let $\mathbf{X}$ be a random vector on $\mathbb{R}^d$. If $\mathbb{E}[X]$ and $\mathbb{E}[\mathbf{X}]$ exist, then:

- (i). $\mathbb{E}[\alpha X + \beta] = \alpha \mathbb{E}[X] + \beta$ for every deterministic $\alpha, \beta \in \mathbb{R}$;
- (ii). $\mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{X}] + \mathbf{b}$ for every deterministic $\mathbf{A} \in \mathbb{R}^{k \times d}$ and $\mathbf{b} \in \mathbb{R}^d$.

Proof. First, we are going to prove (i) using equation (91). Pick arbitrary $\alpha, \beta \in \mathbb{R}$. Let us define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) := \alpha x + \beta$. Since g is continuous, it is measurable. Hence, by equation (91) we have:

$$\mathbb{E}[\alpha X + \beta] = \mathbb{E}[g(X)] = \int_{\mathbb{R}} \alpha X + \beta \, d\mathbb{P} = \alpha \cdot \int_{\mathbb{R}} X \, d\mathbb{P} + \beta = \alpha \cdot \mathbb{E}[X] + \beta. \quad (131)$$

Second, we are going to prove (ii) using the linear functional $L_i : \mathbb{R}^k \rightarrow \mathbb{R}$ defined by $L_i[\mathbf{y}] := \mathbf{e}_i^T \cdot \mathbf{y}$, where $\mathbf{e}_i$ is the i^{th} canonical basis vector of $\mathbb{R}^k$ and $i \in \{1, \dots, k\}$. Pick arbitrary $\mathbf{A} \in \mathbb{R}^{k \times d}$ and $\mathbf{b} \in \mathbb{R}^d$. Hence, the i^{th} element of the expectation of $\mathbf{A}\mathbf{X} \in \mathbb{R}^k$ is given by:

$$\begin{aligned} (\mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}])_i &= L_i[\mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}]] = L_i \left[\begin{bmatrix} \mathbb{E}[\sum_{j=1}^d A_{1j}X_j + b_1] \\ \vdots \\ \mathbb{E}[\sum_{j=1}^d A_{kj}X_j + b_k] \end{bmatrix} \right] = \mathbb{E} \left[\sum_{j=1}^d A_{ij}X_j + b_i \right] \\ &= \sum_{j=1}^d A_{ij} \mathbb{E}[X_j] + b_i \\ &= (\mathbf{A}\mathbb{E}[\mathbf{X}] + \mathbf{b})_i, \end{aligned} \quad (132)$$

So, $\mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{X}] + \mathbf{b}$, for all $\mathbf{A} \in \mathbb{R}^{k \times d}$ and $\mathbf{b} \in \mathbb{R}^d$. □

Observe that (i) is a special case of (ii) with $d = k = 1$. Furthermore, another interesting special is for $k = 1$:

$$\mathbb{E}[\mathbf{a}^T \mathbf{X} + \mathbf{b}] = \mathbf{a}^T \mathbb{E}[\mathbf{X}] + \mathbf{b}, \quad \mathbf{a} \in \mathbb{R}^d. \quad (133)$$

Alternatively, (ii) can also be proven by using the theory of Bochner integrals.

6.3.4.2 Covariance

Let X and Y be two random variables. If $\mathbb{E}[X]$ and $\mathbb{E}[Y]$ exist, then the covariance of X and Y is defined as (van der Vaart, 1997):

$$\text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \quad (134)$$

The variance of X is equal to:

$$\text{Var}(X) = \text{Cov}(X, X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2. \quad (135)$$

Similarly, let $\mathbf{X}$ be a random vector on $\mathbb{R}^d$. If $\mathbb{E}[\mathbf{X}]$ exists, then the covariance matrix of $\mathbf{X}$ is defined as:

$$\begin{aligned} \text{Cov}(\mathbf{X}) &= \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T] = \int_{\mathbb{R}^d} (\mathbf{x} - \mathbb{E}[\mathbf{x}]) (\mathbf{x} - \mathbb{E}[\mathbf{x}])^T \, d\mathbb{P}_{\mathbf{X}}(\mathbf{x}) \\ &= \begin{pmatrix} \text{Cov}(X_1, X_1) & \dots & \text{Cov}(X_1, X_d) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_d, X_1) & \dots & \text{Cov}(X_d, X_d) \end{pmatrix}. \end{aligned} \quad (136)$$

The vector containing the variances of $\mathbf{X}$ component-wise is equal to:

$$\text{Var}(\mathbf{X}) = \text{diag}(\text{Cov}(\mathbf{X})), \quad (137)$$

where $\text{diag}(\cdot)$ extracts the diagonal elements from a matrix.

Next, some important properties regarding covariance functions are given in the following proposition.

Proposition 6.2.

Let X, Y and Z be random variables, and let $\mathbf{X}$ be a random vector on $\mathbb{R}^d$. If $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[Z]$ and $\mathbb{E}[\mathbf{X}]$ exist, then:

- (i). $\text{Cov}(\alpha X + \beta Y, Z) = \alpha \text{Cov}(X, Z) + \beta \text{Cov}(Y, Z)$ for every deterministic $\alpha, \beta \in \mathbb{R}$;
- (ii). $\text{Cov}[\mathbf{A}\mathbf{X} + \mathbf{b}] = \mathbf{A} \text{Cov}(\mathbf{X}) \mathbf{A}^T$ for deterministic $\mathbf{A} \in \mathbb{R}^{k \times d}$ and $\mathbf{b} \in \mathbb{R}^d$;
- (iii). $\text{Cov}(\mathbf{X})$ is symmetric and positive semi-definite.

Proof. First, we are going to prove (i) using Proposition 6.1 (i). Pick arbitrary deterministic $\alpha, \beta \in \mathbb{R}$. Then, by linearity of expectation and equation (134):

$$\begin{aligned} \text{Cov}(\alpha X + \beta Y, Z) &= \mathbb{E}[(\alpha X + \beta Y)Z] - \mathbb{E}[\alpha X + \beta Y]\mathbb{E}[Z] \\ &= \mathbb{E}[\alpha XZ + \beta YZ] - \mathbb{E}[\alpha X + \beta Y]\mathbb{E}[Z] \\ &= \alpha \mathbb{E}[XZ] + \beta \mathbb{E}[YZ] - \alpha \mathbb{E}[X]\mathbb{E}[Z] - \beta \mathbb{E}[Y]\mathbb{E}[Z] \\ &= \text{Cov}(\alpha X + \beta Y, Z) = \alpha \text{Cov}(X, Z) + \beta \text{Cov}(Y, Z). \end{aligned} \quad (138)$$

Second, we going to prove (ii) using Proposition 6.1. Pick arbitrary deterministic $\mathbf{A} \in \mathbb{R}^{k \times d}$ and $\mathbf{b} \in \mathbb{R}^d$. Then, by linearity of expectation and equation (136):

$$\begin{aligned} \text{Cov}(\mathbf{A}\mathbf{X} + \mathbf{b}) &= \mathbb{E}\left[(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}])(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}])^T\right] \\ &= \mathbb{E}\left[(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbf{A}\mathbb{E}[\mathbf{X}] - \mathbf{b})(\mathbf{A}\mathbf{X} + \mathbf{b} - \mathbf{A}\mathbb{E}[\mathbf{X}] - \mathbf{b})^T\right] \\ &= \mathbb{E}\left[(\mathbf{A}\mathbf{X} - \mathbf{A}\mathbb{E}[\mathbf{X}])(\mathbf{A}\mathbf{X} - \mathbf{A}\mathbb{E}[\mathbf{X}])^T\right] \\ &= \mathbb{E}\left[\left(\mathbf{A}(\mathbf{X} - \mathbb{E}[\mathbf{X}])\right)\left(\mathbf{A}(\mathbf{X} - \mathbb{E}[\mathbf{X}])\right)^T\right] \\ &= \mathbb{E}\left[\mathbf{A}(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T \mathbf{A}^T\right] \\ &= \mathbf{A} \mathbb{E}\left[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T\right] \mathbf{A}^T \\ &= \mathbf{A} \text{Cov}(\mathbf{X}) \mathbf{A}^T. \end{aligned} \quad (139)$$

Third, using the fact that the covariance function is symmetric for two random variables X_i and X_j , i.e., $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$, implies that $\text{Cov}(\mathbf{X})^T = \text{Cov}(\mathbf{X})$. So, $\text{Cov}(\mathbf{X})$ is symmetric. Next, observe that for any $\mathbf{a} \in \mathbb{R}^d$, the variance of the inner product $\mathbf{a}^T \mathbf{X} \in \mathbb{R}$ is equal to $\mathbf{a}^T \text{Cov}(\mathbf{X}) \mathbf{a}$:

$$\begin{aligned} \text{Var}(\mathbf{a}^T \mathbf{X}) &= \mathbb{E}\left[(\mathbf{a}^T \mathbf{X} - \mathbb{E}[\mathbf{a}^T \mathbf{X}])(\mathbf{a}^T \mathbf{X} - \mathbb{E}[\mathbf{a}^T \mathbf{X}])^T\right] \\ &= \mathbb{E}\left[(\mathbf{a}^T (\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{a}^T (\mathbf{X} - \mathbb{E}[\mathbf{X}]))^T\right] \\ &= \mathbb{E}\left[\mathbf{a}^T (\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T \mathbf{a}\right] \\ &= \mathbf{a}^T \text{Cov}(\mathbf{X}) \mathbf{a}. \end{aligned} \quad (140)$$

Hence, since Var is always non-negative, $\mathbf{a}^T \text{Cov}(\mathbf{X}) \mathbf{a} \geq 0, \forall \mathbf{a} \in \mathbb{R}^d$. So, $\text{Cov}(\mathbf{X})$ is positive semi-definite. $\square$

Observe that in (i), αX and βY can be viewed as linear functionals on $\mathbb{R}$ defined by $L_\alpha[X] = \alpha X$ and $L_\beta[Y] = \beta Y$. In addition, the proof of (iii) shows that $\text{Var}(\mathbf{a}^T \mathbf{X}) = \mathbf{a}^T \text{Cov}(\mathbf{X}) \mathbf{a}, \forall \mathbf{a} \in \mathbb{R}^d$. This is a special case of (ii) with $k = 1$ and $\mathbf{b} = \mathbf{0}$.

6.3.5 Gaussian Measure

Equivalently, using the Radon-Nikodym derivative Theorem 6.1: G is absolutely continuous with respect to the Lebesgue measure λ and the Radon-Nikodym derivative is the Gaussian PDF:

$$\frac{dG}{d\lambda}(x) = p_X(x). \quad (141)$$

By a slight abuse of notation, since the following expression is not an equation of differentials, the Gaussian measure is often denoted as follows:

$$dG(x) = p_X(x) d\lambda(x) = p_X(x) dx. \quad (142)$$

Hence, by equation (105) and Theorem 6.1:

$$\begin{aligned} \mathbb{E}[g(X)] &= \int_{\mathbb{R}} g(x) d\mathbb{P}_X(x) = \int_{\mathbb{R}} g(x) dG(x) \\ &= \int_{\mathbb{R}} g(x) \cdot p_X(x) dx \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \int_{\mathbb{R}} g(x) \cdot \exp\left\{-\frac{(x-m)^2}{2\sigma^2}\right\} dx, \end{aligned} \quad (143)$$

for any measurable function $g : (\mathbb{R}, \mathcal{B}) \rightarrow (\mathbb{R}, \mathcal{B})$

6.4 Linear Operators

The term operator often refers to a mapping from a space of functions to another space. More generally, however, it can denote any mapping between two spaces (or within the same space). Of particular interest is the special case of a linear operator.

Definition 6.1. *Sterk and de Snoo (2025)*

Let X and Y be linear spaces over $\mathbb{R}$, i.e., vector spaces over $\mathbb{R}$. A linear operator is a map $T : X \rightarrow Y$ that is linear.

A map $T : X \rightarrow Y$ is linear, if T is defined everywhere on X and if the following two conditions are satisfied:

1. $T[(x + y)] = T[x] + T[y];$
2. $T[\lambda \cdot x] = \lambda \cdot (T[x]),$

for all $x, y \in X$ and $\lambda \in \mathbb{R}$.

The collection of all linear operators from X to Y is denoted by $\mathfrak{L}(X, Y)$. Sometimes, the square brackets in $T[\cdot]$ are omitted.

An example of a linear space over $\mathbb{R}$ is the function space $\mathcal{F}_X$, where the elements of $\mathcal{F}_X$ are functions mapping from a space X to $\mathbb{R}$. Note that $\mathbb{R}^d$ is a linear space over $\mathbb{R}$, for all $d \in \mathbb{N}$. Functions mapping elements to $\mathbb{R}$ are called functionals. Hence, $\mathcal{F}_X$ is defined as follows:

$$\mathcal{F}_X := \{\text{all functionals } f : X \rightarrow \mathbb{R}\} \subseteq \mathfrak{L}(X, \mathbb{R}) \quad (144)$$

The X -subscript is sometimes omitted when it is clear from the situation. Observe that (linear) functionals are a special case of (linear) operators with $Y = \mathbb{R}$. A functional f on X is linear if:

$$f(\lambda \cdot x_1 + x_2) = \lambda \cdot f(x_1) + f(x_2), \quad \forall x_1, x_2 \in X, \lambda \in \mathbb{R}. \quad (145)$$

Hence, if every functional in $\mathcal{F}_X$ satisfies this property, then $\mathcal{F}_X$ forms a linear space over $\mathbb{R}$.

Furthermore, an important property of an operator is boundedness. Normed linear spaces are defined in Subsection 6.5.

Definition 6.2.

Let X and Y be normed linear spaces. A linear operator $T : X \rightarrow Y$ is bounded, if there exists a constant $C \in \mathbb{R}_{>0}$ such that:

$$\|T[x]\| \leq C \cdot \|x\|, \quad \forall x \in X. \quad (146)$$

In addition, the collection of all bounded linear operators from X to Y is denoted by $\mathfrak{B}(X, Y) \subseteq \mathfrak{L}(X, Y)$.

An operator T is bounded if and only if it is continuous.

Moreover, a bilinear form $B : X \times X \rightarrow \mathbb{R}$, where X is a linear space over $\mathbb{R}$, is a generalization of a linear functional. Specifically, a bilinear form is a function $B : X \times X \rightarrow \mathbb{R}$ that is linear in each argument separately:

- $B(\lambda_1 \cdot v_1 + \lambda_2 \cdot v_2, u) = \lambda_1 \cdot B(v_1, u) + \lambda_2 \cdot B(v_2, u), \quad \forall v_1, v_2, u \in X \text{ and } \forall \lambda_1, \lambda_2 \in \mathbb{R};$
- $B(v, \lambda_1 \cdot u_1 + \lambda_2 \cdot u_2) = \lambda_1 \cdot B(v, u_1) + \lambda_2 \cdot B(v, u_2), \quad \forall v, u_1, u_2 \in X \text{ and } \forall \lambda_1, \lambda_2 \in \mathbb{R}.$

Additionally, if $B(v, u) = B(u, v)$, $\forall v, u \in X$, then B is a bilinear form.

6.5 Banach & Hilbert Space

This section is based upon [Sterk and de Snoo \(2025\)](#) and [\(De Snoo and Winkler, 2018\)](#).

Banach spaces are normed linear spaces that are complete. Similarly, Hilbert spaces are inner product spaces that are complete. First, we give a definition of normed linear spaces over the field $\mathbb{R}$.

Definition 6.3.

Let $\mathbb{B}$ be a linear space over $\mathbb{R}$. The map $\|\cdot\| : \mathbb{B} \rightarrow \mathbb{R}$ is a norm, if for all $x, y \in \mathbb{B}$ and $\lambda \in \mathbb{R}$ the following axioms are satisfied:

1. $\|x\| \geq 0$;
2. $\|x\| = 0 \iff x = 0$;
3. $\|\lambda \cdot x\| = |\lambda| \cdot \|x\|$;
4. $\|x + y\| \leq \|x\| + \|y\|$.

The pair $(\mathbb{B}, \|\cdot\|)$, often shortened to $\mathbb{B}$, is a normed linear space over $\mathbb{R}$.

The norm $\|\cdot\|$ induces the metric $d(x, y) = \|x - y\|$, $x, y \in \mathbb{B}$. Banach spaces $(\mathbb{B}, \|\cdot\|)$ are normed linear spaces that are complete, i.e., each Cauchy sequence converges to an element in the space according to the norm $\|\cdot\|$. Completeness ensures that limits of convergent sequences, for instance (inside) integrals, remain inside the space $\mathbb{B}$.

A special normed linear space is an inner product space where the norm is induced by an inner product.

Definition 6.4.

Let $\mathbb{H}$ be a linear space over $\mathbb{R}$. The map $\langle \cdot, \cdot \rangle : \mathbb{H} \rightarrow \mathbb{R}$ is a norm, if for all $x, y, z \in \mathbb{H}$ and $\lambda, \theta \in \mathbb{R}$ the following axioms are satisfied:

1. $\langle x, x \rangle \geq 0$;
2. $\langle x, x \rangle = 0 \iff x = 0$;
3. $\langle \lambda \cdot x + \theta \cdot y, z \rangle = \lambda \cdot \langle x, z \rangle + \theta \cdot \langle y, z \rangle$;
4. $\langle x, y \rangle = \langle y, x \rangle$.

The third bullet point is called bilinearity. The pair $(\mathbb{H}, \langle \cdot, \cdot \rangle)$, often shortened to $\mathbb{H}$, is a inner product space over $\mathbb{R}$.

An inner product space that is complete is called a Hilbert space. Hence, a Hilbert space $\mathbb{H}$ is a Banach space whose norm is induced by an inner product:

$$\|x\|_{\mathbb{H}} := \sqrt{\langle x, x \rangle}. \quad (147)$$

In every finite-dimensional normed linear space, all norms equivalent which implies that convergence is independent of the chosen norm. In addition, every finite-dimensional normed space is complete. Therefore, since $\mathbb{R}^d$ is finite dimensional for every $d < \infty$, $\mathbb{R}^d$ with any norm is a Banach space. However, this is not true for infinite-dimensional spaces. An important normed linear function space is the L^p , $1 \leq p \leq \infty$, space (De Snoo and Winkler, 2018).

Definition 6.5.

Let $(\Omega, \mathcal{A}, \nu)$ be a measure space. For $1 \leq p < \infty$, define $\|f\|_p := \left(\int_{\Omega} |f|^p d\nu \right)^{1/p}$ for any measurable function $f : \Omega \rightarrow \mathbb{R}$. The set of all measurable real-valued functions for which $\|f\|_p < \infty$ is defined as:

$$\mathcal{L}^p(\Omega, \nu) := \left\{ \text{measurable } f : \Omega \rightarrow \mathbb{R} \mid \|f\|_p := \left(\int_{\Omega} |f|^p d\nu \right)^{1/p} < \infty \right\} \quad (148)$$

For $p = \infty$:

$$\mathcal{L}^{\infty}(\Omega, \nu) := \left\{ \text{measurable } f : \Omega \rightarrow \mathbb{R} \mid \exists c \geq 0 \text{ such that } |f(\omega)| \leq c \text{ almost everywhere} \right\}. \quad (149)$$

Furthermore, for $f \in \mathcal{L}^{\infty}(\Omega, \nu)$:

$$\begin{aligned} \|f\|_{\infty} &:= \inf \{c \geq 0 \mid |f(\omega)| \leq c \text{ almost everywhere}\} \\ &= \inf \{c \geq 0 \mid \nu(\{\omega \in \Omega : |f(\omega)| > c\}) = 0\} =: \text{ess sup } f. \end{aligned} \quad (150)$$

For any $f \in \mathcal{L}^p(\Omega, \nu)$, f is measurable and $|f|^p$ is integrable with respect to ν . The space $\mathcal{L}^p(\Omega, \nu)$, $1 \leq p \leq \infty$, is a semi-normed linear space of measurable, i.e., a normed linear space where only the implication $\Leftarrow$ holds for point 2. of Definition 6.3, we define the collection of a neutral element in $\mathcal{L}^p(\Omega, \nu)$ by:

$$\mathcal{Z}_p(\nu) := \left\{ f \in \mathcal{L}^p(\Omega, \nu) \mid \|f\|_p = 0 \right\}. \quad (151)$$

This is a linear subspace of $\mathcal{L}^p(\Omega, \nu)$, where the following holds:

$$\|f\|_p = 0 \iff f \equiv 0 \text{ for almost all } \omega \in \Omega. \quad (152)$$

This equivalence ensures that $\|\cdot\|_p$ is a norm, because the $\Rightarrow$ also holds for point 2. of Definition 6.3. Hence, we have a normed linear space with $\|\cdot\|_p$ as norm.

Theorem 6.2.

The quotient space $L^p(\Omega, \nu) := \mathcal{L}^p(\Omega, \nu) / \mathcal{Z}_p(\nu)$ is a Banach space (with $\|\cdot\|_p$ as norm).

$L^p(\Omega, \nu)$ is not a space of functions but of equivalence classes of measurable functions equal almost everywhere, i.e., two functions in the same equivalence class differ only on a set of measure zero. In particular, $f, h \in \mathcal{X}$ are equivalent if $f \sim h \iff f - h \in \mathcal{Z}_p(\nu) \iff \|f - h\| = 0$, where $\sim$ denotes an equivalence relation. Hence, for any $f \in \mathcal{X}$, its equivalence class is defined by:

$$\begin{aligned} [f] &= \{h \in \mathcal{X} : f \sim h\} = \{h \in \mathcal{X} : f - h \in \mathcal{Z}_p(\nu)\} \\ &= \{h : f(x) = h(x) \text{ for almost all } x\} \end{aligned} \quad (153)$$

This means that if $\nu(A) = 0$ for some $A \in \mathcal{A}$ and $f, h \in \mathcal{L}^\infty(\Omega, \nu)$ are functions satisfying $f(x) = h(x)$, $\forall x \in \Omega \setminus A$, then:

$$\|f - h\|_p^p = \int_{\Omega} |f - h|^p \, d\nu = 0. \quad (154)$$

So, from the viewpoint of integration, f and h are the same function.

$$L^p(\Omega, \nu) = \{\text{equivalence classes } [f] : f \in \mathcal{L}(\Omega, \nu)\}, \quad (155)$$

Hence, in $L^p(\Omega, \nu)$ spaces, we do not distinguish between functions that differ only on measure-zero sets. And in practice, one does not distinguish between an equivalence class and any of its elements. Hence, we use the following abuse of notation: $[f] = f$.

Theorem 6.2 states that all $L^p(\Omega, \nu)$ spaces are Banach spaces. However, only $L^2(\Omega, \nu)$ spaces are Hilbert spaces with inner product:

$$\langle f, h \rangle := \int_{\Omega} f(x) \cdot h(x) \, d\nu(x). \quad (156)$$

Hence, the norm on the Hilbert space $L^2(\Omega, \nu)$ is given by:

$$\|f\|_{L^2(\Omega, \nu)} := \sqrt{\langle f, f \rangle} = \left(\int_{\Omega} f(x)^2 \, d\nu(x) \right)^{1/2}. \quad (157)$$

The following theorem, Hölder's inequality, is a fundamental result for $L^p(\Omega, \nu)$ spaces.

Theorem 6.3.

Let $1 \leq p, q \leq \infty$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. If $f \in L^p(\Omega, \nu)$ and $g \in L^1(\Omega, \nu)$, then $fg \in L^1(\Omega, \nu)$ and:

$$\|f \cdot g\|_1 \leq \|f\|_p \cdot \|g\|_1. \quad (158)$$

In particular, for $p = q = 2$, this is the Cauchy-Schwarz inequality.

Next, we provide an important definition and a fundamental theorem regarding Banach and Hilbert spaces.

Definition 6.6.

Let $\mathbb{B}$ be a normed linear space over $\mathbb{R}$. The dual space of $\mathbb{B}$ is the Banach space defined by:

$$\mathbb{B}^* := \mathfrak{B}(\mathbb{B}, \mathbb{R}) = \{\text{all linear bounded functionals } f : \mathbb{B} \rightarrow \mathbb{R}\}. \quad (159)$$

The element of $\mathbb{B}^*$ are the linear functionals $f : \mathbb{B} \rightarrow \mathbb{R}$ which are bounded and whose norm is given by:

$$\|f\| = \sup_{x \neq 0} \frac{|f(x)|}{\|x\|} = \sup_{\|x\|=1} \|T[x]\| = \sup_{\|x\| \leq 1} \|T[x]\|. \quad (160)$$

Note that if X Banach, then X^* also Banach. The next theorem is pivotal for bounded linear functionals on Hilbert spaces.

Theorem 6.4. (Riesz-Fréchet Representation Theorem)

Let $\mathbb{H}$ be a Hilbert space. For $y \in \mathbb{H}$, the bounded linear functional $F_y : \mathbb{H} \rightarrow \mathbb{R}$ defined by:

$$F_y(x) = \langle x, y \rangle, \quad \forall x \in \mathbb{H}, \quad (161)$$

belongs to $\mathbb{H}^*$. In addition:

$$\|F_y\| = \sup_{x \neq 0} \frac{|F_y(x)|}{\|x\|} = \|y\|. \quad (162)$$

In fact, for each $F \in \mathbb{H}^*$ there exists a unique $y \in \mathbb{H}$ such that $F = F_y$.

[Sterk and de Snoo \(2025\)](#) show that the dual $\mathbb{H}^*$ is also a Hilbert space, and it can be identified with the space $\mathbb{H}$ itself.

6.6 Stochastic Processes

A stochastic process is a collection of random variables defined jointly on some common probability space $(\Omega, \mathcal{E}, \mathbb{P})$. The random variables are indexed on some set $\mathfrak{T}$. If $\mathfrak{T} \subseteq \mathbb{Z}_{\geq 0}$ or $\mathfrak{T} \subseteq \mathbb{R}_{\geq 0}$, then a stochastic process is called discrete or continuous, respectively. Furthermore, every random variable takes values in the same measurable space $(S, \mathcal{S})$, where the random variable must be $\mathcal{S}$ -measurable. The space S is called the state space.

Definition 6.7.

A stochastic process with state space S is collection of random variables:

$$\{X(t, \omega) : t \in \mathfrak{T}\}, \quad (163)$$

where $X : \mathfrak{T} \times \Omega \rightarrow S$. At a point $t \in \mathfrak{T}$, $X(t, \omega) = X(t)$ is a random variable representing a value at t .

Historically, t represents a time index, but this does not need to be the case. Often, $X(t, \omega)$ is shortened to $X(t)$ or X_t . Next, we are going to assume that $(S, \mathcal{S}) := (\mathbb{R}, \mathcal{B})$. Furthermore, we define the following (Rosenthal, 2009):

$$\begin{aligned} S^{\mathfrak{T}} &= \mathbb{R}^{\mathfrak{T}} := \{\text{all (possible) functions } \phi : \mathfrak{T} \rightarrow \mathbb{R}\}; \\ \mathcal{S}^{\mathfrak{T}} &= \mathcal{B}^{\mathfrak{T}} := \sigma(\{X(t) \in B \mid t \in \mathfrak{T}, B \in \mathcal{B}\}), \end{aligned} \quad (164)$$

where $\mathcal{B}^{\mathfrak{T}}$ is called the cylinder σ -algebra and $\sigma(\cdot)$ means that it is the smallest that makes all maps $\phi(t)$ measurable. Now, the stochastic process in equation (163) can be equivalently represented as $X : \Omega \rightarrow \mathbb{R}^{\mathfrak{T}}$ on the probability space $(\Omega, \mathcal{E}, \mathbb{P})$. Analogously to the definition of the probability distribution of a random variable in equation (89), the distribution of a stochastic process X is the push-forward measure $\mathbb{P}_X = \mathbb{P} \circ X^{-1}$ with (Billingsley, 1995) :

$$\mathbb{P}_X(B) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) \in B\}), \quad B \in \mathcal{B}. \quad (165)$$

Recall that a real-valued random variable is by definition $\mathcal{B}$ -measurable. A sample function, i.e., a realization, is a single outcome of a stochastic process given by the mapping $X(\cdot, \omega) : \mathfrak{T} \rightarrow S = \mathbb{R}$, for all $\omega \in \Omega$.

We are interested in continuous stochastic processes, which can be viewed as infinite collections of random variables indexed by a continuous domain. However, it is generally intractable to describe the distribution of such an infinite collection directly. Therefore, we would like to study the finite-dimensional distributions of the process, i.e., the joint distributions of the random variables evaluated at finitely many points.

Definition 6.8.

For a stochastic process $X : \Omega \rightarrow \mathbb{R}^{\mathfrak{T}}$ with distribution $\mathbb{P}_X$, its finite-dimensional distribution for distinct $t_1, \dots, t_k \in \mathfrak{T}$, $k \in \mathbb{N}$, is defined as the following probability measure:

$$\mathbb{P}_{t_1, \dots, t_k} := \mathbb{P} \circ (X(t_1), \dots, X(t_k))^{-1}. \quad (166)$$

$\mathbb{P}_{t_1, \dots, t_k}$ can be viewed as the joint distribution of the random vector $(X(t_1), \dots, X(t_k))$, or as a projection of the distribution $\mathbb{P}_X$ onto a finite subset of $\mathfrak{T}$. For $B_k \in \mathcal{B}^k$, $\mathbb{P}_{t_1, \dots, t_k}$ is defined as:

$$\begin{aligned} \mathbb{P}_{t_1, \dots, t_k}(B_k) &= \mathbb{P}(\{\omega \in \Omega \mid (X(t_1, \omega), \dots, X(t_k, \omega)) \in B_k\}) \\ &= \mathbb{P}((X(t_1), \dots, X(t_k)) \in B_k). \end{aligned} \quad (167)$$

Hence, the collection of finite-dimensional distributions for a stochastic process $\{X(t) : t \in \mathfrak{T}\}$ is:

$$\{\mathbb{P}_{t_1, \dots, t_k} \mid k \in \mathbb{N}, t_1, \dots, t_k \in \mathfrak{T}\}. \quad (168)$$

These finite-dimensional distributions satisfy two consistency conditions, for some $k \in \mathbb{N}$ (Rosenthal, 2009):

C1). If $\tau((1 \dots k)) = (\tau(1) \dots \tau(k))$ is a permutation of $\{1, \dots, k\}$, i.e., a bijection $\tau : \{1, \dots, k\} \rightarrow \{1, \dots, k\}$, then:

$$\mathbb{P}_{t_1, \dots, t_k}(B_1 \times \dots \times B_k) = \mathbb{P}_{t_{\tau(1)}, \dots, t_{\tau(k)}}(B_{\tau(1)} \times \dots \times B_{\tau(k)}), \quad (169)$$

for distinct $t_1, \dots, t_k \in \mathfrak{T}$ and for all $B_1, \dots, B_k \in \mathcal{B}$.

C2). For distinct $t_1, \dots, t_k \in \mathfrak{T}$ and for all $B_1, \dots, B_k \in \mathcal{B}$, we have:

$$\mathbb{P}_{t_1, \dots, t_k}(B_1 \times \dots \times B_{k-1} \times \mathbb{R}) = \mathbb{P}_{t_1, \dots, t_{k-1}}(B_1 \times \dots \times B_{k-1}). \quad (170)$$

Condition C1). says that if we permute the indices of t_i and B_i , $i \in \{1, \dots, k\}$, then we do not change the probabilities. For example, for real-valued random variables X_1 and X_2 and $B_1, B_2 \in \mathcal{B}$:

$\mathbb{P}(X_1 \in B_1, X_2 \in B_2) = \mathbb{P}(X_1 \in B_2, X_2 \in B_1)$. Moreover, condition C2). says that allowing X_{t_k} to be anywhere in $\mathbb{R}$ is equivalent to disregarding X_{t_k} in the probability measure. For example, $\mathbb{P}(X_1 \in B_1, X_2 \in \mathbb{R}) = \mathbb{P}(X_1 \in B_1)$.

It turns out that conditions C1) and C2) are sufficient conditions for existence of a stochastic process.

Theorem 6.5. (Kolmogorov's Existence Theorem)

A family of Borel probability measure $\{\mathbb{P}_{t_1, \dots, t_k} \mid k \in \mathbb{N}, \text{ distinct } t_1, \dots, t_k \in \mathfrak{T}\}$, where $\mathbb{P}_{t_1, \dots, t_k}$ is a measure on $\mathbb{R}^k$, satisfies consistency conditions C1). and C2). if and only if there exists a probability space $(\Omega, \mathcal{E}, \mathbb{P})$ and random variables $\{X(t) : t \in \mathfrak{T}\}$ defined on $(\Omega, \mathcal{E}, \mathbb{P})$ such that:

$$\mathbb{P}((X(t_1), \dots, X(t_k)) \in B_k) = \mathbb{P}_{t_1, \dots, t_k}(B_k), \quad (171)$$

for all $k \in \mathbb{N}$, distinct $t_1, \dots, t_k \in \mathfrak{T}$ and $B_k \in \mathcal{B}^k$.

See Billingsley (1995) for a proof.

6.7 Kernels

First, the rational quadratic (RQ) kernel $k_{RQ} : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ is defined by:

$$k_{RQ}(\mathbf{x}, \tilde{\mathbf{x}} \mid \sigma_f^2, l, \alpha) = k_{RQ}(\|\mathbf{x} - \tilde{\mathbf{x}}\|_2 \mid \sigma_f^2, l, \alpha) := \sigma_f^2 \cdot (\mathbf{x}^T \tilde{\mathbf{x}} + c)^p, \quad (172)$$

where $\alpha, l > 0$. The rational quadratic is an infinite sum of squared-exponential kernels with different length-scales. In particular, $\lim_{\alpha \rightarrow -\infty} k_{RQ}(\mathbf{x}, \tilde{\mathbf{x}} \mid \sigma_f^2, l, \alpha) = k_{SE}(\mathbf{x}, \tilde{\mathbf{x}} \mid \sigma_f^2, l)$. Therefore, the heatmaps of the rational quadratic kernel in Figure 51 and the five sample functions drawn from $\mathcal{GP}(0, k_{RQ})$ in Figure 52 appear similar to those obtained from the squared-exponential kernel.

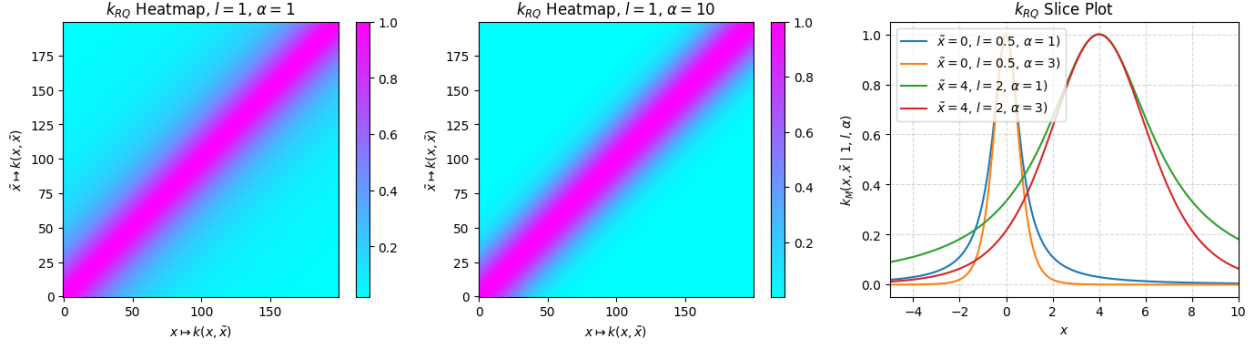


Figure 51: k_P heatmaps and slice plot.

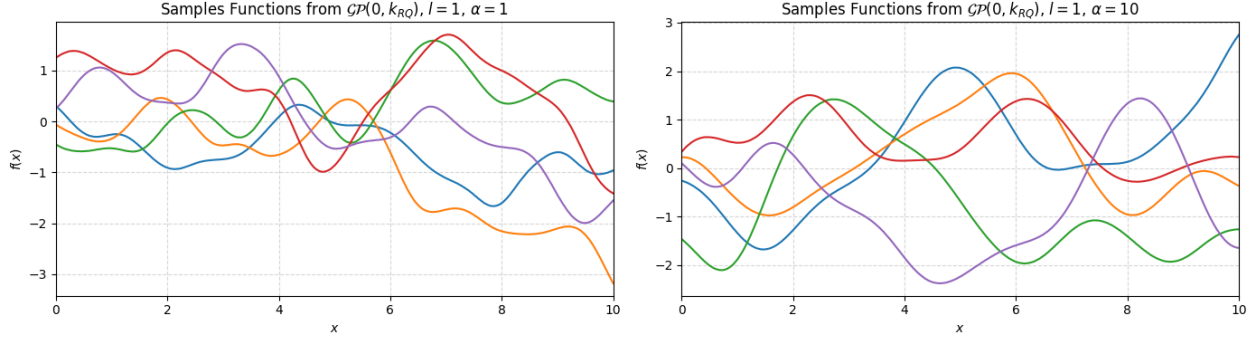


Figure 52: Sample functions drawn from $\mathcal{GP}(0, k_{RQ})$, for $\alpha = 1$ and $\alpha = 10$ with $l = 1$.

Second, the polynomial kernel $k_P : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ is defined by:

$$k_P(\mathbf{x}, \tilde{\mathbf{x}} \mid \sigma_f^2, p, c) := \sigma_f^2 \cdot (\mathbf{x}^T \tilde{\mathbf{x}} + c)^p, \quad (173)$$

where $c \in \mathbb{R}$ is the offset hyperparameter and $p \in \mathbb{N}$ is the degree of the polynomial. This kernel induces a hypothesis space consisting of polynomials, and has been widely-used in machine learning (Kanagawa et al., 2018). Importantly, k_P is non-stationary, since it depends on the inner product $\mathbf{x}^T \tilde{\mathbf{x}}$ rather than only $\mathbf{x} - \tilde{\mathbf{x}}$. Therefore, it belongs to the class of dot product kernels. As a result, the similarity between two input points is not translation-invariant, but depends on their location and scale. In particular, pairs of points with large inner products (typically those with both large absolute values) are considered more similar. In addition, Figure 54 shows sample functions drawn from the Gaussian process prior $\mathcal{GP}(0, k_P)$ with different values of p and a fixed $c = 0$. The resulting sample functions exhibit strong scaling behaviour and polynomial growth, specifically when p increases.

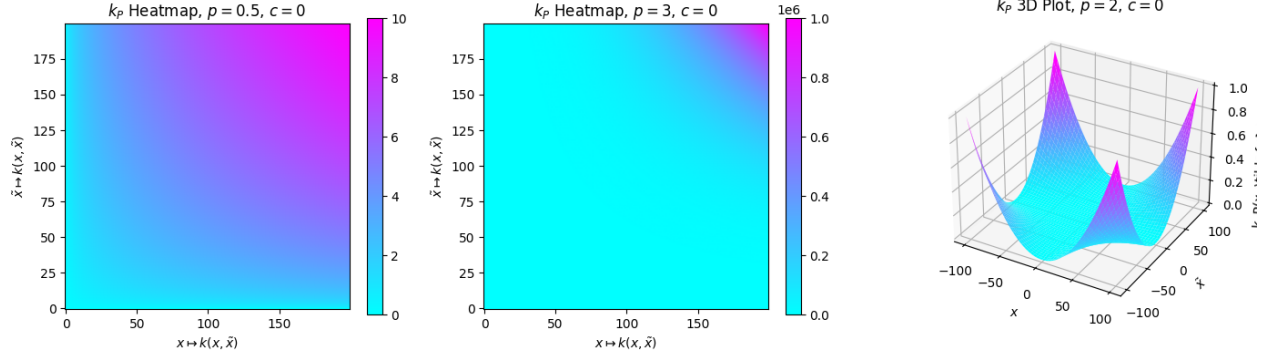


Figure 53: k_P heatmaps and slice plot.

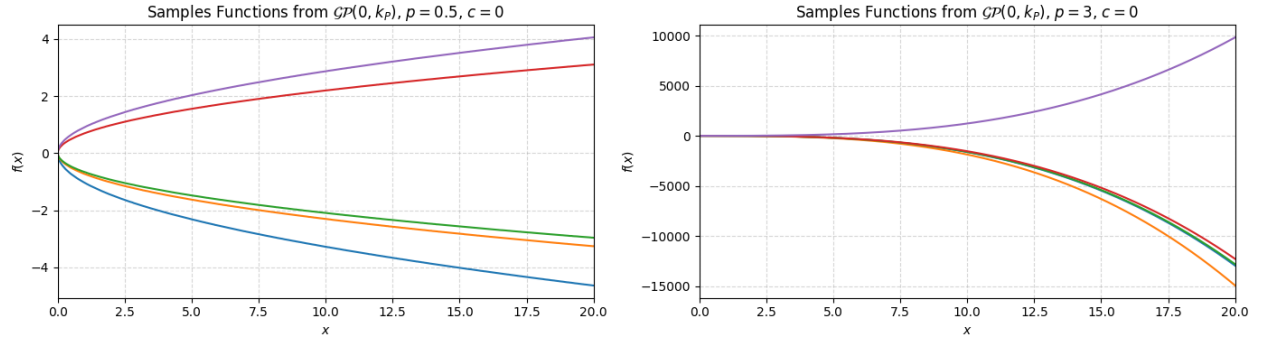


Figure 54: Sample functions drawn from $\mathcal{GP}(0, k_P)$, for $p = 0.5$ and $p = 3$ with $c = 0$.

Finally, Figure 55 shows the plot of the (Hadamard) product of two squared-exponential kernels, i.e., $k(x, \tilde{x}) := k_{SE}(x, \tilde{x} \mid \sigma_f^2, l_1) \odot k_{SE}(x, \tilde{x} \mid \sigma_f^2, l_2)$, for two different combinations of length-scale hyperparameters. The resulting kernels also exhibit a bell shape, but are narrower than the original squared-exponential kernels in Figure 1. The right plot in Figure 55 shows three sample functions drawn from $\mathcal{GP}(0, k_{SE} \odot k_{SE})$ with length-scale hyperparameters $l_1 = \frac{1}{2}$ and $l_2 = 3$.

So far, all plots have involved kernels defined on one-dimensional domains, i.e., $k : \mathbb{X} \times \mathbb{X} \rightarrow \mathbb{R}$ for some $\mathbb{X} \subseteq \mathbb{R}$. However, kernels naturally extend to multi-dimensional domains $\mathbb{X} \subseteq \mathbb{R}^d$. For $d \geq 2$, a common construction is a product kernel over individual inputs:

$$k(\mathbf{x}, \tilde{\mathbf{x}}) = \prod_{i=1}^d k(\mathbf{x}_i, \tilde{\mathbf{x}}_i), \quad \mathbf{x}, \mathbf{x}_i, \tilde{\mathbf{x}}, \tilde{\mathbf{x}}_i \in \mathbb{X} \subseteq \mathbb{R}^d, \quad (174)$$

The corresponding covariance is the Kronecker product of the individual covariance matrices, i.e., $\mathbf{K} = \mathbf{K}_1 \otimes \dots \otimes \mathbf{K}_d$. Since the Kronecker product of positive semi-definite matrices is positive semi-definite, it follows from Definition 2.6 that this product kernel k defined above is a positive semi-definite covariance kernel. Next, let $d = 2$. Similar to the one-dimensional cases above, Figure 56 shows a slice plot of a two-dimensional product kernel of the form $k(\mathbf{x}, \tilde{\mathbf{x}}) = k_{SE}(\mathbf{x}, \tilde{\mathbf{x}} \mid 1, l_1 = \frac{1}{2}) \cdot k_{SE}(\mathbf{x}, \tilde{\mathbf{x}} \mid 1, l_2 = 3)$, where $\mathbf{x} := (x_1, x_2)^T$ and $\tilde{\mathbf{x}} := (\tilde{x}_1, \tilde{x}_2)^T$ with x_2 and $\tilde{x}_2$ set to zero. Additionally, the middle and right plot of 56 display sample functions drawn from $\mathcal{GP}(\mathbf{0}, k_{SE} \cdot k_{SE})$ for $l = \frac{1}{2}$ and $l = 3$, respectively.

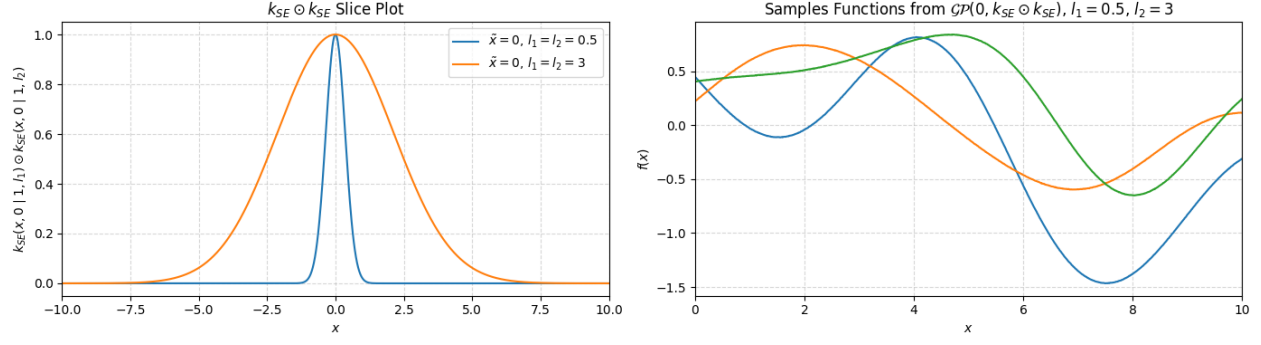


Figure 55: $k_\pi \odot k_M$ slice plot and sample functions drawn from $\mathcal{GP}(0, k_\pi \odot k_M)$.

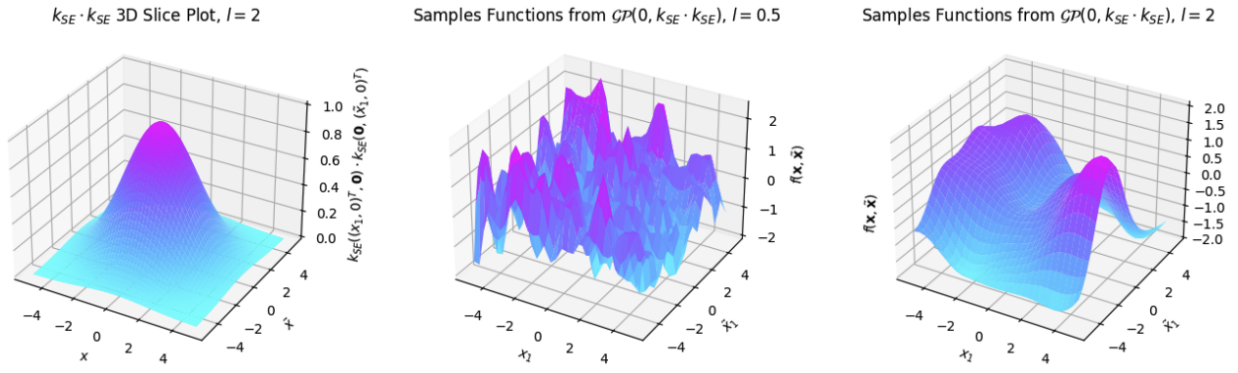


Figure 56: Slice plot of $k_{SE} \cdot k_{SE}$ with two-dimensional input vectors, and sample functions drawn from $\mathcal{GP}(0, k_{SE} \cdot k_{SE})$ for $l = \frac{1}{2}$ and $l = 2$.

6.8 Some Proofs & Derivations

6.8.1 Subsection 2.1 – Expectation and Covariance as Linear Functionals

6.8.1.1 Proof of Proposition 2.1

First, T is well-defined if $\int_Y |\kappa(x, y) \cdot f(y)| \, d\nu_Y(y) < \infty$, $\forall x \in X$. Pick arbitrary $f \in L^p(Y, \nu_Y)$ and $x \in X$. Since we assume that $\kappa(x, \cdot) \in L^{\bar{p}}(Y, \nu_Y)$, we have that $\|\kappa(x, \cdot)\|_{\bar{p}} < \infty$. In addition, $\|f\|_p < \infty$. Hence, since $\frac{1}{p} + \frac{1}{\bar{p}} = 1$, Hölder's inequality gives us the following result:

$$\begin{aligned} \int_Y |\kappa(x, y) \cdot f(y)| \, d\nu_Y(y) &= \|\kappa(x, \cdot) \cdot f(\cdot)\|_1 \\ &\leq \|\kappa(x, \cdot)\|_{\bar{p}} \cdot \|f\|_p < \infty. \end{aligned} \quad (175)$$

Second, T is a linear operator because the two conditions in Definition 6.1 are satisfied:

$$\begin{aligned} (T[\lambda \cdot f + g])(x) &= \int_Y \kappa(x, y) \cdot (\lambda \cdot f + g)(y) \, d\nu_Y(y) \\ &= \lambda \cdot \int_Y \kappa(x, y) \cdot f(y) \, d\nu_Y(y) + \int_Y \kappa(x, y) \cdot g(y) \, d\nu_Y(y) \\ &= \lambda \cdot (T[f])(x) + (T[g])(x), \end{aligned} \quad (176)$$

for all $f, g \in L^p(Y, \nu_Y)$, $x \in X$ and $\lambda \in \mathbb{R}$.

Third, T is bounded, because there exists a constant $C \in \mathbb{R}_{>0}$ such that the condition in Definition 6.2 is satisfied:

$$\begin{aligned} \|T[f]\|_{L^q(X, \nu_X)} &= \left\| \int_Y \kappa(x, y) \cdot f(y) \, d\nu_Y(y) \right\|_{L^q(X, \nu_X)} \\ &\stackrel{(\star)}{\leq} \left\| \|\kappa(x, \cdot)\|_{\bar{p}} \cdot \|f\|_p \right\|_{L^q(X, \nu_X)} \\ &= \left\| \|\kappa(x, \cdot)\|_{\bar{p}} \right\|_{L^q(X, \nu_X)} \cdot \|f\|_{L^p(Y, \nu_Y)} =: C \cdot \|f\|_{L^p(Y, \nu_Y)}. \end{aligned} \quad (177)$$

Note that $C > 0$, since the double norm is zero if and only if $\kappa(x, \cdot) \equiv 0$. Furthermore, observe that, for example, $\|f\|_{L^p(Y, \nu_Y)} = \|f\|_p$, i.e., equivalent norm notation. Moreover, inequality $(\star)$ holds because, by equation (175), the following inequality also holds:

$$|(T[f])(x)| \leq \int_Y |\kappa(x, y) \cdot f(y)| \, d\nu_Y(y) \leq \|\kappa(x, \cdot)\|_{L^{\bar{p}}(Y, \nu_Y)} \cdot \|f\|_{L^p(Y, \nu_Y)}. \quad (178)$$

Therefore, by monotonicity of integrals, we obtain:

$$\begin{aligned} \|T[f]\|_{L^q(X, \nu_X)} &= \left(\int_X |(T[f])(x)|^q \, d\nu_X(x) \right)^{1/q} \\ &\leq \left(\int_X \left| \|\kappa(x, \cdot)\|_{L^{\bar{p}}(Y, \nu_Y)} \cdot \|f\|_{L^p(Y, \nu_Y)} \right|^q \, d\nu_X(x) \right)^{1/q} \\ &= \left\| \|\kappa(x, \cdot)\|_{L^{\bar{p}}(Y, \nu_Y)} \cdot \|f\|_{L^p(Y, \nu_Y)} \right\|_{L^q(X, \nu_X)}. \end{aligned} \quad (179)$$

Since $\|f\|_{L^p(Y, \nu_Y)}$ does not depend on $x \in X$, it can be factored out of the $L^q(X, \nu_X)$ -norm. $\square$

6.8.1.2 Proof of Proposition 2.2

Suppose $1 \leq p, q \leq \infty$ are such that $\frac{1}{p} + \frac{1}{q} = 1$. Define the constant function $g \equiv 1$. Hence, $g \in L^q(\Omega, \nu)$, because $\|g\|_q = (\int_{\Omega} |1|^q \, d\nu)^{1/q} = (\nu(\Omega))^{1/q} = 1^{1/q} = 1 < \infty$. Therefore, if $p < \infty$ then by Hölder's inequality:

$$\|f\|_1 = \int_{\Omega} |f| \, d\nu = \int_{\Omega} |f \cdot g| \, d\nu = \|f \cdot g\|_1 \leq \|f\|_p \cdot \|1\|_q = \|f\|_p. \quad (180)$$

Alternatively, if $p = \infty$ then by definition of the essential supremum:

$$\|f\|_1 = \int_{\Omega} |f| \, d\nu \leq \|f\|_{\infty} \cdot \nu(\Omega) = \|f\|_{\infty}. \quad (181)$$

□

6.8.1.3 Proof of Lemma 2.1

Pick an arbitrary $\mathbf{u} \in \mathbb{H}$. First, we show that $F_{\mathbf{u}}$ is linear. For arbitrary $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{H}$ and $\lambda_1, \lambda_2 \in \mathbb{R}$, and by using Definition 6.4:

$$\begin{aligned} F_{\mathbf{u}}(\lambda_1 \cdot \mathbf{v}_1 + \lambda_2 \cdot \mathbf{v}_2) &= \text{Cov}(\mathbf{u}, \lambda_1 \cdot \mathbf{v}_1 + \lambda_2 \cdot \mathbf{v}_2) \\ &= \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{u} \rangle \langle \mathbf{X} - \mathbf{m}, \lambda_1 \cdot \mathbf{v}_1 + \lambda_2 \cdot \mathbf{v}_2 \rangle] \\ &= \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{u} \rangle \langle \lambda_1 \cdot \mathbf{v}_1 + \lambda_2 \cdot \mathbf{v}_2, \mathbf{X} - \mathbf{m} \rangle] \\ &= \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{u} \rangle (\lambda_1 \cdot \langle \mathbf{v}_1, \mathbf{X} - \mathbf{m} \rangle + \lambda_2 \cdot \langle \mathbf{v}_2, \mathbf{X} - \mathbf{m} \rangle)] \\ &= \lambda_1 \cdot \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{u} \rangle \langle \mathbf{v}_1, \mathbf{X} - \mathbf{m} \rangle] + \lambda_2 \cdot \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{u} \rangle \langle \mathbf{v}_2, \mathbf{X} - \mathbf{m} \rangle] \\ &= \lambda_1 \cdot \text{Cov}(\mathbf{u}, \mathbf{v}_1) + \lambda_2 \cdot \text{Cov}(\mathbf{u}, \mathbf{v}_2). \end{aligned} \quad (182)$$

Second, to show that $F_{\mathbf{u}}$ is bounded, we need to show that the operator norm $\|F_{\mathbf{u}}\| = \sup_{\|\mathbf{v}\|=1} \|F_{\mathbf{u}}(\mathbf{v})\|$ in (160) is bounded. Since $F_{\mathbf{u}}$ is a functional, the norm $\|F_{\mathbf{u}}(\mathbf{v})\|$ is equal to the absolute value $|F_{\mathbf{u}}(\mathbf{v})|$. Therefore, we need to show that there exists a $C \in \mathbb{R}_{\geq 0}$ such that:

$$\|F_{\mathbf{u}}(\mathbf{v})\| = |F_{\mathbf{u}}(\mathbf{v})| \leq C \cdot \|\mathbf{v}\|, \quad \forall \mathbf{v} \in \mathbb{H}. \quad (183)$$

By the Cauchy-Schwarz inequality, i.e., Hölder's inequality in Theorem 6.3 with $p = q = 2$, we have:

$$\begin{aligned} |F_{\mathbf{u}}(\mathbf{v})| &= |\mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{u} \rangle \langle \mathbf{X} - \mathbf{m}, \mathbf{v} \rangle]| \\ &\leq \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{u} \rangle^2]^{\frac{1}{2}} \cdot \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{v} \rangle^2]^{\frac{1}{2}} \\ &= c_{\mathbf{u}} \cdot \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{v} \rangle^2]^{\frac{1}{2}} =: (\star). \end{aligned} \quad (184)$$

Note that $c_{\mathbf{u}} < \infty$ because $\mathbf{X}$ has finite second moment and $\mathbf{u}$ is fixed. [Sterk and de Snoo \(2025\)](#) state the Cauchy-Schwarz for real-valued inner products in a Hilbert space $\mathbb{H}$:

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle, \quad \mathbf{x}, \mathbf{y} \in \mathbb{H}. \quad (185)$$

Consequently:

$$(\star) \leq c_{\mathbf{u}} \cdot \mathbb{E}[\langle \mathbf{X} - \mathbf{m}, \mathbf{X} - \mathbf{m} \rangle \langle \mathbf{u}, \mathbf{u} \rangle]^{\frac{1}{2}} \leq c_{\mathbf{u}} \cdot \mathbb{E}[\|\mathbf{X} - \mathbf{m}\|]^{\frac{1}{2}} \cdot \|\mathbf{v}\| =: C \cdot \|\mathbf{v}\|. \quad (186)$$

So, $F_{\mathbf{u}}$ is a bounded linear functional. □

6.8.2 Subsection 2.2 – Gaussian Measures

6.8.2.1 Proof of Corollary 2.2

We use the following lemma from [van der Vaart \(1997\)](#) in this proof.

Lemma 6.1. $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \mathbf{X})$ if and only if $\mathbf{a}^T \mathbf{X} \sim \mathcal{N}_1(\mathbf{a}^T \boldsymbol{\mu}, \mathbf{a}^T \mathbf{X} \mathbf{a}), \forall \mathbf{a} \in \mathbb{R}^d$.

First, the aim is to prove normality of $\mathbf{A}\mathbf{X}$ using Lemma 6.1. Pick an arbitrary $\mathbf{a} \in \mathbb{R}^d$. Suppose $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \mathbf{X})$. And define a linear operator $L : \mathbb{R}^d \rightarrow \mathbb{R}^m$ by $L[\mathbf{X}] := \mathbf{A}\mathbf{X}$. By associativity:

$$\mathbf{a}^T L[\mathbf{X}] = \mathbf{a}^T (\mathbf{A}\mathbf{X}) = (\mathbf{a}^T \mathbf{A})\mathbf{X} = (\mathbf{A}^T (\mathbf{a}^T)^T)^T \mathbf{X} = (\mathbf{A}^T \mathbf{a})^T \mathbf{X} \in \mathbb{R}. \quad (187)$$

Define $\mathbf{c} := \mathbf{A}^T \mathbf{a} \in \mathbb{R}^d$. Hence, by Lemma 6.1, $\mathbf{c}^T \mathbf{X} \sim \mathcal{N}_1(\mathbf{c}^T \boldsymbol{\mu}, \mathbf{c}^T \boldsymbol{\Sigma} \mathbf{c})$. Since $\mathbf{c}^T \mathbf{X} = \mathbf{a}^T L[\mathbf{X}]$ by equation (187), the preceding is equivalent to:

$$\begin{aligned} \mathbf{a}^T L[\mathbf{X}] &= (\mathbf{A}^T \mathbf{a})^T \mathbf{X} \sim \mathcal{N}_1 \left((\mathbf{A}^T \mathbf{a})^T \boldsymbol{\mu}, (\mathbf{A}^T \mathbf{a})^T \boldsymbol{\Sigma} (\mathbf{A}^T \mathbf{a}) \right) \\ &\stackrel{d}{=} \mathcal{N}_1 \left(\mathbf{a}^T (\mathbf{A} \boldsymbol{\mu}), (\mathbf{a}^T \mathbf{A}) \boldsymbol{\Sigma} (\mathbf{A}^T \mathbf{a}) \right) \\ &\stackrel{d}{=} \mathcal{N}_1 \left(\mathbf{a}^T (\mathbf{A} \boldsymbol{\mu}), \mathbf{a}^T (\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^T) \mathbf{a} \right). \end{aligned} \quad (188)$$

Therefore, invoking Lemma 6.1 in the reverse direction yields:

$$L[\mathbf{X}] = \mathbf{A} \mathbf{X} \sim \mathcal{N}_d(\mathbf{A} \boldsymbol{\mu}, \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^T) \stackrel{d}{=} \mathcal{N}_d(L[\boldsymbol{\mu}], L \boldsymbol{\Sigma} L^*), \quad (189)$$

where $L^* : \mathbb{R}^m \rightarrow \mathbb{R}^d$ is the (Hermitan) adjoint operator. Notice that $\mathbb{R}^n$ is a Hilbert space with the standard inner product, $\forall \eta \in \mathbb{N}$.

Next, since $\mathbf{A}$ and $\mathbf{b}$ are both deterministic, $\mathbb{E}[L[\mathbf{X}] + \mathbf{b}] = L[\mathbb{E}[\mathbf{X}]] + \mathbf{b} = \mathbf{A} \boldsymbol{\mu} + \mathbf{b}$ and $\text{Cov}(L[\mathbf{X}] + \mathbf{b}) = \text{Cov}(L[\mathbf{X}]) = L \text{Cov}(\mathbf{X}) L^* = L \boldsymbol{\Sigma} L^*$. Therefore, the distribution of $\mathbf{A} \mathbf{X} + \mathbf{b}$ is given by:

$$L[\mathbf{X}] + \mathbf{b} = \mathbf{A} \mathbf{X} + \mathbf{b} \sim \mathcal{N}_d(\mathbf{A} \boldsymbol{\mu} + \mathbf{b}, \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^T) \stackrel{d}{=} \mathcal{N}_d(L[\boldsymbol{\mu}] + \mathbf{b}, L \boldsymbol{\Sigma} L^*). \quad (190)$$

□

6.8.2.2 Proof of Theorem 2.4

All the details of the proof can for example be found in [Soch et al. \(2025\)](#). A sketch of the proof is presented here. Assume that $p_{\mathbf{X}_2}(\mathbf{x}_2) > 0$. Since $\boldsymbol{\Sigma}$ is non-singular and $\mathbb{P}_{\mathbf{X}}$ is absolutely continuous with respect to λ_d , the joint PDF $p_{\mathbf{X}_1, \mathbf{X}_2}(\mathbf{x}_1, \mathbf{x}_2)$ exists. Furthermore, since $\mathbb{P}_{\mathbf{X}_1}$ is absolutely continuous with respect to λ_l , the marginal PDF of $\mathbf{X}_1$ exists and is given in equation (119). Hence, the corresponding conditional PDF of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$ exists and equation (122) defines it as:

$$p_{\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2}(\mathbf{x}_1 | \mathbf{x}_2) = \frac{p_{\mathbf{X}_1, \mathbf{X}_2}(\mathbf{x}_1, \mathbf{x}_2)}{p_{\mathbf{X}_2}(\mathbf{x}_2)}. \quad (191)$$

The corresponding probability distribution measure is given by:

$$\mathbb{P}_{\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2}(B) = \int_B p_{\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2}(\mathbf{x}_1 | \mathbf{x}_2) d\mathbf{x}_1, \quad B \in \mathcal{B}^m. \quad (192)$$

Plugging the multivariate Gaussian PDF from equation (28) into the conditional PDF expression and working out the matrix algebra by using the matrix inversion lemma ([Rasmussen and Williams, 2008](#)):

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} (\mathbf{A} - \mathbf{B} \mathbf{D}^{-1} \mathbf{C})^{-1} & -(\mathbf{A} - \mathbf{B} \mathbf{D}^{-1} \mathbf{C})^{-1} \mathbf{B} \mathbf{D}^{-1} \\ -\mathbf{D}^{-1} \mathbf{C} (\mathbf{A} - \mathbf{B} \mathbf{D}^{-1} \mathbf{C})^{-1} & \mathbf{D}^{-1} + \mathbf{D}^{-1} \mathbf{C} (\mathbf{A} - \mathbf{B} \mathbf{D}^{-1} \mathbf{C})^{-1} \mathbf{B} \mathbf{D}^{-1} \end{bmatrix}, \quad (193)$$

yields:

$$p_{\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2}(\mathbf{x}_1 | \mathbf{x}_2) = \frac{1}{(2\pi)^{m/2} \cdot \sqrt{\det(\boldsymbol{\Sigma}_{1|2})}} \cdot \exp \left\{ -\frac{1}{2} \cdot (\mathbf{x}_1 - \boldsymbol{\mu}_{1|2})^T \boldsymbol{\Sigma}_{1|2}^{-1} (\mathbf{x}_1 - \boldsymbol{\mu}_{1|2}) \right\}. \quad (194)$$

□

6.8.3 Subsection 2.3 – Gaussian Process

6.8.3.1 Proof of Existence of Gaussian Processes

If $f \sim \mathcal{GP}(m, k)$, then $f : \Omega \rightarrow \mathbb{R}^{\mathbb{X}}$ is a stochastic process with distribution $\mathbb{P}_f$. By Definition 6.8, the finite-dimensional distribution of f for distinct $x_1, \dots, x_n \in \mathbb{X}$, is defined as the following probability measure on $\mathbb{R}^n$:

$$\mathbb{P}_{x_1, \dots, x_n} := \mathbb{P} \circ \mathbf{f}_{\mathbf{X}}^{-1} = \mathcal{N}_n(\mathbf{m}_{\mathbf{X}}, k_{\mathbf{X}\mathbf{X}}). \quad (195)$$

That is, $\mathbf{f}_{\mathbf{X}} \sim \mathcal{N}_n(\mathbf{m}_{\mathbf{X}}, k_{\mathbf{X}\mathbf{X}})$. For $B_n \in \mathcal{B}^n$:

$$\mathbb{P}_{x_1, \dots, x_n}(B_n) = \mathcal{N}(\mathbf{m}_{\mathbf{X}}, k_{\mathbf{X}\mathbf{X}})(B_n) = \mathbb{P}(\{\omega \in \Omega \mid (f(x_1, \omega), \dots, f(x_n, \omega)) \in B_n\}). \quad (196)$$

Kolmogorov's Existence Theorem, i.e., Theorem 6.5, provides sufficient conditions regarding finite-dimensional distributions for existence of a stochastic process. Hence, if $f \sim \mathcal{GP}(m, k)$ satisfies the conditions C1) and C2) in equations (169) and (170), respectively, then Kolmogorov's Existence Theorem says that the Gaussian process exists.

First, let us show that condition C1) is satisfied. Let τ be any permutation of $\{1, \dots, n\}$, i.e., $\tau((1 \dots n)) = (\tau(1) \dots \tau(n))$. We want to show that

$$\mathbb{P}_{x_1, \dots, x_n}(B_1 \times \dots \times B_n) = \mathbb{P}_{x_{\tau(1)}, \dots, x_{\tau(n)}}(B_{\tau(1)} \times \dots \times B_{\tau(n)}), \quad (197)$$

for distinct $x_1, \dots, x_n \in \mathbb{X}$ and for all $B_1, \dots, B_n \in \mathcal{B}$. Hence, pick arbitrary but distinct $x_1, \dots, x_n \in \mathbb{X}$. Further, define the following use of notation:

$$\tau(\mathbf{X}) := (x_{\tau(1)}, \dots, x_{\tau(n)}), \quad \mathbf{f}_{\tau(\mathbf{X})} := (f(\mathbf{x}_{\tau(1)}), \dots, f(\mathbf{x}_{\tau(n)}))^T = \mathbf{P}_{\tau} \mathbf{f}_{\mathbf{X}}, \quad (198)$$

for the permutation matrix $\mathbf{P}_{\tau}$ corresponding to τ . A permutation matrix is a square matrix where exactly one entry has value 1 in each row and column with all the other entries 0. For example, $\mathbf{f}_{\tau(\mathbf{X})}$ with $\tau((123)) := (132)$ is given by:

$$\mathbf{f}_{\tau(\mathbf{X})} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} f(\mathbf{x}_1) \\ f(\mathbf{x}_2) \\ f(\mathbf{x}_3) \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}_1) \\ f(\mathbf{x}_3) \\ f(\mathbf{x}_2) \end{bmatrix}. \quad (199)$$

If $f \sim \mathcal{GP}(m, k)$, then $\mathbf{f}_{\mathbf{X}} \sim \mathcal{N}(\mathbf{m}_{\mathbf{X}}, k_{\mathbf{X}\mathbf{X}})$. Hence, by Corollary 2.2:

$$\mathbf{f}_{\tau(\mathbf{X})} = \mathbf{P}_{\tau} \mathbf{f}_{\mathbf{X}} \sim \mathcal{N}_n(\mathbf{P}_{\tau} \mathbf{m}_{\mathbf{X}}, \mathbf{P}_{\tau} k_{\mathbf{X}\mathbf{X}} \mathbf{P}_{\tau}^T). \quad (200)$$

So, equation (197) is satisfied and hence condition C1) is satisfied for a Gaussian process.

Second, condition C2) is satisfied because a Gaussian distribution possesses the property that it is closed under marginalisation presented in Theorem 2.3.

Thus, Kolmogorov's Existence Theorem ensures us that a Gaussian process $f \sim \mathcal{GP}(m, k)$ exists. $\square$

6.8.3.2 Proof of Equation (54)

$$\begin{aligned} \text{Cov}(h_1, h_2) &= \text{Cov}(\langle f, h_1 \rangle, \langle f, h_2 \rangle) \\ &= \text{Cov} \left(\int_{\mathbb{X}} f(\mathbf{x}) h_1(\mathbf{x}) \, d\nu(\mathbf{x}), \int_{\mathbb{X}} f(\tilde{\mathbf{x}}) h_2(\tilde{\mathbf{x}}) \, d\nu(\tilde{\mathbf{x}}) \right) \\ &= \int_{L^2(\mathbb{X}, \nu)} \langle f, h_1 \rangle \langle f, h_2 \rangle \, d\mathbb{P}_f(f) \\ &= \int_{L^2(\mathbb{X}, \nu)} \left(\int_{\mathbb{X}} f(\mathbf{x}) h_1(\mathbf{x}) \, d\nu(\mathbf{x}) \right) \left(\int_{\mathbb{X}} f(\tilde{\mathbf{x}}) h_2(\tilde{\mathbf{x}}) \, d\nu(\tilde{\mathbf{x}}) \right) \, d\mathbb{P}_f(f) \\ &\stackrel{(1)}{=} \int_{L^2(\mathbb{X}, \nu)} \left(\int_{\mathbb{X}} \int_{\mathbb{X}} f(\mathbf{x}) h_1(\mathbf{x}) f(\tilde{\mathbf{x}}) h_2(\tilde{\mathbf{x}}) \, d\nu(\mathbf{x}) \, d\nu(\tilde{\mathbf{x}}) \right) \, d\mathbb{P}_f(f) \\ &\stackrel{(2)}{=} \int_{\mathbb{X}} \int_{\mathbb{X}} \left(\int_{L^2(\mathbb{X}, \nu)} f(\mathbf{x}) f(\tilde{\mathbf{x}}) \, d\mathbb{P}_f(f) \right) h_1(\mathbf{x}) h_2(\tilde{\mathbf{x}}) \, d\nu(\mathbf{x}) \, d\nu(\tilde{\mathbf{x}}) \\ &\stackrel{(3)}{=} \int_{\mathbb{X}} \int_{\mathbb{X}} k(\mathbf{x}, \tilde{\mathbf{x}}) h_1(\mathbf{x}) h_2(\tilde{\mathbf{x}}) \, d\nu(\mathbf{x}) \, d\nu(\tilde{\mathbf{x}}) \\ &\stackrel{(4)}{=} \int_{\mathbb{X}} \left(\int_{\mathbb{X}} k(\mathbf{x}, \tilde{\mathbf{x}}) h_1(\mathbf{x}) \, d\nu(\mathbf{x}) \right) h_2(\tilde{\mathbf{x}}) \, d\nu(\tilde{\mathbf{x}}) \\ &= \int_{\mathbb{X}} (T_k[h_1])(\tilde{\mathbf{x}}) \cdot h_2(\tilde{\mathbf{x}}) \, d\nu(\tilde{\mathbf{x}}) = \langle T_k h_1, h_2 \rangle_{L^2(\mathbb{X}, \nu)}. \end{aligned} \quad (201)$$

Equalities (1), (2) and (4) hold by Fubini's Theorem 2.2. In particular, for equality (2), Fubini's Theorem is applied on the function $F : (L^2(\mathbb{X}, \nu), \mathcal{B}^d, \nu) \times (\mathbb{X} \times \mathbb{X}, \mathcal{B}^d \otimes \mathcal{B}^d, \nu \otimes \nu) \rightarrow \mathbb{R}$ defined by $F(f, \mathbf{x}, \tilde{\mathbf{x}}) := f(\mathbf{x})f(\tilde{\mathbf{x}})h_1(\mathbf{x})h_2(\tilde{\mathbf{x}})$, where $h_1, h_2 \in L^2(\mathbb{X}, \nu)$. Note that the Borel σ -algebra on $\mathbb{X} \subseteq \mathbb{R}^d$ is sometimes denoted as $\mathcal{B}^d$ instead of $\mathcal{B}(\mathbb{X})$. Moreover, σ -finiteness of $(L^2(\mathbb{X}, \nu), \mathcal{B}^d, \nu)$ is inherited from the underlying measure space $(\mathbb{X}, \mathcal{B}^d, \nu)$, which is assumed to be σ -finite. Equality (3) holds because:

$$\int_{L^2(\mathbb{X}, \nu)} f(\mathbf{x})f(\tilde{\mathbf{x}}) \, d\mathbb{P}_f(f) = \text{Cov}(f(\mathbf{x}), f(\tilde{\mathbf{x}})) =: k(\mathbf{x}, \tilde{\mathbf{x}}). \quad (202)$$

□

6.8.3.3 Proof of Proposition 2.5

First, we prove *i*). Since the k_i 's are covariance kernels, they are positive semi-definite kernels. Hence, each k_i satisfies definition 3 from Definition 2.6: $\mathbf{v}^T \mathbf{K}_{\mathbf{X}\mathbf{X}, i} \mathbf{v} \geq 0$, $\forall \mathbf{v} \in \mathbb{R}^n$, where $\mathbf{K}_{\mathbf{X}\mathbf{X}, i}$ denotes the Gram matrix of k_i for any $\mathbf{X} \in \mathbb{X}^n$. Consequently:

$$\mathbf{v}^T \left(\sum_{i=1}^m a_i \cdot \mathbf{K}_{\mathbf{X}\mathbf{X}, i} \right) \mathbf{v} = \sum_{i=1}^m a_i \cdot \mathbf{v}^T \mathbf{K}_{\mathbf{X}\mathbf{X}, i} \mathbf{v} \geq 0, \quad \forall \mathbf{v} \in \mathbb{R}^n. \quad (203)$$

Hence, $\sum_{i=1}^m a_i \cdot k_i(\mathbf{x}, \tilde{\mathbf{x}})$ is a positive semi-definite kernel and thus a covariance kernel.

Second, we prove *ii*). Often, this is proven by Schur's product theorem which states that the Hadamard product $\odot$ of two positive semi-definite matrices is positive semi-definite; see [Shawe-Taylor and Cristianini \(2004\)](#). The Hadamard product of two matrices is the element-wise product of the two matrices.

Alternatively, Mercer's Theorem can be used to prove this; see [Müller et al. \(2001\)](#). We present a probabilistic proof. Let $f_1 \sim \mathcal{GP}(0, \mathbf{K}_{\mathbf{X}\mathbf{X}, 1})$ and $f_2 \sim \mathcal{GP}(0, \mathbf{K}_{\mathbf{X}\mathbf{X}, 2})$ be independent Gaussian processes.

Note that the assumption $m \equiv 0$ is without loss of generality, since we might replace f_i by $f_i - m_i$ if $m_i \neq 0$. Pick arbitrary $\mathbf{X} := \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \in \mathbb{X}^n$. Next, suppose that

$\mathbf{f}_{\mathbf{X}, 1} := (f_1(\mathbf{x}_1), \dots, f_1(\mathbf{x}_n))^T \sim \mathcal{N}_n(\mathbf{0}, \mathbf{K}_{\mathbf{X}\mathbf{X}, 1})$ and $\mathbf{f}_{\mathbf{X}, 2} := (f_2(\mathbf{x}_1), \dots, f_2(\mathbf{x}_n))^T \sim \mathcal{N}_n(\mathbf{0}, \mathbf{K}_{\mathbf{X}\mathbf{X}, 2})$ are independently distributed. Further, define $\mathbf{F} := \mathbf{f}_{\mathbf{X}, 1} \odot \mathbf{f}_{\mathbf{X}, 2}$, where $\mathbf{F}_i = (f_1 \odot f_2)(\mathbf{x}_i) = f_1(\mathbf{x}_i) \cdot f_2(\mathbf{x}_i)$ for $i \in \{1, \dots, n\}$. Then, the covariance between components of $\mathbf{F}$ is:

$$\begin{aligned} \text{Cov}(\mathbf{F}_i, \mathbf{F}_j) &= \text{Cov}(f_1(\mathbf{x}_i) \cdot f_2(\mathbf{x}_i), f_1(\mathbf{x}_j) \cdot f_2(\mathbf{x}_j)) \\ &= \mathbb{E}[f_1(\mathbf{x}_i) \cdot f_2(\mathbf{x}_i) \cdot f_1(\mathbf{x}_j) \cdot f_2(\mathbf{x}_j)] \\ &= \mathbb{E}[f_1(\mathbf{x}_i) \cdot f_1(\mathbf{x}_j)] \cdot \mathbb{E}[f_2(\mathbf{x}_i) \cdot f_2(\mathbf{x}_j)] \\ &= \text{Cov}(f_1(\mathbf{x}_i), f_1(\mathbf{x}_j)) \cdot \text{Cov}(f_2(\mathbf{x}_i), f_2(\mathbf{x}_j)) \\ &= k_1(\mathbf{x}_i, \mathbf{x}_j) \cdot k_2(\mathbf{x}_i, \mathbf{x}_j) = k(\mathbf{x}_i, \mathbf{x}_j) \end{aligned} \quad (204)$$

where the third equality holds by independence of f_1 and f_2 . Therefore, the Gram matrix of k defined by $\mathbf{K}_{\mathbf{X}\mathbf{X}} = \mathbf{K}_{\mathbf{X}\mathbf{X}, 1} \odot \mathbf{K}_{\mathbf{X}\mathbf{X}, 2} = [k_1(\mathbf{x}_i, \mathbf{x}_j) \cdot k_2(\mathbf{x}_i, \mathbf{x}_j)]_{i,j=1}^n$ is equal to the covariance matrix (operator) of $\mathbf{F}$. By Proposition 6.2, covariance matrices are positive semi-definite. So, since $\mathbf{K}_{\mathbf{X}\mathbf{X}}$ is positive semi-definite, k is a positive semi-definite kernel by Definition 2.6. □

6.8.3.4 Proof of Theorem 2.6

First, we prove that $T[f]$ is a $\mathbb{H}_2$ -valued Gaussian random variable. The generalised version of 2.4 states that $\mathbb{P}_{T[f]}$ is Gaussian, if for every linear functional $\phi \in \mathbb{H}_2^*$, $\mathbb{P}_{T[f]} \circ \phi^{-1}$ is also a Gaussian measure. Note that since f is a $\mathbb{H}_1$ -valued Gaussian random variable, $\mathbb{P}_f \circ \phi_1^{-1}$ is a Gaussian measure for every $\phi_1 \in \mathbb{H}_1^*$. Pick an arbitrary $\phi \in \mathbb{H}_2^* := \{\text{all linear bounded functionals } \phi : \mathbb{H}_2 \rightarrow \mathbb{R}\}$. Then, $\phi \circ T : \mathbb{H}_1 \rightarrow \mathbb{R}$ is also a bounded linear functional on $\mathbb{H}_1$. Hence:

$$\phi(T[f]) = (\phi \circ T)[f], \quad (205)$$

is a one-dimensional Gaussian random variable since f is Gaussian. So, $T[f]$ is a $\mathbb{H}_2$ -valued Gaussian random variable. Second, $\mathbb{E}[T[f]] = T[\mathbb{E}[f]]$ holds by Theorem 2.1 with $\mathbf{X} := f$, $\mathbb{B}_1 := \mathbb{H}_1$ and $\mathbb{B}_2 := \mathbb{H}_2$.

Third, $\Sigma_T := T\Sigma T^*$ is the covariance operator of $T[f]$, if:

$$\text{Cov}(\langle T[f], \mathbf{u}_2 \rangle_{\mathbb{H}_2}, \langle T[f], \mathbf{v}_1 \rangle_{\mathbb{H}_2}) = \text{Cov}(\mathbf{u}_2, \mathbf{v}_2) = \langle \Sigma \mathbf{u}_1, \mathbf{v}_1 \rangle_{\mathbb{H}_2}, \quad \forall \mathbf{u}_2, \mathbf{v}_2 \in \mathbb{H}_2. \quad (206)$$

Pick arbitrary $\mathbf{u}_2, \mathbf{v}_2 \in \mathbb{H}_2$. By definition of the adjoint operator $T : \mathbb{H}_2 \rightarrow \mathbb{H}_1$ of T , the following holds:

$$\langle T[f], \mathbf{u}_2 \rangle_{\mathbb{H}_2} = \langle f, T^*[\mathbf{u}_2] \rangle_{\mathbb{H}_1} \quad \& \quad \langle T[f], \mathbf{v}_2 \rangle_{\mathbb{H}_2} = \langle f, T^*[\mathbf{v}_2] \rangle_{\mathbb{H}_1}. \quad (207)$$

Hence, using the defining property of Σ and the definition of T^* , we obtain:

$$\begin{aligned} \text{Cov}(\langle T[f], \mathbf{u}_2 \rangle_{\mathbb{H}_2}, \langle T[f], \mathbf{v}_1 \rangle_{\mathbb{H}_2}) &= \text{Cov}(\langle f, T^*[\mathbf{u}_2] \rangle_{\mathbb{H}_1}, \langle f, T^*[\mathbf{v}_2] \rangle_{\mathbb{H}_1}) \\ &= \text{Cov}(T^*[\mathbf{u}_2], T^*[\mathbf{v}_2]) \\ &= \langle \Sigma T^*[\mathbf{u}_2], T^*[\mathbf{v}_2] \rangle_{\mathbb{H}_1} \\ &= \langle T[\Sigma T^*[\mathbf{u}_2]], \mathbf{v}_2 \rangle_{\mathbb{H}_2} \\ &= \langle T\Sigma T^* \mathbf{u}_2, \mathbf{v}_2 \rangle_{\mathbb{H}_2}. \end{aligned} \quad (208)$$

So, $\Sigma_T := T\Sigma T^*$ is the covariance operator of $T[f]$. □

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